

Non-Equilibrium Statistical Physics

Lecture notes

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1 Introduction

1.1 From equilibrium to non-equilibrium

In the classical description we start from a Hamiltonian $H = H(q_i, p_i)$, $i = 1, \dots, N$, with N the number of degrees of freedom. Solving

$$\dot{p}_i = -\frac{\partial H}{\partial q_i}, \quad \dot{q}_i = \frac{\partial H}{\partial p_i}$$

gives a trajectory in phase space starting from the initial condition Γ . We do statistics in two senses:

- $N \gg 1$;
- instead of an exact initial condition we consider a whole set of possible initial conditions with certain properties: the initial condition is a “patch” of phase space.

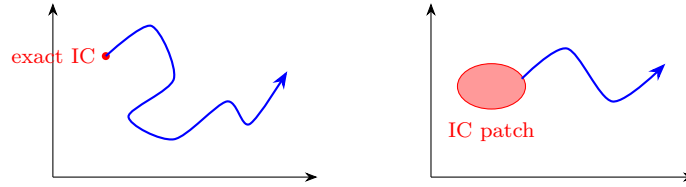


Figure 1.1: A single trajectory from an exact initial condition (left) and the evolution of a patch of initial conditions (right).

From equilibrium statistical mechanics we know that, as $t \rightarrow \infty$, the probability distribution $F(\Gamma, t)$ of finding the state Γ tends to a stationary equilibrium distribution,

$$\lim_{t \rightarrow \infty} F(\Gamma, t) = F_{\text{eq}}(\Gamma).$$

This distribution satisfies $\{F_{\text{eq}}, H\} = 0$ ($[F_{\text{eq}}, H] = 0$ in quantum mechanics), from which

$$F_{\text{eq}}(\Gamma) = F_{\text{eq}}(E, X),$$

where X stands for other conserved quantities. If there are none ($X = \emptyset$), there are two important cases:

1. If the initial condition has a well-defined energy, the microcanonical distribution

$$F_{\text{eq}}(\Gamma) = \frac{1}{\Omega} \delta(H(\Gamma) - E).$$

2. In contact with a heat bath, the Gibbs distribution

$$F_{\text{eq}}(\Gamma) = \frac{1}{Z} e^{-\beta H(\Gamma)}.$$

From $X = \emptyset$ it also follows that:

1. $F_{\text{eq}}(\Gamma) = F(E)$;
2. if Γ_1 and Γ_2 have $H(\Gamma_1) = H(\Gamma_2)$, there is t_0 such that $\Gamma_1(t_0) = \Gamma_2$, and t'_0 such that $\Gamma_2(t'_0) = \Gamma_1$. Otherwise, there would be disconnected pieces of the same energy surface; on a piece not visited, $F = 0$, violating 1.

In fact, 2 holds for all points Γ except for a set of measure zero (an example of such an exception: the Lorentz billiard).

Conclusion. If 2 does not hold, there is another conserved quantity. All of the above is for a system in equilibrium. *Out of equilibrium there is no general expression for $F(\Gamma, t)$* ; nevertheless, we can still make progress.

1.2 Examples of non-equilibrium systems

There are several ways to generate non-equilibrium states.

1. Initial condition. “Typical” states are far more numerous than “atypical” ones. Example: N particles that can be on the left or the right half of a box, $\Gamma = \{S_1, \dots, S_N\}$, $S_i \in \{S_{\text{left}}, S_{\text{right}}\}$. In equilibrium $F_{\text{eq}} = 1/2^N$: all states are equally probable. Define the macroscopic variable Π , the number of particles on the right:

$$P(\Pi) = \frac{1}{2^N} \binom{N}{\Pi},$$

which for $N \gg 1$ tends to a Gaussian centred at $N/2$ with relative width $\sim 1/\sqrt{N}$. A rare initial state is one with $\Pi \neq N/2$. It evolves irreversibly to a state with $\Pi \simeq N/2$, simply because it is more probable to evolve towards it. During this evolution to equilibrium the state is never stationary, $F = F(\Gamma, t)$: in this sense there is no equilibrium.

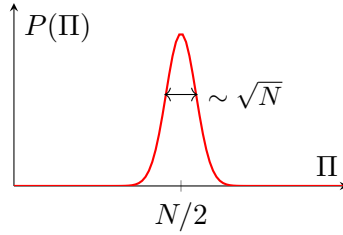


Figure 1.2: Distribution of the number of particles on the right half.

2. Boundary conditions. For example, imposing differences of voltage, temperature, velocity, etc.: a current J between two reservoirs at potentials $\phi_1 \neq \phi_2$, a heat flux between $T_1 \neq T_2$, or a shear flow between a moving plate ($v = v_0 \neq 0$) and a fixed one ($v = 0$), where the difference is generated by different forcings. In these cases one can obtain stationary distributions $F_{\text{ness}}(\Gamma)$ describing **non-equilibrium steady states** (NESS).

3. Non-conservative external forces that do not let the system recover its energy. Example: a cup of coffee with a vortex generated by a spoon has (1) an external force, (2) boundary conditions and (3) a rare initial condition, since vortices are rare compared with the Maxwell distribution.

After the 1980s two other ways of generating non-equilibrium states were found:

4. Granular media: systems with dissipative interactions, e.g. sand. There is no well-defined H , only equations of motion.

5. Active matter: systems that inject and dissipate energy at different scales. For example, a bacterium in a fluid has an internal energy (food) that it dissipates into the fluid as it moves.

In (4) and (5) there are internal degrees of freedom; in grains, for instance, phonons are excited in each collision. There are *explicit* degrees of freedom (velocity and position of the centre of mass) and *implicit* ones (positions and velocities of the atoms, phonon energies, ...). In the processes we study energy is transferred between degrees of freedom, ending up from explicit to internal ones (which increases the entropy). In general, the “temperatures” associated with the energy scales of the explicit and internal degrees of freedom satisfy

$$T_{\text{explicit}} \gg T_{\text{internal}}.$$

1.3 Detailed balance

Warning: the following analysis is only valid in equilibrium. (A proof can be found in van Kampen.) We do a classical analysis: microstates are $\Gamma = \{q_i, p_i\}_{i=1}^N$. Let Γ^t be the time evolution of the initial condition Γ , and $\bar{\Gamma}$ the time reversal, $\bar{\Gamma} := \{q_i, -p_i\}_{i=1}^N$. One can show that

- $\bar{\bar{\Gamma}} = \Gamma$;
- $d\bar{\Gamma} = d\Gamma$;
- $d\Gamma^t = d\Gamma$ (Liouville’s theorem: phase-space volumes are conserved);
- $(\bar{\Gamma}^t)^t = \bar{\Gamma}$: **microscopic reversibility**.

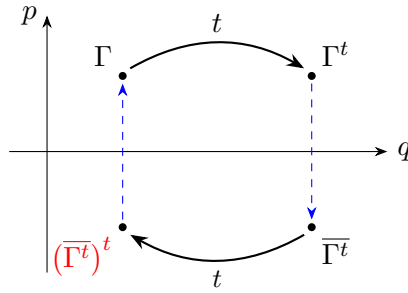


Figure 1.3: Microscopic reversibility: evolving the time-reversed final state for a time t gives the time-reversed initial state.

Examples of state functions: the kinetic energy $A_\Gamma = \sum_i \frac{p_i^2}{2m}$, and the macroscopic density field $A_\Gamma(\vec{r}) = \sum_i \delta(\vec{r} - \vec{r}_i)$. To prove detailed balance we make the simplifications

- $H(\bar{\Gamma}) = H(\Gamma)$: H is even under time reversal (no magnetic fields or Coriolis forces);
- $A_{\bar{\Gamma}} = A_\Gamma$: the observables are even.

Notation: $A_\Gamma(0) = A_{\Gamma(0)} = A_\Gamma$ at the initial condition, $A_\Gamma(t) = A_{\Gamma^t}$. Properties:

$$A_{\bar{\Gamma}}(0) = A_\Gamma(0), \quad A_{\bar{\Gamma}}(t) = A_{(\bar{\Gamma})^t}(0) = A_{\bar{\Gamma}^{-t}}(0) = A_\Gamma(-t).$$

Now take a system in thermal equilibrium such that there are no other conserved quantities, $F_{\text{eq}} = f(H)$, and $F_{\text{eq}}(\bar{\Gamma}) = F_{\text{eq}}(\Gamma)$ (H is even). Define

1. the probability distribution of A ,

$$P_{\text{eq}}(a) := \int d\Gamma F_{\text{eq}}(\Gamma) \delta(A_\Gamma - a);$$

2. the two-time distribution,

$$P_{\text{eq}}(a_1, t_1; a_2, t_2) := \int d\Gamma F_{\text{eq}}(\Gamma) \delta(A_{\Gamma^{t_1}} - a_1) \delta(A_{\Gamma^{t_2}} - a_2).$$

Since we are in equilibrium, this can only depend on $t = t_2 - t_1$:

$$\begin{aligned}
P_{\text{eq}}(a_1, 0; a_2, t) &= \int d\Gamma F_{\text{eq}}(\Gamma) \delta(A_\Gamma - a_1) \delta(A_{\Gamma^t} - a_2) \\
&= \int d\bar{\Gamma} F_{\text{eq}}(\bar{\Gamma}) \delta(A_{\bar{\Gamma}} - a_1) \delta(A_{\bar{\Gamma}^t} - a_2) \\
&= \int d\Gamma F_{\text{eq}}(\Gamma) \delta(A_\Gamma - a_1) \delta(A_{\Gamma^{-t}} - a_2) \\
&= \int d\Gamma F_{\text{eq}}(\Gamma) \delta(A_{\Gamma^t} - a_1) \delta(A_\Gamma - a_2) = P_{\text{eq}}(a_2, 0; a_1, t),
\end{aligned}$$

where in the last line we shifted time (allowed in equilibrium). Defining the conditional probability

$$P_{\text{eq}}(a_2, t|a_1, 0) := \frac{P_{\text{eq}}(a_1, 0; a_2, t)}{P_{\text{eq}}(a_1)},$$

the probability of measuring a_2 after a time t given that a_1 was measured at $t = 0$, we obtain

$$P_{\text{eq}}(a_2, t|a_1, 0) P_{\text{eq}}(a_1) = P_{\text{eq}}(a_1, t|a_2, 0) P_{\text{eq}}(a_2).$$

If A evolves in a Markovian way (the evolution to $t + 1$ depends only on t), this reads

$$W_{\text{eq}}(a_2|a_1) P_{\text{eq}}(a_1) = W_{\text{eq}}(a_1|a_2) P_{\text{eq}}(a_2),$$

with W_{eq} the **transition rate**. This says that there can be no cycles, i.e. no preferred direction of flow: that would require $W(a_2|a_1) \neq 0$ while $W(a_1|a_2) = 0$, which is only possible if $P(a_1) = 0$.

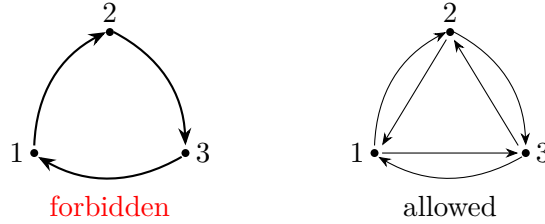


Figure 1.4: Detailed balance forbids net probability circulation around cycles; every transition must be balanced by its reverse.

This gives a sixth way of breaking equilibrium: take rates $W(a_2|a_1)$ satisfying detailed balance, wait until equilibrium is reached, and then change the W that sustained the equilibrium state.

Remark. If A is even and B is odd under time reversal, $A_{\bar{\Gamma}} = A_\Gamma$, $B_{\bar{\Gamma}} = -B_\Gamma$, so that $P_{\text{eq}}(a, b) = P_{\text{eq}}(a, -b)$. Although we only proved it for even A , there is a version of detailed balance for this case:

$$W(a_2, b_2|a_1, b_1) P_{\text{eq}}(a_1, b_1) = W(a_1, -b_1|a_2, -b_2) P_{\text{eq}}(a_2, b_2).$$

1.4 Universality

There are universal laws involving non-universal quantities. For example, all metals obey Ohm's law $\vec{J} = \sigma \vec{E}$ (universal), with σ non-universal. All liquids near the critical point have $c_V \sim c_0 |T - T_c|^{-\alpha}$, with c_0 non-universal and α universal. Statistical physics aims to explain universality and to give tools to compute the non-universal quantities.

Example (Diffusion). All very dilute solutes (no interactions, so the evolution of the density ρ is linear) obey

$$\partial_t \rho = D \nabla^2 \rho.$$

To see this, assume that there is no long memory (no polymers, which take long to relax when they interact with solutes), so that the equation is local in time: $\partial_t \rho$ cannot depend on past times,

$$\partial_t \rho = F[\rho], \quad F \text{ some functional.}$$

Since ρ is conserved and particles do not teleport, the equation must be a conservation law,

$$\partial_t \rho = -\nabla \cdot \vec{J}[\rho],$$

with $\vec{J}[\rho]$ linear. The most general linear form is

$$J_\alpha[\rho](\vec{r}) = \int d\vec{r}' G_\alpha(\vec{r}, \vec{r}') \rho(\vec{r}').$$

In a homogeneous space $G_\alpha = G_\alpha(\vec{x})$ with $\vec{x} = \vec{r} - \vec{r}'$, so $J_\alpha(\vec{r}) = \int d\vec{x} G_\alpha(\vec{x}) \rho(\vec{r} - \vec{x})$, a convolution: in Fourier space

$$\tilde{J}_\alpha(\vec{k}) = \tilde{G}_\alpha(\vec{k}) \tilde{\rho}(\vec{k}).$$

In the limit of long wavelengths, $\vec{k} \rightarrow 0$ (this restricts the regime we describe),

$$\tilde{G}_\alpha(\vec{k}) = G_{\alpha 0} + G_{\alpha\beta} k_\beta + G_{\alpha\beta\gamma} k_\beta k_\gamma + \dots$$

Now add the hypothesis that space is isotropic:

1. $G_{\alpha 0}$ must be a constant isotropic vector; the only option is $G_{\alpha 0} = 0$.
2. $G_{\alpha\beta}$ must be an isotropic tensor: $G_{\alpha\beta} = g_1 \delta_{\alpha\beta}$.
3. $G_{\alpha\beta\gamma} = 0$, since there is no isotropic three-index tensor.
4. $G_{\alpha\beta\gamma\delta} \neq 0$.

Hence $\tilde{G}_\alpha(\vec{k}) = g_1 k_\alpha + g_3 k^2 k_\alpha + \dots$. By dimensional analysis $[g_3/g_1] = L^2$, so $g_3 = a^2 g_1$ with $[a] = L$. This a has to do with the microscopic scales of the system, so we expect it to lie between ångströms and microns. Then

$$\tilde{G}_\alpha(\vec{k}) = g_1 (1 + a^2 k^2) k_\alpha + \dots$$

For every a there is $k^* = 1/a$ such that for all $k \ll k^*$, $ak \ll 1$. In this regime $\tilde{G}_\alpha = g_1 k_\alpha$, and with $g_1 = -iD$ we get $\tilde{J}_\alpha = -iD k_\alpha \tilde{\rho}$; transforming back, $\vec{J} = -D \nabla \rho$ (Fick's law) and

$$\partial_t \rho = D \nabla^2 \rho.$$

Remark. When we say something is universal we mean that it is universal in an *asymptotic* regime.

Another example of universality: long- and short-range correlation functions. Consider the Ising model of magnetism, spins $s_i = \pm 1$ on a lattice at zero field $B = 0$, with magnetization $m = \langle s_i \rangle$. Below T_c there is a ferromagnetic phase with $m \neq 0$ (spontaneous symmetry breaking); above T_c a paramagnetic phase with $m = 0$. Are all the states with $m = 0$ the same? To distinguish them we define the correlation function.

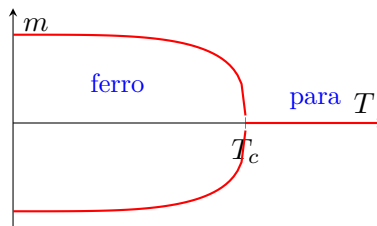


Figure 1.5: Magnetization of the Ising model at $B = 0$.

1.5 Correlation functions

In one dimension, $C_i = \langle S_{i+k} S_k \rangle_k$, averaged over k ; in the continuum

$$C(x) = \langle S(x+x') S(x') \rangle_{x'}.$$

The average $\langle \cdot \rangle$ can be a spatial, temporal or ensemble average. Properties:

- $C(0) = \langle S(x')^2 \rangle_{x'} \geq 0$;
- $C(\infty) = \langle S(\infty+x') \rangle \langle S(x') \rangle = 0$: no correlation at large distances (for $\langle S \rangle = 0$);
- $0 < C(x) < C(0)$ for small x .

So C decays. How does it decay? Two typical scenarios:

1. Exponential decay. $C(x) \sim C(0) e^{-x/\xi}$ for $x > x_0$, with ξ the **correlation length**. Looking from further away, $x \rightarrow \Lambda x$ with $\Lambda > 1$,

$$C(x) \sim C(0) e^{-\Lambda x/\xi} = C(0) e^{-x/(\xi/\Lambda)} :$$

looking from far away is equivalent to reducing the correlation length, $\xi \rightarrow \xi/\Lambda$. From sufficiently far, ξ becomes microscopic (smaller than anything we are interested in), and $C(x) \rightarrow C(0)\delta(x)$: each point is only correlated with itself. This is **short-range correlation**.

2. Algebraic decay. $C(x) \sim A x^{-\eta}$ for $x > x_0$ (for $x < x_0$ the behaviour is not universal). Under $x \rightarrow \Lambda x$,

$$C(x) \rightarrow A(\Lambda x)^{-\eta} = \frac{A}{\Lambda^\eta} x^{-\eta} :$$

it decays in the same way, only with a smaller amplitude. This is **scale-free, long-range correlation**.

In general $C(x) \sim e^{-x/\xi} x^{-\eta}$. For the Ising model, with $C(x) = \langle (S(x') - m)(S(x' + x) - m) \rangle$, the decay is exponential away from T_c (both below and above it, ξ finite) and algebraic at T_c , where ξ diverges. $C(x)$ gives information on spatial memory; $C(t)$ on temporal memory.

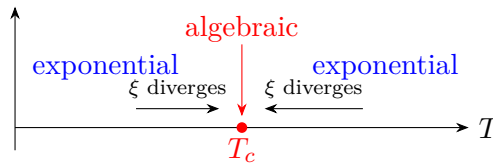


Figure 1.6: Decay of the correlation function of the Ising model as a function of temperature.

1.6 Wiener–Khinchin theorem

Let $\psi(\vec{r})$ be a field with zero mean, $\langle \psi(\vec{r}) \rangle = 0$, and $C(\vec{r}) = \langle \psi(\vec{r}') \psi(\vec{r} + \vec{r}') \rangle$, in a finite system of volume $V = L^3$. With

$$\psi(\vec{r}) = \sum_{\vec{k}} \psi_{\vec{k}} e^{i\vec{k} \cdot \vec{r}}, \quad \psi_{\vec{k}} = \frac{1}{V} \int d^3r \psi(\vec{r}) e^{-i\vec{k} \cdot \vec{r}},$$

$$C(\vec{r}) = \sum_{\vec{k}, \vec{q}} \langle \psi_{\vec{k}} \psi_{\vec{q}} \rangle e^{i\vec{q} \cdot \vec{r}} e^{i(\vec{k} + \vec{q}) \cdot \vec{r}'}.$$

Hypothesis: the system is statistically homogeneous, so C does not depend on $\vec{r}'$ and we can average over it, $\langle \cdot \rangle_{\vec{r}'} = \frac{1}{V} \int d\vec{r}'(\cdot)$, which gives $\delta_{\vec{k}+\vec{q},0}$:

$$C(\vec{r}) = \sum_{\vec{k}} \langle \psi_{\vec{k}} \psi_{-\vec{k}} \rangle e^{-i\vec{k} \cdot \vec{r}} = \sum_{\vec{k}} \langle |\psi_{\vec{k}}|^2 \rangle e^{-i\vec{k} \cdot \vec{r}}.$$

The **structure factor** (or power spectrum) is

$$S(\vec{k}) := \langle |\psi_{\vec{k}}|^2 \rangle.$$

C and S contain the same information (though not always all the information one needs).

1.7 Coarsening and response functions

Coarsening. Water and oil (like micellar water) separate: small droplets merge into larger ones over time. The same happens in magnetism, where in the ferromagnetic phase domains of different orientations grow. To characterize this we define an order parameter

$$\rho(\vec{r}) = \begin{cases} 1, & \text{water,} \\ 0, & \text{oil,} \end{cases} \quad \delta\rho(\vec{r}) = \rho(\vec{r}) - \langle \rho(\vec{r}) \rangle, \quad C(\vec{r}) = \langle \delta\rho(\vec{r} + \vec{r}') \delta\rho(\vec{r}') \rangle.$$

$C(r)$ decays over the typical droplet size

$$R := \frac{\int dr r C(r)}{\int dr C(r)},$$

and the structure factor $S(k)$ decays for $k \gtrsim k_0 \sim 2\pi/R$.

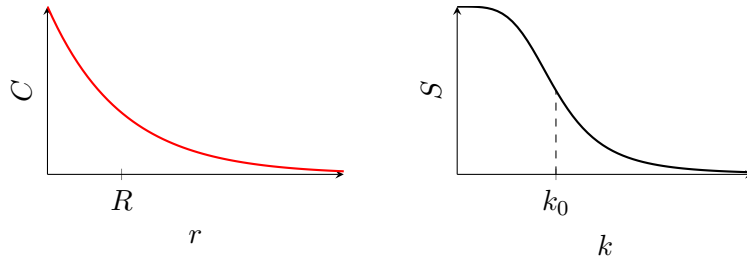


Figure 1.7: Correlation function and structure factor during coarsening.

One studies how $R(t)$ grows; typically (after short times) $R \sim t^n$, where n depends on the dimension of space, on whether ψ is conserved or not (conserved order parameters coarsen more slowly, since material has to be brought from further away), on spontaneous symmetry breaking, and on hydrodynamics. Further details: the global shape of the domains (circular, fingers, labyrinths, sponges, fractals), whether the boundary is smooth or rough, the monotonicity of $R(t)$, and the dynamics of the interface. For a 2D droplet described by $R(\theta, t)$, one studies $C(\theta, t) = \langle R(\theta, t) R(\theta', t') \rangle$, i.e. $\langle |\tilde{R}(\kappa, \omega)|^2 \rangle$ in terms of Fourier modes $e^{im\theta}$.

Response functions. How does a system in some state respond to a perturbation?

1. A solid to which a point charge q is added at $t = 0$: during the transient a charge density $\rho(\vec{r}, t)$ is induced (screening, which eventually becomes perfect). For an external charge density ρ_{ext} and a linear response,

$$\rho_{\text{ind}}(\vec{r}, t) = \int d\vec{r}' \int dt' G(\vec{r}, t; \vec{r}', t') \rho_{\text{ext}}(\vec{r}', t'),$$

where G is the **response function**.

2. A fluid between two plates; at $t = 0$ the upper plate is set in motion with velocity V . The force needed, $F(t)$, and the velocity field $v(y, t)$ are response functions. Asymptotic regimes: small t (the lower plate does not yet notice the motion) and large t (stationary regime).

Remark. An example of an exotic phase: between the solid and the liquid in two dimensions there is the *hexatic* phase, where correlations change more than expected (Kosterlitz–Thouless–Halperin–Nelson–Young theory).

2 Stochastic processes and Brownian motion

2.1 The Langevin equation

In 1827 R. Brown observed the erratic motion of pollen grains in water; Einstein explained it in 1905. The key is that the pollen particle is *mesoscopic* ($\sim 1 \mu\text{m}$):

- it is classical: $m\dot{\vec{v}} = \vec{F}$;
- the fluid exerts a viscous drag $\vec{F} = -\gamma\vec{v}$; for a sphere, Stokes' law gives $\gamma = 6\pi\eta a$, with η the viscosity and a the radius;
- but it is not that large: it feels the individual molecular collisions.

Its area is $A = 4\pi a^2 \sim 10 \mu\text{m}^2$, while a molecule occupies $A_{\text{molec}} \sim 1 \text{\AA}^2$, so

$$N_{\text{molec}} \sim \frac{4\pi a^2}{1 \text{\AA}^2} \sim \frac{10 \cdot 10^{-12}}{10^{-20}} \sim 10^9$$

molecules hit it at any time, with fluctuations $\sim \sqrt{N_{\text{molec}}} \sim 10^4$ – 10^5 : small but measurable. Hence $\vec{F} = -\gamma\vec{v}$ is only the average, and in reality

$$\vec{F} = -\gamma\vec{v} + \vec{F}_{\text{fl}}(t), \quad \langle \vec{F}_{\text{fl}}(t) \rangle = 0 \quad \forall t.$$

Properties of $\vec{F}_{\text{fl}}(t)$:

1. It results from many collisions, so it is modelled as a *Gaussian stochastic process* (information is lost in this description).
2. It fluctuates on a correlation time τ_c , the collision time,

$$\tau_c \sim \frac{d_{\text{molec}}}{v_{\text{molec}}} \sim \frac{1 \text{\AA}}{300 \text{ m/s}} \sim 10^{-13} \text{ s} \sim \text{ps}, \quad v_{\text{molec}} \simeq \sqrt{k_B T/m}.$$

There is another time scale, the Brownian (viscous braking) time

$$\tau_B = \frac{m}{\gamma} \sim 10^{-6} \text{ s} \sim \mu\text{s}$$

for a $1 \mu\text{m}$ particle. The time scales are well separated: molecular fluctuations (10^{-13} s) $\ll$ viscous braking (10^{-6} s) $\ll$ observation ($\sim 1 \text{ s}$). The theory is valid from some t^* with $\tau_c \ll t^* \ll \tau_B$, i.e. at mesoscopic scales, so we may set $\tau_c = 0$.

Summary. At the mesoscale,

$$m\dot{v} = -\gamma v + F_{\text{fl}}(t), \quad \langle F_{\text{fl}}(t) \rangle = 0, \quad \langle F_{\text{fl}}(t) F_{\text{fl}}(t') \rangle = \Gamma \delta(t - t'),$$

the **Langevin equation** with **white noise**. If $\tau_c \neq 0$ the noise is called **coloured**: $\langle F_{\text{fl}}(t) F_{\text{fl}}(t') \rangle = C(t - t')$, a function that decays over τ_c .

2.2 Solution

Conceptually, we are given one realization of the noise $F_{\text{fl}}(t)$ and integrate to obtain one realization of the solution. Rewrite the equation as

$$\dot{v} = -\lambda v + \xi(t), \quad \lambda = \frac{\gamma}{m} = \tau_B^{-1}, \quad \xi = \frac{F_{\text{fl}}}{m}, \quad \langle \xi(t)\xi(t') \rangle = \frac{\Gamma}{m^2} \delta(t - t').$$

Its solution is

$$v(t) = v(t_0)e^{-\lambda(t-t_0)} + \int_{t_0}^t ds e^{-\lambda(t-s)} \xi(s),$$

and letting $t_0 \rightarrow -\infty$,

$$v(t) = \int_{-\infty}^t ds e^{-\lambda(t-s)} \xi(s).$$

Properties:

1. Since ξ is Gaussian (at each step it takes a value drawn from a Gaussian), v is a Gaussian stochastic process: it is a sum (integral) of Gaussian variables. Only the first two moments are needed.
2. Averaging over noise realizations (many experiments),

$$\langle v(t) \rangle = \int_{-\infty}^t ds e^{-\lambda(t-s)} \langle \xi(s) \rangle = 0.$$

3.

$$\langle v^2(t) \rangle = \int_{-\infty}^t ds_1 \int_{-\infty}^t ds_2 e^{-\lambda(t-s_1)} e^{-\lambda(t-s_2)} \langle \xi(s_1)\xi(s_2) \rangle = \frac{\Gamma}{m^2} \int_0^\infty ds e^{-2\lambda s} = \frac{\Gamma}{2\lambda m^2} \neq 0 :$$

the particle moves.

Taking $t_0 \rightarrow -\infty$ amounts to saying that the Brownian particle has reached thermal equilibrium with the molecular bath. If T is the bath temperature,

$$\frac{m}{2} \langle v^2 \rangle = \frac{1}{2} k_B T = \frac{\Gamma}{2\lambda m^2} \frac{m}{2} \Rightarrow \Gamma = 2k_B T \gamma,$$

the **fluctuation–dissipation relation**. Similarly, $\langle v(t)v(t') \rangle = \langle v^2 \rangle e^{-|t-t'|/\tau_B}$.

Trajectory. By kinematics,

$$\Delta x = x(t) - x_0 = \int_0^t v(s) ds,$$

which is a Gaussian stochastic process with $\langle \Delta x(t) \rangle = 0$ for all t and

$$\begin{aligned} \langle \Delta x^2(t) \rangle &= \int_0^t \int_0^t ds_1 ds_2 \langle v(s_1)v(s_2) \rangle = \langle v^2 \rangle \int_0^t \int_0^t ds_1 ds_2 e^{-\lambda|s_1-s_2|} \\ &= 2 \langle v^2 \rangle \int_0^t ds_1 \int_0^{s_1} ds_2 e^{-\lambda(s_1-s_2)} = 2 \langle v^2 \rangle \left[\frac{t}{\lambda} - \frac{1 - e^{-\lambda t}}{\lambda^2} \right], \end{aligned}$$

the **mean squared displacement** (MSD). For $\lambda t \ll 1$, $1 - e^{-\lambda t} \simeq \lambda t - \frac{1}{2} \lambda^2 t^2$ and

$$\langle \Delta x^2(t) \rangle \simeq \langle v^2 \rangle t^2 :$$

at short times the motion is *ballistic*, $\Delta x = vt$. For $\lambda t \gg 1$ ($t \gg \tau_B$),

$$\langle \Delta x^2(t) \rangle \simeq 2 \langle v^2 \rangle \left[\frac{t}{\lambda} - \frac{1}{\lambda^2} \right] \simeq \frac{2 \langle v^2 \rangle t}{\lambda} :$$

the motion is *diffusive*, $\sqrt{\langle x^2 \rangle} \sim \sqrt{t}$.

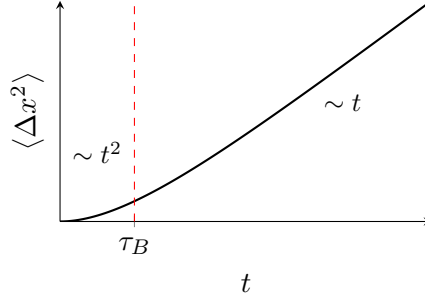


Figure 2.1: Mean squared displacement: ballistic for $t \ll \tau_B$, diffusive for $t \gg \tau_B$.

The diffusion law is $\langle \Delta x^2(t) \rangle = 2Dt$. Comparing,

$$D = \frac{\langle v^2 \rangle}{\lambda} = \frac{m \langle v^2 \rangle}{2} \frac{2}{m\lambda},$$

and using $\frac{m}{2} \langle v^2 \rangle = \frac{1}{2} k_B T$ we arrive at the **Einstein relation**

$$D = \frac{k_B T}{\gamma}.$$

This is the first equation in which Boltzmann's constant appears by itself, and hence Avogadro's number. For a sphere, $D = k_B T / (6\pi\eta a)$ is the **Stokes–Einstein relation** (η the viscosity of the fluid; for a suspension, $\eta_{\text{susp}} = \eta_{\text{water}}(1 + \frac{5}{2}\phi)$ with ϕ the volume fraction of the suspended particles).

Remark (Coarse-grained velocity field). For a velocity field $v(x)$ coarse-grained over boxes of size Δx containing $N_i = n\Delta x \gg 1$ particles, the probability of each box is $P_i \sim e^{-mN_i v_i^2 / 2k_B T}$ (canonical = grand canonical when $N_i \gg 1$), so

$$P \simeq \prod_i P_i = \exp \left(-\frac{mn}{2k_B T} \int dx v^2(x) \right).$$

This requires a Δx small enough that v is constant in each box, but large enough that $N_i \gg 1$: the *mesoscopic approximation*.

2.3 Diffusion equation

Consider one or several Brownian particles, non-interacting with each other, in a fluid. Let $n(\vec{r}, t)$ be the particle density, or the probability that a tagged particle is at $\vec{r}$. It obeys the **diffusion equation**

$$\partial_t n = D \nabla^2 n.$$

For a single Brownian particle starting with certainty at the origin, $n(\vec{r}, 0) = \delta(\vec{r})$ (so $\int d^d r n(\vec{r}, 0) = 1$), the solution (by Fourier transform) is a Gaussian

$$n(\vec{r}, t) = \frac{1}{(4\pi Dt)^{d/2}} e^{-r^2/4Dt}, \quad d = 1, 2, 3.$$

Then $\langle \vec{r}(t) \rangle = \int d^d r \vec{r} n(\vec{r}, t) = 0$ by isotropy, and

$$\langle r^2(t) \rangle = \int d^d r (x^2 + y^2 + z^2) n(\vec{r}, t) = d \int dx x^2 \frac{e^{-x^2/4Dt}}{(4\pi Dt)^{1/2}} = 2dDt,$$

which grows linearly with t (Langevin only gives this at long times). Hence the **Einstein formula**

$$D = \frac{\langle r^2(t) \rangle}{2dt}.$$

A Brownian (Langevin) particle diffuses at long times; for $t \lesssim \tau_B$, $\partial_t n = D \nabla^2 n$ does not hold. A more useful form, especially in numerical computations, is

$$D = \lim_{t \rightarrow \infty} \frac{\langle \Delta r^2(t) \rangle}{2dt} = \lim_{t \rightarrow \infty} \frac{d}{dt} \left(\frac{\langle \Delta r^2(t) \rangle}{2d} \right).$$

From the Langevin MSD, $D = \lim_{t \rightarrow \infty} \frac{1}{2} \frac{\langle \Delta x^2 \rangle}{t} = \frac{\langle v^2 \rangle}{\lambda} = \frac{k_B T}{m\lambda} = \frac{k_B T}{\gamma}$.

2.4 Green–Kubo relation

Write $\Delta \vec{r}(t) = \vec{r}(t) - \vec{r}(0) = \int_0^t ds \vec{v}(s)$, so

$$\langle \Delta r^2(t) \rangle = \int_0^t \int_0^t ds_1 ds_2 \langle \vec{v}(s_1) \cdot \vec{v}(s_2) \rangle = \int_0^t \int_0^t ds_1 ds_2 C_v(s_1 - s_2),$$

with C_v the velocity correlation function, which is even (symmetric under $s_1 \leftrightarrow s_2$). Then

$$\frac{d}{dt} \langle \Delta r^2 \rangle = \int_0^t ds_2 C_v(t - s_2) + \int_0^t ds_1 C_v(s_1 - t) = 2 \int_0^t ds C_v(s),$$

and with $D = \lim_{t \rightarrow \infty} \frac{1}{2d} \frac{d}{dt} \langle \Delta r^2 \rangle$,

$$D = \frac{1}{d} \int_0^\infty ds C_v(s),$$

the **Green–Kubo relation**. It relates a transport coefficient of an irreversible, non-equilibrium process (diffusion) to a correlation function of the stationary equilibrium state.

Application where Langevin does not apply: self-diffusion of water. Tag one water molecule; for long times its density obeys $\partial_t n = D \nabla^2 n$, and Green–Kubo gives $D = \frac{1}{3} \int_0^\infty dt C_v(t)$. A naive estimate is $C_v(t) = \langle \vec{v}(t) \cdot \vec{v}(0) \rangle \sim \langle v^2 \rangle e^{-t/\tau}$ with $\langle v^2 \rangle = 3k_B T/m$ and $\tau \sim 1 \text{ \AA}/v_{\text{th}}$:

$$D \sim \frac{k_B T}{m} \tau = v_{\text{th}}^2 \frac{1 \text{ \AA}}{v_{\text{th}}} \sim v_{\text{th}} \cdot 1 \text{ \AA} \sim 3 \times 10^{-8} \text{ m}^2/\text{s},$$

while experimentally $D_{\text{H}_2\text{O}} \sim 2.3 \times 10^{-9} \text{ m}^2/\text{s}$. The discrepancy signals new physics: in liquids $C_v(t)$ has a negative minimum, because the molecule is trapped in a “cage” formed by its neighbours and reverses its direction of motion (*caging*); eventually it jumps out of the cage. The negative part reduces D .

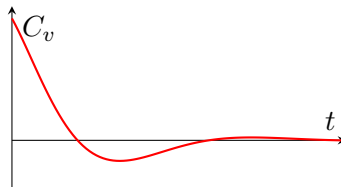


Figure 2.2: Velocity autocorrelation in a liquid: the negative minimum reflects caging.

Exercise. (i) *Single-file diffusion*: particles moving in 1D that cannot cross each other have $\langle \Delta x^2(t) \rangle \sim t^{1/2}$: the motion is not diffusive but subdiffusive. (ii) A Brownian particle in a potential U has the stationary distribution $p(x, v) \propto e^{-U/k_B T} e^{-mv^2/2k_B T}$; even in an asymmetric periodic potential (a ratchet), the equilibrium diffusion is symmetric.

2.5 Fokker–Planck equation

A set of non-interacting Brownian particles is equivalent to an ensemble of systems with one Brownian particle each. The Langevin equation for $\vec{r}, \vec{v}$ is a stochastic differential equation; the Fokker–Planck (FP) description is probabilistic: its object is $P(\vec{r}, \vec{v}, t)$, the probability of having position $\vec{r}$ and velocity $\vec{v}$.

Take Δt with $\tau_c \ll \Delta t \ll \tau_B$. Then (master equation)

$$P(\vec{r}, \vec{v}, t + \Delta t) = P(\vec{r} - \vec{v}\Delta t, \vec{v}, t) + \int d\vec{v}' P(\vec{r}, \vec{v}', t) w(\vec{v}', \vec{v}, \Delta t) - \int d\vec{v}' P(\vec{r}, \vec{v}, t) w(\vec{v}, \vec{v}', \Delta t),$$

where the first term is the deterministic change of $\vec{r}$ and the others are probabilistic: $w(\vec{v}, \vec{v}', \Delta t)$ is the probability of going from $\vec{v}$ to $\vec{v}'$ in Δt . (The positions in the integrals could be shifted too, but those terms are subdominant.) Without fluctuations, $\vec{v} \rightarrow \vec{v} - \lambda\vec{v}\Delta t$ (viscous drag) and $w(\vec{v}, \vec{v}', \Delta t) = \delta(\vec{v}' - (\vec{v} - \lambda\vec{v}\Delta t))$; with fluctuations w is a peaked function around $\vec{v} - \lambda\vec{v}\Delta t$, very peaked in the Brownian limit $\tau_B/\tau_c \rightarrow \infty$.

Define $\hat{w}(\vec{v}, \vec{u}, \Delta t) := w(\vec{v}, \vec{v} + \vec{u}, \Delta t)$, with $\vec{u} = \Delta\vec{v}$ the jump. It is smooth in the initial velocity $\vec{v}$ (which allows a Taylor expansion) and peaked in $\vec{u}$. Then

$$P(\vec{r}, \vec{v}, t + \Delta t) = P(\vec{r} - \vec{v}\Delta t, \vec{v}, t) + \int d\vec{u} [P(\vec{r}, \vec{v} - \vec{u}, t) \hat{w}(\vec{v} - \vec{u}, \vec{u}, \Delta t) - P(\vec{r}, \vec{v}, t) \hat{w}(\vec{v}, \vec{u}, \Delta t)],$$

and expanding in $\vec{u}$,

$$P(\vec{r}, \vec{v}, t + \Delta t) = P(\vec{r} - \vec{v}\Delta t, \vec{v}, t) - \int d\vec{u} \vec{u} \cdot \frac{\partial}{\partial \vec{v}} [P\hat{w}] + \frac{1}{2} \int d\vec{u} \vec{u} \vec{u} : \frac{\partial}{\partial \vec{v}} \frac{\partial}{\partial \vec{v}} [P\hat{w}] + \dots$$

Expanding the left-hand side and the first term on the right, $P + \Delta t \partial_t P = P - \Delta t \vec{v} \cdot \partial_{\vec{r}} P + \dots$, so

$$\partial_t P = -\vec{v} \cdot \partial_{\vec{r}} P - \partial_{\vec{v}} \cdot [P\vec{a}_1] + \frac{1}{2} \nabla_{\vec{v}}^2 [Pa_2],$$

with the drift and diffusion coefficients

$$\vec{a}_1 := \frac{1}{\Delta t} \int d\vec{u} \vec{u} \hat{w}(\vec{v}, \vec{u}, \Delta t) = \left\langle \frac{\vec{u}}{\Delta t} \right\rangle = -\lambda\vec{v}, \quad a_2 := \frac{1}{d} \frac{1}{\Delta t} \int d\vec{u} u^2 \hat{w}(\vec{v}, \vec{u}, \Delta t) = \frac{\langle u^2 \rangle}{d \Delta t},$$

the latter due to the fluctuations (noise). One can show that if the process is diffusive ($u^2 \sim \Delta t$), then a_1, a_2 are finite as $\Delta t \rightarrow 0$ and a_3 and higher vanish.

Where does d come from? $\partial_{\vec{v}} \partial_{\vec{v}} : \int \vec{u} \vec{u} \dots = \sum_{\alpha\beta} \partial_{v_\alpha} \partial_{v_\beta} \int u_\alpha u_\beta \dots$ involves a rank-2 tensor $A_{\alpha\beta}$. If there is no privileged direction, $A_{\alpha\beta} = A \delta_{\alpha\beta}$ (with a magnetic field, for instance, one could have $A_{\alpha\beta} = a \delta_{\alpha\beta} + b B_\alpha B_\beta$). Taking the trace, $A_{\alpha\alpha} = dA$, i.e. $u^2 = dA$ and $A = u^2/d$, so $\sum_{\alpha\beta} \partial_{v_\alpha} \partial_{v_\beta} A_{\alpha\beta} = \nabla_{\vec{v}}^2 (\frac{1}{d} \int u^2 \dots)$.

In general a_2 depends on $\vec{v}$, $a_2(\vec{v}) = a_{20} + a_{21}v + a_{22}v^2 + \dots$; near equilibrium a_{20} dominates, and we define Γ by $a_{20} = \Gamma/m^2$. This gives the **Fokker–Planck equation**

$$\partial_t P + \vec{v} \cdot \frac{\partial P}{\partial \vec{r}} = \frac{\gamma}{m} \frac{\partial}{\partial \vec{v}} \cdot (\vec{v} P) + \frac{\Gamma}{2m^2} \nabla_{\vec{v}}^2 P.$$

With an external force,

$$\partial_t P + \vec{v} \cdot \nabla P + \frac{\vec{F}_{\text{ext}}}{m} \cdot \frac{\partial P}{\partial \vec{v}} = \frac{\gamma}{m} \frac{\partial}{\partial \vec{v}} \cdot (\vec{v} P) + \underbrace{\frac{\Gamma}{2m^2} \nabla_{\vec{v}}^2 P}_{\text{diffusive term}}.$$

Solutions.

1. Without external force, homogeneous and stationary:

$$\frac{\partial}{\partial v} \left[\frac{\gamma}{m} v P + \frac{\Gamma}{2m^2} \frac{\partial P}{\partial v} \right] = 0.$$

The bracket is constant, and the boundary condition at $v \rightarrow \infty$ sets the constant to zero, so

$$P_0 = A e^{-m\gamma v^2/\Gamma}.$$

Einstein's criterion: stationary $\Rightarrow$ thermal equilibrium, $P_0 = A e^{-mv^2/2k_B T}$, so $\gamma/\Gamma = 1/(2k_B T)$, i.e. $\Gamma = 2k_B T \gamma$.

2. With a conservative external force $F_{\text{ext}} = -U'(x)$, stationary, in 1D: try $P(x, v) = R(x)V(v)$. Then

$$v R' V - \frac{U'}{m} R V' = \frac{\gamma}{m} R (v V)' + \frac{k_B T \gamma}{m^2} R V''.$$

Dividing by $R V v$,

$$\frac{R'}{R} - \frac{U'}{m} \frac{V'}{v V} = \frac{\gamma}{m v V} \left(V + v V' + \frac{k_B T}{m} V'' \right).$$

The right-hand side depends only on v and is solved, as before, by $V(v) = e^{-mv^2/2k_B T}$, for which it vanishes and $V'/(vV) = -m/k_B T$. The left-hand side then gives

$$\frac{R'}{R} + \frac{U'}{k_B T} = 0 \Rightarrow R(x) = e^{-U(x)/k_B T},$$

and

$$P(x, v) = A e^{-U(x)/k_B T} e^{-mv^2/2k_B T},$$

the equilibrium Gibbs distribution, which does not depend on γ .

In general, equilibrium properties do not depend on the non-equilibrium (transport) coefficients; the dynamics of relaxation does depend on γ .

2.6 From Langevin to Fokker–Planck

If the Langevin equation is

$$\dot{x} = A(x) + \xi(t), \quad \langle \xi(t) \xi(t') \rangle = \Gamma \delta(t - t'),$$

one can show that it is equivalent to

$$\frac{\partial P}{\partial t} = -\frac{\partial}{\partial x} (A(x)P) + \frac{\Gamma}{2} \frac{\partial^2 P}{\partial x^2}.$$

(Langevin $\rightarrow$ FP by computing $P(\dots, t + \Delta t)$; FP $\rightarrow$ Langevin since the process is Gaussian.)
With several variables,

$$\dot{\vec{x}} = \vec{f}(\vec{x}) + \vec{\xi}, \quad \langle \xi_i(t) \xi_j(t') \rangle = \Gamma_{ij} \delta(t - t') \iff \frac{\partial P(\vec{x}, t)}{\partial t} = -\frac{\partial}{\partial \vec{x}} \cdot (\vec{f}(\vec{x})P) + \sum_{ij} \frac{\Gamma_{ij}}{2} \frac{\partial^2 P}{\partial x_i \partial x_j}.$$

For example, for $\vec{x} = (x, n_x)$ with $\dot{x} = n_x$, $\dot{n}_x = \xi$, $\Gamma = \text{diag}(0, 1)$:

$$\frac{\partial f}{\partial t} = -n_x \frac{\partial f}{\partial x} + \frac{1}{2} \frac{\partial^2 f}{\partial n_x^2}.$$

Multiplicative noise. Sometimes

$$\dot{x} = A(x) + C(x)\xi(t).$$

Since ξ has infinite variance, $x_{t+\Delta t} = \dots + C(x)\sqrt{\Gamma\Delta t}\mathcal{N}$ (with $\mathcal{N}$ a standard normal variable) is ambiguous: at which x is C evaluated, before or during the step? The equation must be supplemented with a prescription:

$$\text{It\^o: } x(t + \Delta t) = x(t) + A(x(t))\Delta t + C(x(t))\sqrt{\Gamma\Delta t}\mathcal{N},$$

$$\text{Stratonovich: } x(t + \Delta t) = x(t) + A(x(t))\Delta t + \frac{1}{2}[C(x(t)) + C(x(t + \Delta t))]\sqrt{\Gamma\Delta t}\mathcal{N}.$$

The corresponding Fokker–Planck equations are

$$\text{It\^o: } \frac{\partial P}{\partial t} = -\frac{\partial}{\partial x}(AP) + \frac{\Gamma}{2}\frac{\partial^2}{\partial x^2}(C^2P),$$

$$\text{Stratonovich: } \frac{\partial P}{\partial t} = -\frac{\partial}{\partial x}(AP) + \frac{\Gamma}{2}\frac{\partial}{\partial x}\left(C\frac{\partial}{\partial x}(CP)\right).$$

3 Statistical field theory: Landau theory and relaxational dynamics

References for this part: Politi–Livi, and Kardar.

3.1 Landau theory of phase transitions

Example: the paramagnetic–ferromagnetic transition, controlled by the temperature T and the external magnetic field B ; we measure the magnetization m . The transition line is $B = 0$, $T < T_c$.

1. $T > T_c$: paramagnetic response; $m(B)$ is smooth, with slope at the origin given by the susceptibility.
2. $B = 0$: spontaneous magnetization (ferromagnetism) below T_c , i.e. spontaneous symmetry breaking.
3. $T < T_c$: hysteresis in $m(B)$.

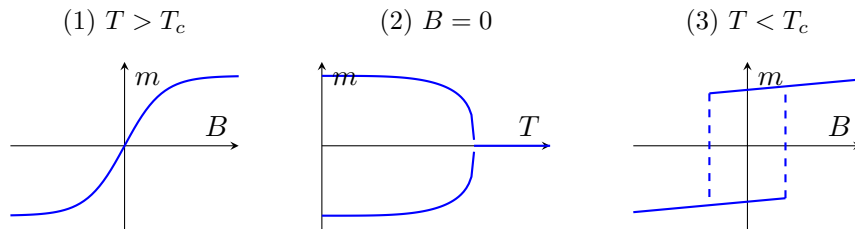


Figure 3.1: Magnetic response: paramagnetic, spontaneous magnetization, and hysteresis.

How is the system characterized?

- *Answer 1 (microscopic):* $H = H(\text{spins}, B)$, $F = -k_B T \ln \sum_{\text{spins}} e^{-\beta H}$, $M \propto \partial F / \partial B$; the susceptibility is related to the variance of m . This can be solved exactly only for the 2D Ising model at $B = 0$.
- *Answer 2 (Landau, phenomenological):* study a fictitious system in which a demon fixes m ; the constrained free energy is $F = F(T, B, m)$.

Take the 1D case ($B, m \in \mathbb{R}$). Thermodynamics says m_{eq} minimizes F , $\partial F/\partial m = 0$. Near T_c (where the theory applies) m is small, and we Taylor expand:

$$F(T, B, m) = F_0(T, B) + F_1(T, B)m + F_2(T, B)m^2 + F_3(T, B)m^3 + F_4(T, B)m^4.$$

Symmetry analysis: F is a scalar while $\vec{B}$ and $\vec{m}$ are vectors; under $\vec{x} \rightarrow -\vec{x}$, $\vec{m} \rightarrow -\vec{m}$, $\vec{B} \rightarrow -\vec{B}$, $F \rightarrow F$. Hence F_0, F_2, F_4 are even in B and F_1, F_3 are odd in B ; at $B = 0$, $F_1 = F_3 = 0$:

$$F(T, 0, m) = F_0(T, 0) + F_2(T, 0)m^2 + F_4(T, 0)m^4.$$

Thermodynamic stability requires a minimum, so $F_4(T, 0) > 0$ (tricritical transitions need F_6). Three cases: $F_2 > 0$ (paramagnet, single minimum at $m = 0$), $F_2 = 0$ (transition) and $F_2 < 0$ (ferromagnet, two minima).

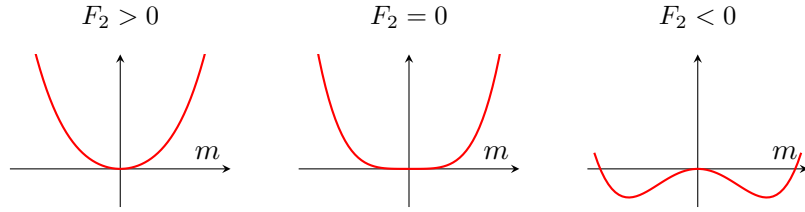


Figure 3.2: Landau free energy above, at and below T_c .

Near the critical point, to the first non-trivial order,

$$F_0(T, B \simeq 0) \simeq F_0, \quad F_4(T, B \simeq 0) \simeq F_4, \quad F_2(T, B \simeq 0) \simeq a(T - T_c)$$

(the coefficient a could vanish, but experimentally this is not usual). Then

$$F(T, 0, m) = F_0 + a(T - T_c)m^2 + F_4m^4, \quad \left. \frac{\partial F}{\partial m} \right|_{m_{\text{eq}}} = 2a(T - T_c)m + 4F_4m^3 = 0,$$

so $m_{\text{eq}} = 0$ or

$$m_{\text{eq}} = \pm \sqrt{\frac{a(T_c - T)}{2F_4}}, \quad T < T_c.$$

We will see later that this exponent ($\frac{1}{2}$) is wrong.

With a field (B near zero),

$$F = F_0 + a(T - T_c)m^2 + F_4m^4 - F_1Bm + \text{subdominant}, \quad F_1 > 0.$$

The linear term tilts the free energy. For $T < T_c$ the system can stay in the metastable minimum until the field makes it disappear (the minimum merges with the maximum at an inflection point, the spinodal), which explains hysteresis (something not captured by 19th-century thermodynamics).

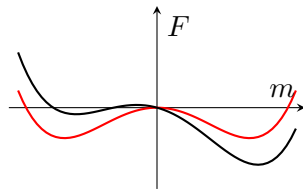


Figure 3.3: Ferromagnetic free energy without (red) and with (black) a field $B > 0$.

3.2 Variational (relaxational) dynamics

Out of equilibrium: how does the system reach the minimum? Consider $T < T_c$, $B > 0$, starting from $m = 0$. We propose

$$\dot{m} = -\lambda \frac{\partial F}{\partial m}.$$

In general, for $F(\text{thermodynamic control parameters}, \psi)$, with ψ the order parameter (which cannot be controlled),

$$\dot{\psi} = -\lambda \frac{\partial F}{\partial \psi}.$$

Property:

$$\frac{dF}{dt} = \frac{\partial F}{\partial \psi} \frac{d\psi}{dt} = -\lambda \left(\frac{\partial F}{\partial \psi} \right)^2 \leq 0,$$

which satisfies the second law if $\lambda > 0$: the order parameters evolve so as to minimize the free energy. F decreases and the evolution stops when $\partial F / \partial \psi = 0$ (equilibrium). Taking λ constant (valid near equilibrium) is **linear irreversible thermodynamics**.

This does not explain two phenomena: (1) spontaneous symmetry breaking (when a control parameter changes so that $\psi = 0$ becomes a maximum, the system would stay there), and (2) metastability (the finite lifetime of a metastable state). *The fluctuations are missing*. We therefore propose

$$\dot{\psi} = -\lambda \frac{\partial F}{\partial \psi} + \xi, \quad \langle \xi(t) \xi(t') \rangle = \Gamma \delta(t - t').$$

The corresponding Fokker–Planck equation for $P(\psi, t)$ is

$$\frac{\partial P}{\partial t} = -\frac{\partial}{\partial \psi} \left(-\lambda \frac{\partial F}{\partial \psi} P \right) + \frac{\Gamma}{2} \frac{\partial^2 P}{\partial \psi^2}$$

(for quadratic F the solutions involve Hermite polynomials). It is a conservation equation,

$$\frac{\partial P}{\partial t} = -\frac{\partial J}{\partial \psi}, \quad J = -\lambda \frac{\partial F}{\partial \psi} P - \frac{\Gamma}{2} \frac{\partial P}{\partial \psi} \quad (\text{probability current}).$$

For a stationary solution J is constant, and with $P(\psi = \pm\infty) = 0$, $J = 0$: detailed balance, which in addition means equilibrium. Then

$$\frac{\Gamma}{2} \frac{\partial P}{\partial \psi} = -\lambda P \frac{\partial F}{\partial \psi} \Rightarrow \frac{\partial}{\partial \psi} \ln P = -\frac{2\lambda}{\Gamma} \frac{\partial F}{\partial \psi} \Rightarrow P_{\text{eq}} = A e^{-2\lambda F(\psi)/\Gamma} \stackrel{!}{=} A e^{-F(\psi)/k_B T},$$

where the last equality is imposed. Hence

$$\Gamma = 2\lambda k_B T \quad (\text{fluctuation–dissipation theorem}).$$

Summary. The order parameter ψ promotes $F(\text{control parameters})$ to $F(\text{control parameters}, \psi)$. Thermodynamics: ψ_{eq} minimizes F . Dynamics:

$$\dot{\psi} = -\lambda \frac{\partial F}{\partial \psi} + \xi(t), \quad \langle \xi(t) \xi(t') \rangle = 2\lambda k_B T \delta(t - t') :$$

the drift takes the system to the minimum, and the noise makes it fluctuate around it with $P(\psi) = P_0 e^{-F(\psi)/k_B T}$.

3.3 Several order parameters

(The demon is equivalent to an initial condition.) With $F = F(\psi_i)$ we propose

$$\dot{\psi}_i = -\lambda_{ik} \frac{\partial F}{\partial \psi_k} + \xi_i.$$

Without noise,

$$\dot{F} = \frac{\partial F}{\partial \psi_i} \frac{d\psi_i}{dt} = -\lambda_{ik} \frac{\partial F}{\partial \psi_i} \frac{\partial F}{\partial \psi_k} = -\lambda_{ik}^s \frac{\partial F}{\partial \psi_i} \frac{\partial F}{\partial \psi_k},$$

since only the symmetric part λ^s contributes. Thermodynamics requires $\dot{F} \leq 0$ always, with $\dot{F} = 0$ iff in equilibrium: λ^s must be positive definite, while the antisymmetric part λ^a can be anything.

Example (Damped harmonic oscillator). One particle in a fluid in a harmonic potential, $F = H = \frac{p^2}{2m} + \frac{1}{2}\kappa x^2$, with order parameters (not controlled) x, p . The equations of motion are $\dot{x} = p/m$, $\dot{p} = -\kappa x - \gamma p/m$, where the friction is external. With $\partial F/\partial x = \kappa x$ and $\partial F/\partial p = p/m$,

$$\begin{pmatrix} \dot{x} \\ \dot{p} \end{pmatrix} = - \underbrace{\begin{pmatrix} 0 & -1 \\ 1 & \gamma \end{pmatrix}}_{\lambda} \begin{pmatrix} \kappa x \\ p/m \end{pmatrix} = - \underbrace{\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}}_{\text{antisymmetric}} \begin{pmatrix} \kappa x \\ p/m \end{pmatrix} - \underbrace{\begin{pmatrix} 0 & 0 \\ 0 & \gamma \end{pmatrix}}_{\text{symmetric}} \begin{pmatrix} \kappa x \\ p/m \end{pmatrix}.$$

The antisymmetric part is symplectic (Poisson brackets) and gives the conservative dynamics; the symmetric part gives the dissipative dynamics. The phase-space trajectory spirals into the origin.

3.4 Linear response

What happens if we perturb the system? Consider a Hamiltonian-type (i.e. thermodynamic) perturbation: with free energy $F_0(\psi)$, apply an external field h ,

$$F(\psi) = F_0(\psi) - h\psi \quad (\text{e.g. } F = \dots - Bm).$$

How does ψ change? In equilibrium we know the solution:

$$P(\psi) = \frac{e^{-F(\psi)/k_B T}}{\int d\psi e^{-F/k_B T}}, \quad \langle \psi \rangle_h = \frac{\int d\psi \psi e^{-F_0/k_B T} e^{h\psi/k_B T}}{\int d\psi e^{-F_0/k_B T} e^{h\psi/k_B T}}.$$

Dividing numerator and denominator by $\int d\psi e^{-F_0/k_B T}$,

$$\langle \psi \rangle_h = \frac{\langle \psi e^{h\psi/k_B T} \rangle_0}{\langle e^{h\psi/k_B T} \rangle_0} :$$

averages with fields can be computed with averages of the free theory. For small h (linear response), $e^{h\psi/k_B T} \simeq 1 + h\psi/k_B T$ and

$$\langle \psi \rangle_h \simeq \frac{\langle \psi \rangle_0 + h \langle \psi^2 \rangle_0 / k_B T}{1 + h \langle \psi \rangle_0 / k_B T} \simeq \langle \psi \rangle_0 + \left[\langle \psi^2 \rangle_0 - \langle \psi \rangle_0^2 \right] \frac{h}{k_B T},$$

so the susceptibility is proportional to the variance of the unperturbed field:

$$\chi = \lim_{h \rightarrow 0} \frac{\partial \langle \psi \rangle}{\partial h} = \frac{1}{k_B T} \left[\langle \psi^2 \rangle_0 - \langle \psi \rangle_0^2 \right].$$

Out of equilibrium. Let $F = F_0$ for $t < 0$ and switch on $h(t)$ for $t > 0$: $F = F_0 - h(t)\Theta(t)\psi$. Initially $P(\psi) = P_{\text{eq}}(\psi) = e^{-F_0/k_B T} / \int d\psi e^{-F_0/k_B T}$. For $t > 0$ the Langevin equation

$$\dot{\psi} = -\lambda \left(\frac{\partial F_0}{\partial \psi} - h \right) + \xi(t)$$

gives the Fokker–Planck equation (with $\Gamma = 2\lambda k_B T$)

$$\frac{\partial P}{\partial t} = \lambda \frac{\partial}{\partial \psi} \left(\frac{\partial F_0}{\partial \psi} P \right) + \lambda k_B T \frac{\partial^2 P}{\partial \psi^2} - \lambda h \frac{\partial P}{\partial \psi}.$$

Changing variables $\tilde{t} = \lambda t$ (and then dropping the tilde),

$$\frac{\partial P}{\partial t} = \underbrace{\frac{\partial}{\partial \psi} \left(\frac{\partial F_0}{\partial \psi} P \right)}_{\mathcal{L}_0 P} + k_B T \frac{\partial^2 P}{\partial \psi^2} - \underbrace{h \frac{\partial P}{\partial \psi}}_{\mathcal{L}_1 P},$$

with $\mathcal{L}_0$ the unperturbed operator (as in quantum mechanics). Define the inner product $(P, Q) := \int d\psi P(\psi)Q(\psi)$ and the adjoint by $(P, \mathcal{L}_0 Q) = (\mathcal{L}_0^\dagger P, Q)$. Properties: $[\mathcal{L}_0, \mathcal{L}_1] \neq 0$, and $\mathcal{L}_0 P_{\text{eq}} = 0$.

Linear perturbative solution. Write $P(\psi, t) = P_0(\psi) + P_1(\psi, t)$, with P_0 the equilibrium distribution (independent of t ; it is the initial condition) and $P_1 = \mathcal{O}(h)$. Then

$$\frac{dP_1}{dt} = \mathcal{L}_0 P_1 + \mathcal{L}_1 P_0,$$

an equation in function space, with $P_1(t=0) = 0$. Its solution is

$$P_1(t) = \int_0^t ds e^{(t-s)\mathcal{L}_0} (\mathcal{L}_1 P_0)(s) = - \int_0^t ds e^{(t-s)\mathcal{L}_0} \frac{\partial P_0}{\partial \psi} h(s).$$

Then

$$\langle \psi \rangle(t) = \int d\psi \psi P_0 + \int d\psi \psi P_1 = \langle \psi \rangle_0 + \int_0^t \chi(t-s) h(s) ds,$$

with the **response (or memory) function**

$$\chi(t-s) := - \int d\psi \psi e^{(t-s)\mathcal{L}_0} \frac{\partial P_0}{\partial \psi}.$$

Fluctuation–dissipation theorem. With $r = t - s$ (χ does not depend on h : linear response) and $P_0 = e^{-F_0/k_B T} / Z$, $-\partial_\psi P_0 = F'_0(\psi) P_0 / k_B T$:

$$\chi(r) = \frac{1}{k_B T} \int d\psi \psi e^{r\mathcal{L}_0} F'_0 P_0 = \frac{1}{k_B T} (\psi, e^{r\mathcal{L}_0} F'_0 P_0) = \frac{1}{k_B T} (e^{r\mathcal{L}_0^\dagger} \psi, F'_0 P_0).$$

What is $\mathcal{L}_0^\dagger$? For a probability distribution $P(\psi, t)$ and an observable $A(\psi)$,

$$\langle A \rangle(t) = \int d\psi A(\psi) P(\psi, t) = \int d\psi A(\psi) e^{\mathcal{L}_0 t} P(\psi, 0) = (e^{\mathcal{L}_0^\dagger t} A, P(\psi, 0))$$

(Schrödinger picture on the left, Heisenberg picture on the right). So $\mathcal{L}_0^\dagger$ is the free evolution operator of observables (without h), while $\mathcal{L}_0$ evolves the probability distribution. Therefore

$$\chi(r) = \frac{1}{k_B T} \int d\psi \psi(r) F'_0(\psi) P_0(\psi) = \frac{1}{k_B T} \langle \psi(r) F'_0(\psi(0)) \rangle_0,$$

a correlation between two instants, evolved with $\mathcal{L}_0^\dagger$ (without h) and computed in equilibrium. $F'_0(\psi)$ is called the “thermodynamic force”. Example: $F_0 = a\psi^2/2$, $F'_0 = a\psi$,

$$\chi(r) = \frac{a}{k_B T} \langle \psi(r) \psi(0) \rangle_0,$$

the time autocorrelation function.

Several observables. With a single external field coupled through constants $g_i = \mathcal{O}(1)$,

$$F(\psi_1, \dots, \psi_n) = F_0(\psi_1, \dots, \psi_n) - h(t) \sum_i g_i \psi_i, \quad \langle \psi_i \rangle(t) = \langle \psi_i \rangle_0 + \int_0^t ds \chi_{ik}(t-s) h(s) g_k,$$

with $\chi_{ik}(s) \sim \langle \psi_i(s) \psi_k(0) \rangle$, correlation matrices.

Phenomenology. For a switch-on perturbation, $h(t) = h_0$ for $t > 0$ and 0 for $t < 0$,

$$\langle \psi(t) \rangle = \langle \psi \rangle_0 + h_0 \int_0^t dr \chi(r).$$

Possible responses:

1. Monotonic approach to $\langle \psi \rangle_0 + h_0 \int_0^\infty \chi$: $\chi_{\text{model}} = \chi_0 e^{-s/s_0}$, a single exponential (one field, $\dot{\psi} = -\psi/s_0$).
2. Overshooting: $\chi_{\text{model}}(s) = \chi_1 e^{-s/s_1} - \chi_2 e^{-s/s_2}$ (overdamped), two exponentials (two fields, $\frac{d}{dt}(\psi_1, \psi_2) = -M(\psi_1, \psi_2)$).
3. Damped oscillations: $\chi_{\text{model}}(s) = e^{-s/s_0} (\chi_c e^{i\omega s} + \chi_c^* e^{-i\omega s})$.
4. If χ decays slowly (very long range), the integral of the correlation function diverges.

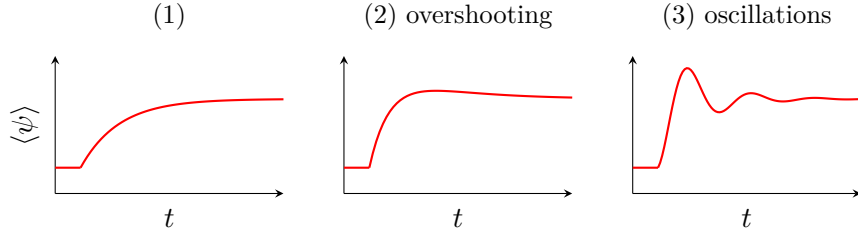


Figure 3.4: Typical responses to a switch-on perturbation.

3.5 Fields: Ginzburg–Landau theory

Experiment: a ferromagnet initially at $T > T_c$, $B = 0$, so $M_{\text{eq}} = 0$. At $t = 0$ the temperature is lowered to $T < T_c$ (a quench), still with $B = 0$. Then $M \rightarrow M_0 \neq 0$, with a free direction, and domains of different orientations form. In Landau theory the order parameter M now becomes a field $m(\vec{r})$ and $F(m)$ a functional.

Attempt 1. $F[m] = \int d^3r f(m)$ with $f(\vec{m}) = a(T - T_c)m^2 + f_4 m^4$, whose minima are $\vec{m}_{\text{eq}} = \sqrt{a(T_c - T)/2f_4} \hat{m}$. This is not the solution: since there is no cost for changing from point to point, in 1D $m(x)$ could jump arbitrarily between $\pm m_{\text{eq}}$. Going back to the microscopic level, in the Ising model $H = -J \sum_{\langle ij \rangle} S_i S_j$ the spins want to be parallel to their neighbours: there is an energy cost for differing from the neighbour, i.e. for $\nabla m \neq 0$.

Attempt 2. Ginzburg proposed the **Ginzburg–Landau free energy**

$$F[\vec{m}] = \int d^d r \left[f(\vec{m}(\vec{r})) + \frac{\kappa}{2} (\nabla \vec{m})^2 \right], \quad (\nabla \vec{m})^2 = \sum_{i=1}^d \sum_{k=1}^D (\partial_i m_k)^2,$$

with d the dimension of space and D the dimension of $\vec{m}$ (a “derivation” can be found in Kardar).

1D case ($d = D = 1$). $F[m] = \int dx \left[f(m(x)) + \frac{\kappa}{2} \left(\frac{\partial m}{\partial x} \right)^2 \right]$. To minimize, let $m(x) \rightarrow m(x) + \delta m$ and demand $\delta F = F[m + \delta m] - F[m] = 0$ for all δm :

$$F[m + \delta m] = \int dx \left(f(m) + \frac{\partial f}{\partial m} \delta m + \frac{\kappa}{2} (m'^2 + 2m' \delta m') \right) = F[m] + \int dx \left(\frac{\partial f}{\partial m} \delta m + \kappa m' \delta m' \right).$$

Integrating by parts (the boundary term vanishes),

$$\delta F = \int dx \left(\frac{\partial f}{\partial m} - \kappa m'' \right) \delta m = 0 \Rightarrow \frac{\partial f}{\partial m} - \kappa m'' = 0.$$

With the Landau form $f = f_0 + f_2 m^2 + f_4 m^4$,

$$2f_2 m + 4f_4 m^3 - \kappa m'' = 0,$$

a differential equation for m_{eq} . Solutions (for $f_2 < 0$):

1. $m(x) = \sqrt{-f_2/2f_4}$;
2. $m(x) = -\sqrt{-f_2/2f_4}$;
3. with boundary conditions $m(\pm\infty) = \pm\sqrt{-f_2/2f_4}$, the **kink**

$$m(x) = \sqrt{\frac{-f_2}{2f_4}} \tanh\left(\frac{x - x_0}{\Delta}\right), \quad \Delta := \sqrt{\frac{\kappa}{-f_2}}.$$

(For $f_2 > 0$ the only solution is $m = 0$.) Reversing the boundary conditions gives the *anti-kink*. With $f_2 = a(T - T_c)$,

$$\Delta = \sqrt{\frac{\kappa}{a(T_c - T)}},$$

so the width of the kink diverges at $T = T_c$.

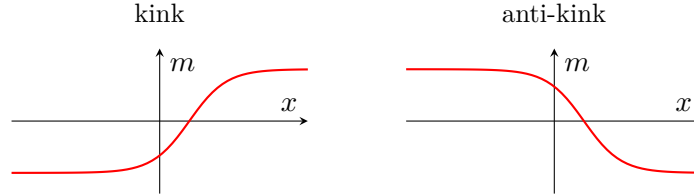


Figure 3.5: Kink and anti-kink solutions.

Cost of a kink. Both F_{kink} and $F_{\text{homogeneous}}$ are infinite, but their difference is finite:

$$\Delta F_{\text{kink}} = \int dx \left[f(m_{\text{kink}}) - f(m_{\text{hom}}) + \frac{\kappa}{2} (m'_{\text{kink}})^2 \right] = \frac{2}{3} \frac{|f_2|^{3/2} \kappa^{1/2}}{f_4} > 0,$$

the energy being localized where m changes. $\Delta F_{\text{kink}} \rightarrow 0$ at $T = T_c$.

2D and 3D. For $m : \mathbb{R}^d \rightarrow \mathbb{R}$ (Ising; materials with an easy axis, which can only be magnetized along one direction), with boundary conditions $m = -m_0$ on one side and $m = +m_0$ on the other (periodic in the other directions), the kink is a wall, and

$$\Delta F_{\text{kink}} = L_y L_z \cdot \left(\frac{2}{3} \frac{|f_2|^{3/2} \kappa^{1/2}}{f_4} \right) :$$

the area times a **surface tension**.

4 Topological defects and slow dynamics

4.1 The Ising universality class and topological defects

The gas–liquid transition (in the P – T diagram, ending at the critical point) is equivalent to Ising-type magnetism ($D = 1$): the order parameter $\psi = \rho - \bar{\rho}$ (density minus its critical value) plays the role of $\psi = m$.

The kink as a topological defect. The kink is topologically protected. Starting from an initial condition with a kink, to remove it one would have to flip a half-infinite region over the barrier between the two minima, which costs $F = (\text{barrier}) \times L$, with L very large: it would take an infinite time. A kink cannot be transformed into “no kink” while keeping all parameters fixed.

A kink–anti-kink pair, in contrast, consists of two defects that are not topologically protected: they can annihilate. How?

1. With a lot of noise, kinks and anti-kinks perform random walks and annihilate when they meet: the *symmetric annihilation process* (SAP).
2. With little noise, m evolves minimizing F . The pair has $\Delta F_{K-AK} = 2\Delta F_K - \Delta F_{\text{int}}(L)$, with an attractive interaction $\Delta F_{\text{int}} \sim e^{-L/\Delta}$ that decays with their separation L (an example: geology).

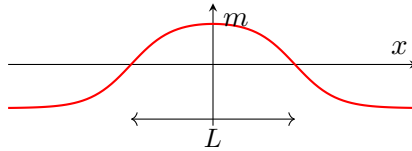


Figure 4.1: A kink–anti-kink pair separated by L .

$\vec{m} : \mathbb{R} \rightarrow \mathbb{R}^2$. With boundary conditions $\vec{m}(-\infty) = -\hat{z}$, $\vec{m}(+\infty) = \hat{z}$, there are two kinds of solution: (1) the magnitude of $\vec{m}$ passes through zero (a kink along z), which costs energy; (2) $\vec{m}$ rotates at fixed modulus $|\vec{m}| = \sqrt{-f_2/2f_4}$, following the circle of minima of f . In the second case the width can grow without bound, so that $\kappa(\nabla\vec{m})^2 \rightarrow 0$.

If T is very low (“deep into the ordered phase”), ΔF_{kink} is large and the only accessible solution is the rotation: $\vec{m} : \mathbb{R} \rightarrow \mathbb{R}^2$ effectively becomes $\hat{m} : \mathbb{R} \rightarrow S^1$ (the unit circle), with $|\vec{m}|$ fixed and $\hat{m}$ the order parameter. The windings cannot be undone: there are topological defects in S^1 but not in $\mathbb{R}^2$, and only at low T . In two dimensions these lead to *skyrmions*.

4.2 Relaxational dynamics of fields

Given $F[\psi]$, equilibrium is the minimum of F : $\delta F = F[m + \delta m] - F[m] = 0$ for all δm . Define the **functional derivative** by

$$\delta F = \int \frac{\delta F}{\delta m(\vec{r})} \delta m(\vec{r}) d\vec{r}.$$

From the 1D computation above, $\delta F = \int dx (f'(m)\delta m + \kappa \partial_x m \partial_x \delta m)$, so

$$\frac{\delta F}{\delta m} = f'(m) - \kappa \frac{\partial^2 m}{\partial x^2},$$

and equilibrium means $\delta F/\delta m(\vec{r}) = 0$ for all $\vec{r}$. We propose the variational dynamics

$$\frac{\partial \psi}{\partial t} = -\lambda \frac{\delta F}{\delta \psi}.$$

1. In equilibrium $\delta F/\delta\psi = 0 \Rightarrow \partial_t\psi = 0$.
2. By the chain rule, $\frac{dF}{dt} = \int \frac{\delta F}{\delta\psi(\vec{r})} \frac{\partial\psi}{\partial t} d\vec{r} = -\lambda \int \left(\frac{\delta F}{\delta\psi}\right)^2 d\vec{r} \leq 0$.
3. If F is the Ginzburg–Landau free energy,

$$\frac{\partial m}{\partial t} = -\lambda \left[2f_2 m + 4f_4 m^3 - \kappa \nabla^2 m \right] = -2\lambda f_2 m - 4\lambda f_4 m^3 + \lambda \kappa \nabla^2 m,$$

the **Ginzburg–Landau equation** ($f_4, \lambda, \kappa > 0$, $f_2 \geq 0$).

With a field B , a kink moves, $x_0 = x_0(B)$: the spins far from the wall cannot flip (that would require crossing the barrier), but those inside the wall can, so the wall moves with $\dot{x}_0 = -\tilde{\lambda} \partial F / \partial x_0$.

4.3 Kink dynamics: multiple time scales

Consider 1D Ising magnetism ($m : \mathbb{R} \rightarrow \mathbb{R}$) without field and $T < T_c$. The kink $m_K(x) = m_0 \tanh((x - x_0)/\Delta)$, with m_0 and Δ functions of the parameters and x_0 arbitrary, reflects spontaneous symmetry breaking (translations). What happens if $B > 0$ (with $|B| < B_{\text{spinodal}}$)? The idea is that x_0 moves.

After rescaling variables and units (for $T < T_c$),

$$\frac{\partial m}{\partial t} = m - m^3 + m'' + B,$$

so the kink is $m_K = \tanh((x - x_0)/\sqrt{2})$. With a small dimensionless parameter $\varepsilon \ll 1$, $B \rightarrow \varepsilon b$:

$$\frac{\partial m}{\partial t} = m - m^3 + m'' + \varepsilon b,$$

with initial condition $m(x, 0) = m_K(x; x_0)$. We use singular perturbation theory:

1. $m = m_0 + \varepsilon m_1 + \varepsilon^2 m_2 + \dots$;
2. several time scales: $t_0 = t$, $t_1 = \varepsilon t$, $t_2 = \varepsilon^2 t$, $\dots$, and $m(x, t) = m(x, t_0, t_1, \dots)$.

Interpretation: if $t \sim \mathcal{O}(1)$, then $t_0 \sim \mathcal{O}(1)$ is active while $t_1 \sim \mathcal{O}(\varepsilon)$ and $t_2 \sim \mathcal{O}(\varepsilon^2)$ have not yet “switched on”. If $t \sim \mathcal{O}(\varepsilon^{-1})$, then $t_0 \sim \mathcal{O}(\varepsilon^{-1}) \rightarrow \infty$ (its dynamics is over), $t_1 \sim \mathcal{O}(1)$ is active and $t_2 \sim \mathcal{O}(\varepsilon)$ is not yet. The separation is arbitrary, and we require that $m(x, t_0, t_1, \dots)$ be regular as $t_i \rightarrow \infty$ and $t_i \rightarrow 0$. Then

$$\frac{\partial}{\partial t} = \frac{\partial t_0}{\partial t} \frac{\partial}{\partial t_0} + \frac{\partial t_1}{\partial t} \frac{\partial}{\partial t_1} + \dots = \sum_{n \geq 0} \varepsilon^n \frac{\partial}{\partial t_n},$$

and

$$\left(\frac{\partial}{\partial t_0} + \varepsilon \frac{\partial}{\partial t_1} + \dots \right) (m_0 + \varepsilon m_1 + \dots) = (m_0 + \varepsilon m_1 + \dots) - (m_0 + \varepsilon m_1 + \dots)^3 + (m_0 + \varepsilon m_1 + \dots)'' + \varepsilon b.$$

Order zero. $\partial_{t_0} m_0 = m_0 - m_0^3 + m_0''$, the G–L equation without b , so

$$m_0(x, t_0, t_1, \dots) = m_K(x - x_0(t_1, \dots)) + \mathcal{O}(e^{-t_0}),$$

where x_0 is a constant for t_0 (the correction vanishes if the initial condition is exactly a kink, and decays as e^{-t_0} if it is close to one).

Order one.

$$\frac{\partial m_0}{\partial t_1} + \frac{\partial m_1}{\partial t_0} = m_1 - 3m_0^2 m_1 + m_1'' + b,$$

with $\partial_{t_1} m_0 = -m_K' \partial_{t_1} x_0$, where the velocity $\partial_{t_1} x_0$ is what we want. Hence

$$\frac{\partial m_1}{\partial t_0} = m_1 - 3m_0^2 m_1 + m_1'' + m_K' \frac{\partial x_0}{\partial t_1} + b. \quad (4.1)$$

The linearized G–L operator. Define

$$\mathcal{L}m := m - 3m_K^2(x)m + m'',$$

a Schrödinger-like operator. Properties: (1) it is linear; (2) $\mathcal{L} = \mathcal{L}^\dagger$ with respect to $(f, g) = \int f^* g dx$; (3) it is not invertible.

Proof of 3. Define the non-linear G–L operator $\mathcal{N}m = m - m^3 + m''$. We know $\mathcal{N}m_K = 0$ for all x_0 (translation symmetry), so $\mathcal{N}m_K(x; x_0 + \delta) - \mathcal{N}m_K(x; x_0) = 0$. Taylor expanding in δ ,

$$-\delta m_K' + 3\delta m_K^2 m_K' - \delta m_K''' = -\delta \mathcal{L}m_K' = 0,$$

so $m_K' \in \ker \mathcal{L}$ and $\mathcal{L}$ is not invertible. The zero mode m_K' reflects the translation invariance.

Solvability condition. Equation (4.1) has the structure $\dot{\vec{x}} = A\vec{x} + \vec{c}$ with $\vec{x} = m_1$, $A = \mathcal{L}$, $\vec{c} = m_K' \partial_{t_1} x_0 + b$, and A not invertible. In a 2×2 example, diagonalizable ($\mathcal{L}$ is Hermitian) and in the diagonal basis,

$$\frac{d}{dt} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} c_1 \\ c_2 \end{pmatrix},$$

with $a < 0$ (the kink is stable; this can be shown). Then $x_1(t) = e^{-|a|t}(x_1(0) - c_1/|a|) + c_1/|a|$ relaxes, while $\dot{x}_2 = c_2$ gives $x_2 = c_2 t$, which grows without bound: *secular growth*. Then $m_1 \sim c_2 t_0$ would grow without bound, violating the initial hypothesis $|m_1| \ll |m_0|$. This is avoided only if $c_2 = 0$: the component of $\vec{c}$ along the null eigenvector must vanish, $\hat{e}_0 \cdot \vec{c} = 0$.

In general $\mathcal{L}$ may be neither invertible nor self-adjoint. Let $e_r \neq 0$ with $\mathcal{L}e_r = 0$ ($e_r \in \ker \mathcal{L}$) and $e_l \neq 0$ with $\mathcal{L}^\dagger e_l = 0$ ($e_l \in \ker \mathcal{L}^\dagger$). Projecting $\partial_{t_0} m_1 = \mathcal{L}m_1 + c$ onto e_l ,

$$\frac{\partial}{\partial t_0}(e_l, m_1) = (e_l, \mathcal{L}m_1) + (e_l, c) = (\mathcal{L}^\dagger e_l, m_1) + (e_l, c) = (e_l, c) \Rightarrow (e_l, m_1) = (e_l, c) t_0,$$

a secular divergence. To avoid it we require the **solvability condition**

$$(e_l, c) = 0 : \quad c \perp \ker \mathcal{L}^\dagger.$$

Here $\mathcal{L}$ is Hermitian and $e_l = e_r = m_K'$:

$$\int dx m_K'(x - x_0) \left(m_K' \frac{\partial x_0}{\partial t_1} + b \right) = 0 \Rightarrow \frac{\partial x_0}{\partial t_1} = -b \frac{\int dx m_K'(x - x_0)}{\int dx m_K'^2(x - x_0)}.$$

With $m_K = \tanh(x/\sqrt{2})$, $\int m_K' = 2$ and $\int m_K'^2 = \frac{2\sqrt{2}}{3}$, so

$$\frac{\partial x_0}{\partial t_1} = -\frac{3}{\sqrt{2}}b =: -\nu b,$$

with mobility $\nu = 3/\sqrt{2}$: the kink moves to the left, so that the region with $m > 0$ (favoured by $b > 0$) grows. One can then solve for m_1 and find that the kink deforms as it travels.

Kinks as particles. The free energy is

$$F = \int dx \left[-\frac{m^2}{2} + \frac{m^4}{4} + \frac{1}{2}m'^2 - bm \right].$$

Moving the kink by δ (from x_0 to $x_0 + \delta$) does not change the first three terms, and

$$\Delta F = -b \int dx [m_K(x - x_0 - \delta) - m_K(x - x_0)] = 2b\delta \Rightarrow \frac{\partial F}{\partial x_0} = 2b.$$

Comparing with the particle-like relaxational dynamics of the first part of the course,

$$\frac{\partial x_0}{\partial t_1} = -\nu b = -\lambda \frac{\partial F}{\partial x_0} \Rightarrow \lambda = \frac{\nu}{2} = \frac{3}{2\sqrt{2}}.$$

The field dynamics reduces to particle dynamics for the kink position.

Exercise. For a kink–anti-kink pair at x_1 and $x_2 = x_1 + L$, propose $m_0(x)$ as the superposition of a kink and an anti-kink. Substituting into the G–L equation $\partial_t m = m - m^3 + m''$ gives 0 plus a remainder, since it is not an exact solution; the remainder is of order $e^{-L/\Delta}$, which plays the role of ε . One finds $\partial x_1/\partial t_1 = -\lambda \partial F/\partial x_1$ and $\partial x_2/\partial t_1 = -\lambda \partial F/\partial x_2$ with the same $\lambda = \nu/2$.

Summary. On the time scale t_0 the G–L dynamics takes a generic initial condition m to a sequence of kinks and anti-kinks ($\partial_{t_0} m = m - m^3 + \dots$: small values grow as e^{t_0} and saturate at ± 1). On the slow scale t_1 , the kinks move with particle-like dynamics

$$\frac{dx_i}{dt} \simeq -\lambda \frac{\partial F}{\partial x_i},$$

except when they annihilate ($L \rightarrow 0$, $e^{-L/\Delta} \sim \mathcal{O}(1)$), where the description fails and one must go back to the G–L equation. On the scale t_1 ,

$$m(x, t) = m_{K-AK}(x; x_1(t), x_2(t), \dots, x_N(t)) :$$

the time dependence is implicit, m is slaved to the motion of the kinks. This is a **dimensional reduction**, from fields to particles, which here occurs because of the existence of topological defects. (Another example of dimensional reduction: Navier–Stokes describes $\sim 10^{23}$ particles with three fields, P, ρ, T .)

4.4 Defects in nematic liquid crystals

Nematic liquid crystals consist of elongated molecules in a liquid phase. At low T the system minimizes its energy by aligning them: the *nematic phase*. What is the Landau order parameter? The director $\hat{n}$, but with $\hat{n} \equiv -\hat{n}$: only the axis matters. This allows more types of defects. Following Ginzburg, $\hat{n}$ tends to be homogeneous. Types of defects:

1. Vortex A: the director circulates around the defect.
2. Vortex B: the director points radially from the defect.

The **topological charge** or winding number (in 2D)

$$q := \frac{1}{2\pi} \oint d\theta$$

around the defect is an integer or a half-integer (there is nothing in between). Both vortices above have $q = 1$; a loop that does not enclose a defect has $q = 0$. In nematics $q = \pm \frac{1}{2}$ defects are also possible, since $\hat{n} \equiv -\hat{n}$.

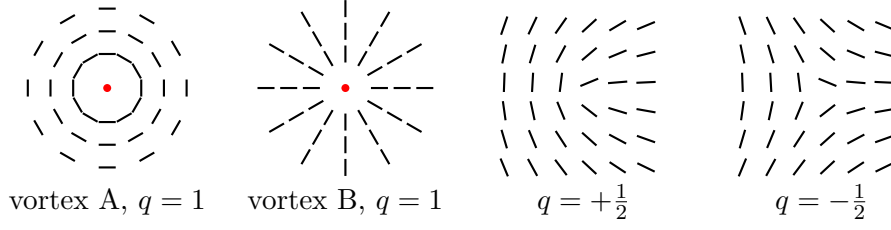


Figure 4.2: Director fields around defects of charge 1 (two kinds) and $\pm\frac{1}{2}$ (only possible because $\hat{n} \equiv -\hat{n}$).

4.5 Binary mixtures and conserved dynamics

Two types of atoms, A and B, with attractive AA and BB interactions and repulsive AB interactions (e.g. water and oil). The control parameters are T and the composition $z = c_A/(c_A + c_B)$. In the phase diagram, below a dome-shaped curve with a maximum at the critical point the mixture separates; above it the phase is homogeneous. This is similar to the gas–liquid coexistence region in the T – ρ plane. (P and ρ are dual: they cannot be controlled at the same time; similarly, in magnetism M plays the role of z .)

Landau theory. The order parameter is the relative concentration $c(\vec{r}) := c_A(\vec{r}) - c_B(\vec{r})$, and $F[c] = F_{\text{GL}}[c]$ (with suitable changes of variables so that the minima are at $\pm$, and a a length), with a double-well f whose minima are the B-rich and A-rich phases. Equilibrium, $\delta F/\delta c = 0$, has a kink solution at x_0 , and F does not change if x_0 moves. But moving x_0 means converting oil into water! This fails because $\int c_A dx$ and $\int c_B dx$ are conserved. The solution is a Lagrange multiplier:

$$F^*[c] = F[c] - \mu \int dx c,$$

with μ the chemical potential; equilibrium is $\delta F^*/\delta c = 0$ with $\int dx c = C$ constant. Now a droplet (a kink–anti-kink pair) is a stable solution.

Dynamics. A first proposal, $\partial_t c = -\lambda \delta F^*/\delta c$, does not conserve mass: integrating, $\partial_t \int c dx = 0 \neq -\lambda \int dx \delta F^*/\delta c$. Second proposal: c is conserved and molecules do not teleport, so

$$\frac{\partial c}{\partial t} = -\nabla \cdot \vec{J},$$

with $\vec{J}$ depending on $\delta F/\delta c$. We propose $\vec{J} := -\lambda \nabla \left(\frac{\delta F}{\delta c} \right)$, so

$$\frac{\partial c}{\partial t} = \lambda \nabla^2 \frac{\delta F}{\delta c}.$$

Properties: (1) c is conserved; (2) integrating by parts,

$$\dot{F} = \int \frac{\delta F}{\delta c} \frac{\partial c}{\partial t} = \lambda \int \frac{\delta F}{\delta c} \nabla^2 \frac{\delta F}{\delta c} = -\lambda \int \left(\nabla \frac{\delta F}{\delta c} \right)^2 = -\frac{1}{\lambda} \int J^2 \leq 0,$$

and $\dot{F} = 0$ iff $\vec{J} = 0$ everywhere, i.e. $\nabla(\delta F/\delta c) = 0$, $\delta F/\delta c = \mu$, which is $\delta F^*/\delta c = 0$.

4.6 Models A and B

In the Halperin–Hohenberg classification, for an order parameter ψ with free energy $F[\psi]$:

- **Model A** (non-conserved order parameter):

$$\frac{\partial \psi}{\partial t} = -\lambda \frac{\delta F}{\delta \psi}.$$

If $F = F_{\text{GL}}[\psi] = \int d\vec{r} \left[\frac{a}{2} \psi^2 + \frac{b}{4} \psi^4 + \frac{\kappa}{2} (\nabla \psi)^2 \right]$, this is the G–L equation

$$\frac{\partial \psi}{\partial t} = -\lambda(a\psi + b\psi^3) + \lambda\kappa \nabla^2 \psi.$$

- **Model B** (conserved order parameter): $\partial_t \psi = -\nabla \cdot \vec{J}$, $\vec{J} = -\lambda \nabla (\delta F / \delta \psi)$,

$$\frac{\partial \psi}{\partial t} = \lambda \nabla^2 \frac{\delta F}{\delta \psi}.$$

ψ evolves with $\dot{F} \leq 0$ and stops when $\dot{F} = 0 \Leftrightarrow \vec{J} = 0 \forall \vec{r} \Leftrightarrow \delta F / \delta \psi = \mu$. For $F = F_{\text{GL}}$ this is the **Cahn–Hilliard equation**

$$\frac{\partial \psi}{\partial t} = \nabla^2 (\lambda a \psi + \lambda b \psi^3) - \lambda \kappa \nabla^2 \nabla^2 \psi$$

(note that $\nabla^2 \nabla^2 \neq \sum_i \partial^4 / \partial x_i^4$).

4.7 Dynamics near equilibrium

Let $F[\psi] = \int d\vec{r} [f(\psi) + \frac{\kappa}{2} (\nabla \psi)^2]$ and ψ_{eq} with $f'(\psi_{\text{eq}}) = 0$, $f''(\psi_{\text{eq}}) > 0$. Near equilibrium, $\psi = \psi_{\text{eq}} + \psi_1(\vec{r}, t)$ and

$$F \simeq \int d\vec{r} \left[f(\psi_{\text{eq}}) + \frac{1}{2} f''(\psi_{\text{eq}}) \psi_1^2 + \frac{\kappa}{2} (\nabla \psi_1)^2 \right].$$

Model A.

$$\frac{\partial \psi_1}{\partial t} = -\lambda \frac{\delta F}{\delta \psi_1} = -\lambda [f''(\psi_{\text{eq}}) \psi_1 - \kappa \nabla^2 \psi_1]$$

(linear dynamics). For Fourier modes $\psi_1(\vec{r}, t) = \psi_{1k}(t) e^{i\vec{k} \cdot \vec{r}}$,

$$\frac{d\psi_{1k}}{dt} = -(\lambda f''(\psi_{\text{eq}}) + \kappa \lambda k^2) \psi_{1k} =: -\Lambda_k \psi_{1k},$$

with relaxation rate Λ_k and relaxation time $T_k := 1/\Lambda_k$. There is a maximum relaxation time $T_k^{\text{max}} = 1/\lambda f''(\psi_{\text{eq}})$: the system reaches equilibrium within at most T_k^{max} , so the dynamics is fast. But if $f''(\psi_{\text{eq}}) \rightarrow 0$ (at the critical point), $\Lambda_k = \kappa \lambda k^2$ and $T_k^{\text{max}} \rightarrow \infty$: there is always a small k for which T_k is large (macroscopic), and the dynamics is slow. With $k_{\text{min}} = 2\pi/L_{\text{sys}}$,

$$T^{\text{max}} = \frac{L_{\text{sys}}^2}{(2\pi)^2 \lambda \kappa} \sim t_0 \left(\frac{L_{\text{macro}}}{l_{\text{micro}}} \right)^2 \sim 10^{-13} \text{ s} \left(\frac{1 \text{ m}}{10^{-10} \text{ m}} \right)^2 \sim 10^7 \text{ s},$$

by dimensional analysis: slow because the system is very large, not because it is intrinsically slow. So model A is fast except at the critical point (which is why systems near critical points are studied).

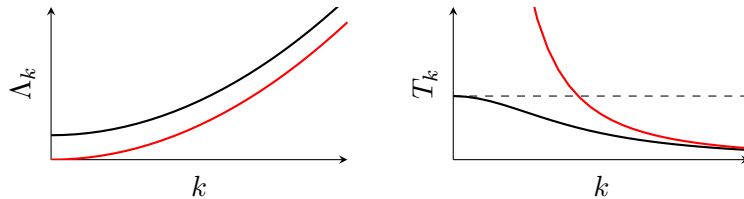


Figure 4.3: Model A relaxation rates and times away from (black) and at (red) the critical point.

Model B.

$$\begin{aligned}\frac{\partial \psi_1}{\partial t} &= \lambda \nabla^2 \frac{\delta F}{\delta \psi_1} = \lambda f''(\psi_{\text{eq}}) \nabla^2 \psi_1 - \kappa \lambda \nabla^2 \nabla^2 \psi_1, \\ \frac{d\psi_{1k}}{dt} &= -\left[\lambda f''(\psi_{\text{eq}}) k^2 + \kappa \lambda k^4\right] \psi_{1k} = -\Lambda_k \psi_{1k}.\end{aligned}$$

Now $\Lambda_k \rightarrow 0$ as $k \rightarrow 0$ always: there is no T^{max} , and the dynamics is always slow.

Preliminary summary. Slow dynamics arises in: model A at the critical point; model B (associated with conservation laws); the dynamics of defects; and, as we see next, Goldstone modes.

4.8 Goldstone modes

XY model (magnetism with $\vec{m} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$). With $F_{\text{GL}}[\vec{m}]$ and $f_2 < 0$ very negative, the barrier $\Delta f \gg k_B T$ cannot be crossed except through defects. The equilibrium state is $\vec{m} = m_0 \hat{m}$, with m_0 the minimum of f and $\hat{m} = (\cos \theta, \sin \theta)$ arbitrary, $\theta \in [0, 2\pi)$: continuous spontaneous symmetry breaking. At small but finite T , or with external forces, there are fluctuations around equilibrium:

1. $|\vec{m}| \rightarrow m_0 + \delta m$, with cost $\Delta F = \frac{1}{2} f''(m_0) \delta m^2 \times (\text{size})$;
2. $\theta \rightarrow \theta + \delta \theta$, with zero cost. A uniform $\delta \theta$ is hard to produce; it is easier to have $\delta \theta = \delta \theta(\vec{r}) = \theta_0 e^{i\vec{k} \cdot \vec{r}}$, a *spin wave*. Since f does not change,

$$\Delta F = F[\vec{m} \text{ perturbed}] - F[\vec{m}_{\text{hom}}] = \frac{\kappa}{2} \int d\vec{r} (\nabla \vec{m})^2,$$

with $\nabla \vec{m} = m_0(-\sin \theta, \cos \theta) \nabla \theta$, $(\nabla \vec{m})^2 = m_0^2 (\nabla \theta)^2 = m_0^2 \theta_0^2 k^2$. Hence $\Delta F = \frac{\kappa}{2} m_0^2 \theta_0^2 k^2 \cdot V$.

The dynamics (per unit volume) is

$$\frac{d\theta_0}{dt} = -\lambda \frac{\partial}{\partial \theta_0} \frac{\Delta F}{V} = -\lambda \kappa m_0^2 k^2 \theta_0 :$$

slow dynamics. Continuous spontaneous symmetry breaking leads to slow dynamics: the **Goldstone modes**. If the symmetry were explicitly broken (e.g. by a field), Λ_k would not vanish at $k = 0$ and the dynamics would be fast.

Liquid–crystalline solid transition. The order parameter is the 1D density profile (de Gennes): uniform in the liquid, periodic with period a in the solid. Better: the amplitude ρ_k of the Fourier mode $k^* = 2\pi/a$, which jumps from a finite value in the solid to zero in the liquid (first-order transition). The broken symmetry is where the first atom is placed, $x_0 \in [0, a)$, or more precisely on a periodic torus $T_{[0,a)}$ (in 2D, $\vec{r}_0 \in T_{[0,a) \times [0,a)}$), equivalent to the phase of the Fourier transform.

Goldstone theorem. If $\vec{r}_0$ is modulated in space, $\Delta F \propto k^2$: longitudinal (density) waves $\vec{r}_0 = \epsilon \hat{x} e^{ikx}$ and transverse (shear) waves $\vec{r}_0 = \epsilon \hat{y} e^{ikx}$ cost energy $\propto k^2$; this is linear elasticity. In a liquid there is no spontaneous symmetry breaking and the only order parameter is the density: perturbations $\rho = \rho_{\text{eq}} + \rho_1 e^{i\vec{k} \cdot \vec{r}}$ (density waves) cost $\Delta F \sim \frac{f''_{\text{eq}}}{2} \rho_1^2$ and, since the density is conserved, relax as $\partial_t \rho_1 = \nabla^2 \rho_1$. In a liquid shear does not cost energy: that is why we sink in water, while a solid resists shear and holds us up.

5 Quantum non-equilibrium statistical mechanics

Goals: linear response theory, the fluctuation–dissipation theorem and the Green–Kubo relations in quantum mechanics. The language is the density matrix ρ .

5.1 Density matrix

In ordinary quantum mechanics a state is $|\psi\rangle \in \mathcal{H}$ (Hilbert space) and expectation values are $\langle\psi|O|\psi\rangle$. If the state is not known we need quantum statistical mechanics, whose postulate is that *all the information is in ρ* . Simple case: an orthonormal basis $|\psi_k\rangle$, with p_k the probability of being in $|\psi_k\rangle$, and $\langle A \rangle := \sum_k p_k \langle\psi_k|A|\psi_k\rangle$. Define the operator on $\mathcal{H}$

$$\rho := \sum_k p_k |\psi_k\rangle \langle\psi_k|.$$

Then

$$\langle A \rangle = \sum_{k,l} p_k \langle\psi_k|\psi_l\rangle \langle\psi_l|A|\psi_k\rangle = \sum_l \langle\psi_l|A\rho|\psi_l\rangle = \text{Tr}\{A\rho\}.$$

Properties (postulates: there exists ρ with): (1) $\rho \geq 0$ (positive semi-definite); (2) $\text{Tr } \rho = 1$; (3) $\rho = \rho^\dagger$.

Dynamics. Heisenberg: $dA/dt = \frac{i}{\hbar}[H, A]$; Schrödinger: $\frac{d}{dt}|\psi\rangle = -\frac{i}{\hbar}H|\psi\rangle$. For an isolated system with Hamiltonian H and constant p_k (an ensemble of systems),

$$\frac{d\rho}{dt} = \sum_k p_k \left(\frac{d}{dt} |\psi_k\rangle \right) \langle\psi_k| + \sum_k p_k |\psi_k\rangle \frac{d}{dt} \langle\psi_k| = -\frac{i}{\hbar} \sum_k p_k (H|\psi_k\rangle \langle\psi_k| - |\psi_k\rangle \langle\psi_k| H),$$

i.e.

$$\frac{d\rho}{dt} = -\frac{i}{\hbar}[H, \rho]$$

(note the sign relative to the Heisenberg equation). Define the **Liouvillian**

$$\mathcal{L} := \frac{i}{\hbar}[H, \cdot],$$

where the factor i makes it anti-Hermitian. Then

$$\frac{dA}{dt} = \mathcal{L}A \quad (\text{Heisenberg}), \quad \frac{d\rho}{dt} = -\mathcal{L}\rho \quad (\text{von Neumann}).$$

Since $\mathcal{L}^\dagger = -\mathcal{L}$, its eigenvalues are imaginary, and the solution of the von Neumann equation, $\rho(t) = e^{-\mathcal{L}t}\rho_0 \sim \sum_k e^{-i\omega_k t}\rho_0$, is a rotation of phases. (When the system is not isolated this does not hold, and the evolution can converge, for instance, to the Gibbs distribution.) Equivalently,

$$\rho(t) = e^{-iHt/\hbar}\rho(0)e^{iHt/\hbar}, \quad A(t) = e^{iHt/\hbar}A(0)e^{-iHt/\hbar}.$$

Equilibrium solutions satisfy $[H, \rho] = 0$, so $\rho = \rho(H, X)$ with X other conserved quantities. With a heat bath, $\rho_{\text{eq}} = Z^{-1}e^{-\beta H}$, $Z = \text{Tr } e^{-\beta H}$.

Properties of $\mathcal{L}$.

1. $\mathcal{L}(AB) = (\mathcal{L}A)B + A(\mathcal{L}B)$: $\mathcal{L}$ acts as a derivative.
2. $e^{\mathcal{L}t}(AB) = e^{iHt/\hbar}ABe^{-iHt/\hbar} = (e^{\mathcal{L}t}A)(e^{\mathcal{L}t}B)$ (inserting $\mathbb{K} = e^{-iHt/\hbar}e^{iHt/\hbar}$), so $(AB)(t) = A(t)B(t)$.
3. $\mathcal{L}\rho_{\text{eq}} = 0 \Rightarrow e^{-\mathcal{L}t}\rho_{\text{eq}} = \rho_{\text{eq}}$.
4. Averages can be computed in either picture, $\langle A \rangle(t) = \text{Tr}\{A(t)\rho(0)\} = \text{Tr}\{A(0)\rho(t)\}$, by cyclicity of the trace.

Define the inner product $(A, B) := \text{Tr}\{AB\rho_{\text{eq}}\}$; with it, $\mathcal{L}^\dagger = -\mathcal{L}$.

5.2 Equilibrium correlation functions

For two observables A and B define

$$C_{AB}(t, t') := \text{Tr}\{A(t)B(t')\rho_{\text{eq}}\},$$

which is an inner product. Properties:

1. Stationarity: moving $e^{-iHt'/\hbar}$ around the trace (it commutes with $e^{-\beta H}$),

$$C_{AB}(t, t') = \text{Tr}\{A(t - t')B(0)\rho_{\text{eq}}\} = C_{AB}(t - t', 0),$$

since ρ_{eq} is stationary. We write $C_{AB}(t) := C_{AB}(t, 0)$.

2. Using cyclicity and $e^{-\beta H} = e^{-\beta H}e^{\beta H}e^{-\beta H}$,

$$\begin{aligned} C_{AB}(t) &= \frac{1}{Z} \text{Tr}\{e^{iHt/\hbar} A e^{-iHt/\hbar} B e^{-\beta H}\} \\ &= \frac{1}{Z} \text{Tr}\{e^{\beta H} e^{-iHt/\hbar} B e^{-\beta H} e^{iHt/\hbar} A e^{-\beta H}\} \\ &= \text{Tr}\{B(-t - i\beta\hbar) A \rho_{\text{eq}}\}, \end{aligned}$$

i.e.

$$C_{AB}(t) = C_{BA}(-t - i\beta\hbar).$$

In Fourier space, $\hat{C}_{AB}(\omega) := \frac{1}{2\pi} \int_{-\infty}^{\infty} dt e^{i\omega t} C_{AB}(t)$, and with $s = -(t + i\beta\hbar)$,

$$\hat{C}_{AB}(\omega) = \frac{1}{2\pi} \int ds e^{i\omega(-s - i\beta\hbar)} C_{BA}(s) = e^{\beta\hbar\omega} \hat{C}_{BA}(-\omega).$$

The factor $e^{\beta\hbar\omega}$ is the **Nyquist factor** (in the classical limit $\hbar \rightarrow 0$ it equals 1).

5.3 Linear response theory

Consider $H(t) = H_0 + H_1 B(t)$, with $B(t) = 0$ for $t < 0$; H, H_0, H_1 are operators and $B(t)$ a function. H and H_0 are Hermitian; H_1 may not be (e.g. $H = H_0 + b(e^{i\omega t} H_1 + e^{-i\omega t} H_1^\dagger)$).

Example (Electrical conduction). In a crystal with classical fixed ions and quantum electrons, an electric field induces $H_1 = \sum_i q\phi(\vec{r}_i)$, with $q = -e$. For a homogeneous field $\vec{E}(\vec{r}, t) = \vec{E}(t)$, $\phi(\vec{r}, t) = -\vec{E}(t) \cdot \vec{r}$, so

$$H_1 = -\vec{E}(t) \cdot \sum_i q\vec{r}_i,$$

a number times an operator. If the observable is the electric current,

$$A = \frac{1}{V} \sum_i q\dot{\vec{r}}_i = \frac{1}{V} \sum_i q \frac{d\vec{r}_i}{dt} \sim \dot{H}_1 :$$

the *current-type* case, $A \sim \dot{H}_1$. For insulators the observable is the polarization, $A = \frac{1}{V} \sum_i q(\vec{r}_i - \vec{r}_{i,\text{eq}}) \sim H_1$: the *polarization-type* case, $A \sim H_1$.

Following Pottier's *Nonequilibrium Statistical Physics* (who writes $H = H_0 - H_1 a(t)$), consider $H = H_0 + H_1 B(t)$ with initial condition $\rho(t=0) = \rho_0 = Z^{-1} e^{-\beta H_0}$ and $B(t) = 0$ for $t < 0$. (Another possibility is $\rho(t=-\infty) = \rho_0$ and $B(t) = e^{\eta t} \hat{B}(t)$ with $\eta \rightarrow 0^+$: an adiabatic switching on.) Now

$$\mathcal{L} = \frac{i}{\hbar} [H, \cdot] = \mathcal{L}_0 + \mathcal{L}_1 B(t), \quad \frac{d\rho}{dt} = -\mathcal{L}\rho.$$

To first order, $\rho = \rho_0 + \rho_1(t)$ with

$$\frac{d\rho_1}{dt} = -\mathcal{L}_0\rho_1 - \mathcal{L}_1\rho_0 B(t) + \mathcal{O}(2), \quad \mathcal{L}_0\rho_0 = 0, \quad \rho_1(0) = 0,$$

whose solution is

$$\rho_1(t) = - \int_0^t e^{-\mathcal{L}_0(t-s)} (\mathcal{L}_1\rho_0) B(s) ds.$$

For an observable with $\langle A \rangle_{\text{eq}} = \text{Tr}\{A\rho_0\} = 0$,

$$\langle A \rangle(t) = \text{Tr}\{A\rho_1(t)\} = \int_0^t ds \underbrace{\left(-\text{Tr}\{Ae^{-\mathcal{L}_0(t-s)}\mathcal{L}_1\rho_0\} \right)}_{\phi(t-s)} B(s), \quad \langle A \rangle(t) = \int_0^t ds \phi(t-s) B(s),$$

with ϕ the **memory or response function**. Using $e^{-\mathcal{L}_0 t}(\mathcal{L}_1\rho_0) = (e^{-\mathcal{L}_0 t}\mathcal{L}_1)\rho_0$ (since $e^{-\mathcal{L}_0 t}\rho_0 = \rho_0$ and $\mathcal{L}_0$ is a derivation), the inner product and $\mathcal{L}_0^\dagger = -\mathcal{L}_0$,

$$\phi(t) = -(A, e^{-\mathcal{L}_0 t}\mathcal{L}_1) = -(e^{\mathcal{L}_0 t}A, \mathcal{L}_1) = -\text{Tr}\{A(t)\mathcal{L}_1\rho_0\},$$

where $A(t)$ is the free (interaction-picture) evolution; $\mathcal{L}_1\rho_0 = \frac{i}{\hbar}[H_1, \rho_0]$. Hence

$$\phi(t) = -\frac{i}{\hbar} \text{Tr}\{A(t)[H_1, \rho_0]\} = -\frac{i}{\hbar} \text{Tr}\{[A(t), H_1]\rho_0\},$$

the **Kubo formula**

$$\phi(t) = -\frac{i}{\hbar} \langle [A(t), H_1] \rangle_0,$$

with free evolution (without the perturbation) and the average taken with ρ_0 .

Spectral representation. Let $H_0|\varphi_n\rangle = E_n|\varphi_n\rangle$ and assume a diagonal density matrix, $\langle\varphi_k|\rho_0|\varphi_n\rangle = p_n\delta_{nk}$ (e.g. $p_n = Z^{-1}e^{-\beta E_n}$). With $A_{nl} := \langle\varphi_n|A|\varphi_l\rangle$ and $A(t)_{nl} = e^{i(E_n-E_l)t/\hbar}A_{nl}$,

$$\phi(t) = -\frac{i}{\hbar} \sum_{n,l} \left[e^{i(E_n-E_l)t/\hbar} A_{nl} H_{1,ln} - e^{i(E_l-E_n)t/\hbar} H_{1,nl} A_{ln} \right] p_n,$$

and exchanging $n \leftrightarrow l$ in the second term,

$$\phi(t) = -\frac{i}{\hbar} \sum_{n,l} (p_n - p_l) e^{i(E_n-E_l)t/\hbar} A_{nl} H_{1,ln}.$$

With few degrees of freedom the sum is finite and $\phi(t)$ does not decay. With many degrees of freedom the Bohr frequencies $\omega_{nl} = (E_n - E_l)/\hbar$ are incommensurable (there is no rational ratio between them; classically, these are the action-angle variables). With few of them there is an approximate common period: e.g. $\omega_1 = 2\pi$, $\omega_2 = 2\pi/\sqrt{2}$, periods $T_1 = 1$, $T_2 = \sqrt{2} \simeq 1.41$; at $t = 141$, ω_1 has made 141 turns and ω_2 100. With many degrees of freedom the approximate common period goes to $+\infty$ (the function is quasi-periodic, effectively irreversible) and $\phi(t)$ decays. Irreversibility ($\phi(t) \rightarrow 0$ as $t \rightarrow \infty$) is reached in the limit

$$\lim_{t \rightarrow \infty} \lim_{N \rightarrow \infty},$$

in this order (not the reverse).

Example (Atom in an electric field). $H = H_0 - q\vec{r} \cdot \vec{E}$ with $\vec{E} = E\hat{z}$, so $H = H_0 - qzE$: $H_1 = -qz$, $B = E$. We measure the polarization $A = qz$ (polarization case). At $T = 0$, $p_0 = 1$ and $p_i = 0$ for $i > 0$ (ground state). With $\omega_{nl} = (E_n - E_l)/\hbar$,

$$\begin{aligned} \phi(t) &= \frac{iq^2}{\hbar} \sum_{n,l} (\delta_{n0} - \delta_{l0}) e^{i\omega_{nl}t} z_{nl} z_{ln} = \frac{iq^2}{\hbar} \left[\sum_l e^{i\omega_{0l}t} |z_{0l}|^2 - \sum_n e^{i\omega_{n0}t} |z_{n0}|^2 \right] \\ &= \frac{iq^2}{\hbar} \sum_l |z_{0l}|^2 (e^{-i\omega_{l0}t} - e^{i\omega_{l0}t}) = \frac{2q^2}{\hbar} \sum_l |z_{0l}|^2 \sin(\omega_{l0}t), \end{aligned}$$

which can be computed and oscillates without decaying.

5.4 Kubo transform and the classical limit

We had $\phi(t) = -\frac{i}{\hbar} \langle [A(t), H_1] \rangle = -\text{Tr}\{A(t)\mathcal{L}_1\rho_0\}$. Define the operator-valued function

$$\Phi(\beta) := [H_1, e^{-\beta H_0}]e^{\beta H_0} = H_1 - e^{-\beta H_0} H_1 e^{\beta H_0}.$$

Properties: (1) $\Phi(0) = 0$; (2) $\frac{d\Phi}{d\beta} = e^{-\beta H_0} [H_0, H_1] e^{\beta H_0}$. Together,

$$\Phi(\beta) = \int_0^\beta d\lambda e^{-\lambda H_0} [H_0, H_1] e^{\lambda H_0}.$$

Then

$$\mathcal{L}_1\rho_0 = \frac{i}{\hbar} Z^{-1} [H_1, e^{-\beta H_0}] = \frac{i}{\hbar} \Phi(\beta)\rho_0 = \int_0^\beta d\lambda e^{-\lambda H_0} \underbrace{\frac{i}{\hbar} [H_0, H_1] e^{\lambda H_0}}_{\mathcal{L}_0 H_1} \rho_0.$$

Define the **Kubo transform**

$$\tilde{A} := \frac{1}{\beta} \int_0^\beta d\lambda e^{-\lambda H_0} A e^{\lambda H_0} = \frac{1}{\beta} \int_0^\beta d\lambda A(i\hbar\lambda).$$

Then $\mathcal{L}_1\rho_0 = \beta(\mathcal{L}_0\tilde{H}_1)\rho_0$ and

$$\phi(t) = -\beta \text{Tr}\{A(t) \mathcal{L}_0\tilde{H}_1 \rho_0\} = -\beta \langle A(t) \dot{\tilde{H}}_1 \rangle, \quad \mathcal{L}_0\tilde{H}_1 = \dot{\tilde{H}}_1,$$

a correlation function between A and $\dot{\tilde{H}}_1$. In the classical limit $\tilde{A} = A$ (the operators commute), and

$$\phi(t) = -\beta \langle A(t) \dot{H}_1 \rangle.$$

For the current-type case $A = \vec{J} = -\frac{1}{V} \dot{H}_1$ (see below), $\phi(t) \propto +\beta \langle J(t)J(0) \rangle$.

5.5 Electric conductivity

A solid with an electric field $\vec{E} = E(t)\hat{z}$ that varies smoothly (no magnetic field):

$$H = H_0 - \sum_i q\vec{r}_i \cdot \vec{E} = H_0 - \underbrace{\sum_i qz_i}_{H_1} \underbrace{E(t)}_{B(t)}, \quad q = -e,$$

where H_0 contains the kinetic and potential energy of the electrons. The observable is the current density $A = J_z = \frac{1}{V} \sum_i qv_{iz} = -\frac{1}{V} \dot{H}_1$. Then

$$\langle J \rangle(t) = \int_0^t \phi(t-s)E(s)ds, \quad \phi(t) = \beta V \langle J(t)\tilde{J}(0) \rangle, \quad \tilde{J} = \frac{1}{\beta} \int_0^\beta d\lambda J(i\hbar\lambda).$$

Parity of $\phi(t)$.

$$\begin{aligned} \phi(-t) &= \beta V \langle J(-t)\tilde{J}(0) \rangle = V \int_0^\beta d\lambda \langle J(-t)J(i\hbar\lambda) \rangle \\ &= \frac{V}{Z} \int_0^\beta d\lambda \text{Tr}\{e^{-iHt/\hbar} J e^{iHt/\hbar} e^{-\lambda H} J e^{\lambda H} e^{-\beta H}\}. \end{aligned}$$

Using cyclicity and the change of variables $\mu = \beta - \lambda$,

$$\phi(-t) = \frac{V}{Z} \int_0^\beta d\mu \text{Tr}\{e^{iHt/\hbar} J e^{-iHt/\hbar} e^{-\mu H} J e^{\mu H} e^{-\beta H}\} = \phi(t) :$$

ϕ is even, because an equilibrium distribution with the same H_0 is used. (If the system starts in equilibrium with H_0 and is then evolved with $H_1 = H_0 + \epsilon(\dots)$, the problem is particularly difficult and $\phi(t)$ is no longer necessarily symmetric.)

Static case. $\langle J(t) \rangle = \int_0^t ds \phi(s) E(t-s)$. For $E(t) = E_0 \Theta(t)$, $\langle J(t) \rangle = \int_0^t ds \phi(s) E_0$, and for large t

$$\langle J \rangle_\infty = \underbrace{\int_0^\infty ds \phi(s)}_{\sigma_0} E_0,$$

Ohm's law $J = \sigma_0 E$ with σ_0 the electrical conductivity.

Oscillatory case. For $E(t) = E_\omega e^{-i\omega t} \Theta(t)$,

$$\langle J \rangle(t) = \int_0^t ds \phi(s) e^{-i\omega(t-s)} E_\omega \xrightarrow{t \rightarrow \infty} \underbrace{\left[\int_0^\infty ds \phi(s) e^{i\omega s} \right]}_{\sigma_\omega} E_\omega e^{-i\omega t},$$

with σ_ω the complex conductivity (a Fourier–Laplace transform), $\sigma_0 = \sigma_{\omega \rightarrow 0}$. We write $\sigma_\omega = \sigma'_\omega + i\sigma''_\omega$: the real part corresponds to the ohmic resistance and Joule dissipation, the imaginary part to the capacitive impedance (energy storage). Since ϕ is even,

$$\sigma'_\omega = \int_0^\infty ds \phi(s) \cos(\omega s) = \frac{1}{2} \int_{-\infty}^\infty ds \phi(s) e^{i\omega s} = \frac{V}{2} \int_0^\beta d\lambda \int_{-\infty}^\infty ds \langle J(s - i\hbar\lambda) J(0) \rangle e^{i\omega s}.$$

With $z = s - i\hbar\lambda$, and since the function is analytic (no poles; to be shown using eigenfunctions), the contour can be shifted back to the real axis:

$$\sigma'_\omega = \frac{V}{2} \int_0^\beta d\lambda e^{-\omega\hbar\lambda} \int_{-\infty}^\infty dz \langle J(z) J(0) \rangle e^{i\omega z} = \frac{V}{2} \frac{1 - e^{-\beta\hbar\omega}}{\hbar\omega} \tilde{C}_{JJ}(\omega),$$

where $\tilde{C}_{JJ}(\omega)$ is the Fourier transform of the equilibrium current correlation function. This is the **fluctuation–dissipation theorem**:

$$\sigma'_\omega = V \left(\frac{1 - e^{-\beta\hbar\omega}}{2\hbar\omega} \right) \tilde{C}_{JJ}(\omega) :$$

dissipation (left) is given by equilibrium fluctuations (right).

Kramers–Kronig. Since $\sigma(\omega)$ is analytic (in the upper half plane),

$$\sigma''(\omega) = -\frac{1}{\pi} \text{P} \int_{-\infty}^\infty dz \frac{\sigma'(z)}{z - \omega},$$

with P the principal value, $\lim_{t \rightarrow \infty} \int_{-t}^t$ excluding the pole symmetrically.

Classical limit ($\hbar \rightarrow 0$): $\sigma'_\omega = \frac{\beta V}{2} \tilde{C}_{JJ}(\omega)$; the static conductivity ($\omega \rightarrow 0$) is $\sigma'_0 = \frac{\beta V}{2} \tilde{C}_{JJ}(0)$.

Is σ extensive? $J = \frac{1}{V} \sum_i q v_i = \mathcal{O}(N/V)$ is intensive, with $\langle J \rangle_{\text{eq}} = 0$. Its standard deviation scales as $1/\sqrt{N}$ (for an extensive quantity, as $\sqrt{N}$), so $\text{Var}(J) \sim 1/N$ and $\sigma \sim V \text{Var}(J) \sim \mathcal{O}(V/N)$: σ is intensive. Equivalently, $\sigma'_0 = \frac{\beta}{2V} \tilde{C}_{XX}(0)$ with $X = \sum_i q v_i$.

5.6 Estimate for a crystalline solid

Electrons are fermions, $\langle n_k \rangle = f_{\text{FD}}(\varepsilon_k) = 1/(1 + e^{\beta(\varepsilon_k - \mu)})$. For free non-relativistic electrons in 3D, $\varepsilon_k = \hbar^2 k^2/2m$, $\vec{k} = 2\pi\vec{n}/L$, $\vec{n} \in \mathbb{Z}^3$, and

$$N = \sum_{\vec{k}, \text{spin}} \langle n_k \rangle = V \int d\varepsilon g(\varepsilon) f_{\text{FD}}(\varepsilon),$$

with $Vg(\varepsilon)d\varepsilon$ the number of levels and g intensive; for free electrons $g(\varepsilon) = g_0\sqrt{\varepsilon}$. Electrons in a crystalline solid have bands separated by gaps; if μ lies in a gap the solid is an insulator.

The current is

$$J = \frac{1}{V} \sum_{i \in e^-} qv_i = \frac{1}{V} \sum_{\vec{k}, \text{spin}} qv_k n_k = \int d\varepsilon q v n g$$

(from first to second quantization). In equilibrium $\langle J \rangle = 0$ by isotropy; $\langle J^2 \rangle = \frac{q^2}{V^2} \sum_{k,l} v_k v_l \langle n_k n_l \rangle$ and $\langle J \rangle^2 = 0$. If the electrons are independent, $\langle n_k n_l \rangle - \langle n_k \rangle \langle n_l \rangle = (\langle n_k^2 \rangle - \langle n_k \rangle^2) \delta_{kl}$, so

$$\langle J^2 \rangle - \langle J \rangle^2 = \frac{q^2}{V^2} \sum_k v_k^2 [\langle n_k^2 \rangle - \langle n_k \rangle^2].$$

Since $n_k \in \{0, 1\}$ with $P(n_k = 1) = p = f_{\text{FD}}$, $\langle n_k \rangle = \langle n_k^2 \rangle = p$ and

$$\langle n_k^2 \rangle - \langle n_k \rangle^2 = p(1 - p) = f_{\text{FD}}(\varepsilon)(1 - f_{\text{FD}}(\varepsilon)),$$

which is peaked around μ : the fluctuations occur near the Fermi energy. In an insulator $\langle J^2 \rangle = 0$, so $C_{JJ} = 0$ and $\sigma = 0$. In a conductor $\langle J^2 \rangle \sim g(\mu)$, and roughly $C_{JJ}(t) = \langle J^2 \rangle e^{-t/\tau}$, with $\langle J^2 \rangle$ from equilibrium and τ from scattering: the conductivity is good if τ is large. An insulator is not a bad conductor: a bad conductor has small τ , while a non-conductor has $\langle J^2 \rangle = 0$.

5.7 Quantum kinetic theory

Goal: the electrical conductivity. We start with quantum gases: $H = H_0 + H_1$, with H_0 for independent particles (thermodynamics) and a small interaction H_1 (dynamics), in a box of volume $V = L^3$ with periodic boundary conditions; $H_0 = \sum_i p_i^2/2m$ or $\sum_i c|p_i|$. The eigenstates are symmetric or antisymmetric (bosons or fermions) products $|\vec{k}_1 s_{z1}\rangle \otimes \cdots \otimes |\vec{k}_N s_{zN}\rangle$, $\vec{k}_i = 2\pi\vec{n}/L$, $\vec{n} \in \mathbb{Z}^3$. In second quantization, $|\psi\rangle = |n_1, n_2, \dots, n_i, \dots\rangle$ with $i = (\vec{k}, s_z)$ and occupation numbers $n_i \in \{0, 1\}$ (fermions) or $\{0, 1, 2, \dots\}$ (bosons). In equilibrium

$$\langle n_i \rangle = \begin{cases} \frac{1}{e^{\beta(\varepsilon_i - \mu)} + 1} = f_{\text{FD}}(\varepsilon_i), & \text{fermions,} \\ \frac{1}{e^{\beta(\varepsilon_i - \mu)} - 1} = f_{\text{BE}}(\varepsilon_i), & \text{bosons.} \end{cases}$$

In the thermodynamic limit $L \rightarrow \infty$ the $\vec{k}$ become quasi-continuous, and

$$\langle N \rangle = \sum_i \langle n_i \rangle = \sum_{s_z} \sum_{\vec{k}} f_{\text{FD}}(\varepsilon(\vec{k})) = \int d1 f_{\text{FD}}(1), \quad \int di := V \sum_{s_z} \int \frac{d^3 k}{(2\pi)^3},$$

with $i = |\vec{k}, s_z\rangle$.

Perturbation theory. Time-independent perturbation theory (H_1 changes ε_n and $|\psi_n\rangle$) is not useful here. Time-dependent perturbation theory gives transitions; in the quasi-continuous limit the transition rate is given by Fermi's golden rule,

$$\bar{W}_{i \rightarrow f} = \frac{2\pi}{\hbar} |\langle f | H_1 | i \rangle|^2 \delta(\varepsilon_f - \varepsilon_i).$$

For a two-body interaction $H_1 = \sum_{ij} \phi(\vec{r}_i - \vec{r}_j)$, the transitions are collisions $\vec{k}_a, \vec{k}_b \rightarrow \vec{k}'_a, \vec{k}'_b$ between (symmetrized or antisymmetrized) states $|i\rangle = |\vec{k}_1, \dots, \vec{k}_a, \dots, \vec{k}_b, \dots\rangle$ and $|f\rangle = |\vec{k}_1, \dots, \vec{k}'_a, \dots, \vec{k}'_b, \dots\rangle$.

Object of study. The idea is to describe $\langle n_i \rangle$ for many i by a *distribution function* $f(i) = \langle n_i \rangle_{\text{CG}}$ (coarse-grained: averaged and smoothed; hypothesis: a smooth function). The average number of events $1, 2 \rightarrow 1', 2'$ per unit time, assuming no correlations, would be $\bar{W}(1, 2; 1', 2') f(1) f(2)$. For fermions this does not respect the Pauli exclusion principle; one needs the probability that the final states are empty:

$$\frac{\#\text{events}}{\Delta t} = \bar{W}(1, 2; 1', 2') f(1) f(2) [1 - f(1')][1 - f(2')].$$

For bosons, $a_i |\dots, n_i, \dots\rangle = \sqrt{n_i} |\dots, n_i - 1, \dots\rangle$ and $a_i^\dagger |\dots, n_i, \dots\rangle = \sqrt{n_i + 1} |\dots, n_i + 1, \dots\rangle$: creating a boson is more likely if there are already bosons. With $H_1 = \sum_{ijkl} a_i^\dagger a_j^\dagger a_k a_l V_{ijkl}$ and $\bar{W}(i \rightarrow f) \propto |\langle f | H_1 | i \rangle|^2$ (for $\vec{k}_1 \neq \vec{k}_2 \neq \vec{k}_3 \neq \vec{k}_4$, $|\langle f | V | i \rangle|^2 \propto (1 \pm n_{k_1})(1 \pm n_{k_2})n_{k_3}n_{k_4}$), the correct calculation gives

$$\frac{\#\text{events}}{\Delta t} = \bar{W}(1, 2; 1', 2') f(1) f(2) [1 + f(1')][1 + f(2')].$$

In summary, $df(1)/dt$ has a loss term (collisions $1, 2 \rightarrow 1', 2'$) and a gain term (collisions $1', 2' \rightarrow 1, 2$), plus $\mathcal{O}(\text{something} \cdot H_0)$, which vanishes since the states $|\vec{k}s\rangle$ are eigenstates of H_0 :

$$\begin{aligned} \frac{df(1)}{dt} = \int d2 d1' d2' & \left[\bar{W}(1', 2'; 1, 2) f(1') f(2') (1 \pm f(1)) (1 \pm f(2)) \right. \\ & \left. - \bar{W}(1, 2; 1', 2') f(1) f(2) (1 \pm f(1')) (1 \pm f(2')) \right]. \end{aligned}$$

If H_1 is Hermitian and time independent, $W_{i \rightarrow f} \propto |\langle f | H_1 | i \rangle|^2$ gives $W_{i \rightarrow f} = W_{f \rightarrow i}$ (microscopic reversibility). For identical particles, $\bar{W}(1, 2; 1', 2') = \bar{W}(1', 2'; 1, 2) = \bar{W}(2, 1; 1', 2') = \bar{W}(2', 1'; 1, 2)$, and we obtain the **quantum Boltzmann equation** (Uehling–Uhlenbeck–Nordheim equation)

$$\begin{aligned} \frac{df(1)}{dt} = \int d2 d1' d2' & \bar{W}(1, 2; 1', 2') \left\{ f(1') f(2') [1 \pm f(1)] [1 \pm f(2)] \right. \\ & \left. - f(1) f(2) [1 \pm f(1')] [1 \pm f(2')] \right\}, \end{aligned}$$

with the upper sign for bosons and the lower sign for fermions.

5.8 H-theorem and equilibrium

Define the function (not the Hamiltonian)

$$H(t) := \int d1 \{ f(1, t) \ln f(1, t) \mp (1 \pm f(1, t)) \ln (1 \pm f(1, t)) \}.$$

Then

$$\frac{dH}{dt} = \int d1 \frac{df(1)}{dt} \{ \ln f(1) + 1 - [\ln(1 \pm f(1)) + 1] \} = \int d1 \frac{df(1)}{dt} \{ \ln f(1) - \ln(1 \pm f(1)) \},$$

with df/dt given by the UUN equation, so $dH/dt = \int d1 d2 d1' d2' \bar{W}(1, 2; 1', 2')(\dots)$. Using the symmetry $1 \leftrightarrow 2$, and the exchange of primed and unprimed variables (which changes the sign), there are four equivalent expressions, with $\ln \frac{f(1)}{1 \pm f(1)}$, $\ln \frac{f(2)}{1 \pm f(2)}$, $-\ln \frac{f(1')}{1 \pm f(1')}$ and $-\ln \frac{f(2')}{1 \pm f(2')}$. Averaging them,

$$\begin{aligned} \frac{dH}{dt} = & -\frac{1}{4} \int d1 d2 d1' d2' \bar{W}(1, 2; 1', 2') (1 \pm f(1))(1 \pm f(2))(1 \pm f(1'))(1 \pm f(2')) \\ & \times (x - y)(\ln x - \ln y), \quad x = \frac{f(1')}{1 \pm f(1')} \frac{f(2')}{1 \pm f(2')}, \quad y = \frac{f(1)}{1 \pm f(1)} \frac{f(2)}{1 \pm f(2)}. \end{aligned}$$

For all $x, y > 0$, $(x - y)(\ln x - \ln y) \geq 0$, with equality iff $x = y$. Therefore $dH/dt \leq 0$: the evolution is irreversible (**H-theorem**). It stops when $dH/dt = 0$, i.e. $x = y$ for all $1, 2, 1', 2'$ for which a collision is possible ($\bar{W} \neq 0$):

$$\ln \frac{f(1)}{1 \pm f(1)} + \ln \frac{f(2)}{1 \pm f(2)} = \ln \frac{f(1')}{1 \pm f(1')} + \ln \frac{f(2')}{1 \pm f(2')}.$$

So $\ln \frac{f}{1 \pm f}$ is a collisional invariant; since the conserved quantities are ε , $\vec{p}$ and m ,

$$\ln \frac{f(1)}{1 \pm f(1)} = \alpha m + \vec{\beta} \cdot \vec{p} + \gamma \varepsilon =: z \Rightarrow f(1) = e^z (1 \pm f(1)) \Rightarrow f(1) = \frac{1}{e^{-\alpha m - \vec{\beta} \cdot \vec{p} - \gamma \varepsilon} \mp 1},$$

the Bose–Einstein / Fermi–Dirac distribution, which depends only on H_0 (through the asymptotic energies of the collision).

6 Electronic transport

6.1 Electronic structure of solids

In a crystalline solid the ions form a lattice with spacing a (unit cell). The electrons are either core electrons (bound to the ions) or valence electrons (free). The Hamiltonian for the valence electrons is

$$H = \sum_i \frac{p_i^2}{2m} + \sum_i V(\vec{r}_i),$$

where $V(\vec{r})$, the interaction with the ions and the core electrons, is periodic with period a .

Theorem (Bloch). If V is periodic, the eigenfunctions have the form

$$\psi_{n,\vec{k}}(\vec{r}) = \frac{e^{i\vec{k} \cdot \vec{r}}}{\sqrt{V}} \varphi_{n,\vec{k}}(\vec{r}), \quad n \in \mathbb{N}, \quad \vec{k} \in \text{FBZ},$$

with $\varphi_{n,\vec{k}}$ periodic and FBZ the first Brillouin zone; the factor $1/\sqrt{V}$ makes the thermodynamic limit better defined, $\langle \psi_i | \psi_j \rangle = \delta_{ij}$.

- The eigenenergies $\varepsilon_{n,\vec{k}}$ are periodic in the FBZ.
- The velocity operator has $\langle \psi_{n\vec{k}} | \vec{v} | \psi_{n\vec{k}} \rangle = \frac{1}{\hbar} \frac{\partial \varepsilon_{n\vec{k}}}{\partial \vec{k}}$, the group velocity.
- $\vec{k}$ -space is periodic, so $\hbar \vec{k}$ is not momentum but *quasi-momentum*: it is conserved modulo the FBZ.

1D case. For a real lattice of spacing a the FBZ is $[-\pi/a, \pi/a)$. The bands $\varepsilon_n(k)$ are separated by gaps where there are no states. With periodic boundary conditions in a volume of length L , $\psi(x=L) = \psi(x=0)$ gives $e^{ikL} = 1$, $k = 2\pi m/L$, $m \in \mathbb{Z}$: quasi-continuous. Since k -space is periodic, the number of different k is

$$\#k = \frac{2\pi/a}{2\pi/L} = \frac{L}{a} = N,$$

the number of unit cells: *each band has N states*. If each cell contributes p valence electrons there are Np electrons in the solid. With two electrons per level (spin; statistical mechanics at $T=0$), if p is even, $p/2$ bands are filled (insulator: adding a little energy does nothing, since all the levels are full); if p is odd, the last band is half filled (conductor). Reality is more complex: sometimes bands overlap, and there is the Peierls instability (in 1D the unit cell doubles and contains more electrons).

More seriously, let $g(\varepsilon)$ be the density of states, $Vg(\varepsilon)d\varepsilon =$ number of states, with g intensive and $\int d\varepsilon g(\varepsilon) = 2N/V$ (spin). For fermions at temperature T the equilibrium distribution is $f_{\text{FD}}(\varepsilon)$, which drops from 1 to 0 around μ over a width $k_{\text{B}}T$. The chemical potential is fixed by

$$N_e = V \int_0^\infty d\varepsilon g(\varepsilon) f_{\text{FD}}(\varepsilon; \mu).$$

At $T=0$, $f_{\text{FD}}(\varepsilon) = \Theta(\mu - \varepsilon)$ and $\mu(T=0) =: \varepsilon_F$ is the **Fermi energy**:

$$\frac{N_e}{V} = \int_0^{\varepsilon_F} d\varepsilon g(\varepsilon);$$

ε_F depends only on N_e/V and on the band structure. Define the Fermi temperature $T_F := \varepsilon_F/k_{\text{B}}$ for conductors, and the gap Δ for insulators.

6.2 Semiclassical kinetic equation

Out of equilibrium, the distribution, which in equilibrium depends only on ε , becomes $f_n(\vec{k})$ (with n the band index, relevant for semiconductors). Bloch waves $\psi_{n\vec{k}} = e^{i\vec{k}\cdot\vec{r}} \varphi_{n\vec{k}}(\vec{r})$ can be combined into wave packets localized in both $\vec{k}$ and $\vec{r}$: one can give $\vec{r}$ and $\vec{k}$ simultaneously, though not with infinite precision (a formal theory was developed by Wigner). The object of study is $f_n(\vec{k}, \vec{r}, t)$, coarse-grained over the (quasi-continuous) bands and over mesoscopic cells Δr (larger than nm and smaller than mm).

What is $\partial f/\partial t$? The processes are:

1. $H_0 = \sum_i p_i^2/2m + \sum_i V_i(\vec{r}) = \sum_k \varepsilon_k n_k$ generates $\psi_{n\vec{k}}, \varepsilon_{n\vec{k}}$. In eigenstates the occupation numbers are constant, $[\hat{n}_k, \hat{H}_0] = 0$, so f would be constant. (The classical picture of electrons bouncing off the ions is false.)
2. Electron–electron interactions: (i) the hard case, a “strong” interaction (highly correlated systems, occurring in 2D and quantum phases), which changes the band structure and the excitations; (ii) the easy case, a weaker interaction treated with time-dependent perturbation theory: the levels do not change and there is scattering, described by the collision term above, which leads to $f \rightarrow f_{\text{FD}}$ (H-theorem).
3. Impurities or point defects: $V = V_0 + V_1$, with V_0 periodic and V_1 not, which scatters $\vec{k} \rightarrow \vec{k}'$.

Exercise. In the classical case, $\partial_t f(\vec{r}, \vec{v}_1, t) = \dots \int (f(\vec{v}_1') f(\vec{v}_2') - f(\vec{v}_1) f(\vec{v}_2))$. For the Maxwell–Boltzmann distribution $f_{\text{MB}}(\vec{v}) = e^{-m(\vec{v}-\vec{V})^2/2k_{\text{B}}T}$, by conservation of energy and momentum,

$$f(\vec{v}_1') f(\vec{v}_2') = e^{-m\beta[(\vec{v}_1'-\vec{V})^2 + (\vec{v}_2'-\vec{V})^2]/2} = e^{-m\beta[(\vec{v}_1-\vec{V})^2 + (\vec{v}_2-\vec{V})^2]/2} = f(\vec{v}_1) f(\vec{v}_2),$$

so the collision term vanishes; the same holds for the quantum case.

Scattering by impurities. Write $H = H_0 + V$ with $H_0 = \sum_i \frac{p_i^2}{2m} + \sum_i V_{\text{per}}(\vec{r}_i)$ (bands) and $V = \sum_i V(\vec{r}_i)$ a non-periodic perturbation (e.g. impurities). Perturbation theory gives $W_{i \rightarrow f} = \frac{2\pi}{\hbar} |\langle f|V|i \rangle|^2 \delta(\varepsilon_i - \varepsilon_f)$. Since V is a sum of one-particle operators,

$$\langle f|V|i \rangle = \left(\text{statistical factors } \sqrt{1 \pm n_i} \text{ or } \sqrt{n_i} \right) \cdot \langle \vec{k}_f | V | \vec{k}_i \rangle,$$

and with Bloch functions normalized as $e^{i\vec{k} \cdot \vec{r}} \varphi_{n\vec{k}} / \sqrt{V}$, $W_{i \rightarrow f} \propto 1/V$. Moreover, with N_{imp} impurities, $V_{\text{imp}}(\vec{r}_i) = \sum_j \phi(\vec{r}_i - \vec{r}_j)$, and $W_{i \rightarrow f} \propto N_{\text{imp}}/V$. In summary,

$$W_{\vec{k} \rightarrow \vec{k}'} = \frac{2\pi}{\hbar} n_{\text{imp}} \delta(\varepsilon(\vec{k}) - \varepsilon(\vec{k}')) \left| \hat{\phi}(\vec{k}, \vec{k}') \right|^2, \quad \hat{\phi}(\vec{k}, \vec{k}') = \int d^3r e^{i(\vec{k} - \vec{k}') \cdot \vec{r}} \varphi_{\vec{k}}^*(\vec{r}) \phi(\vec{r}) \varphi_{\vec{k}'}(\vec{r}),$$

with n_{imp} the impurity density and $\hat{\phi}$ a “Fourier transform of the potential” between Bloch functions.

Electromagnetic field. Classically, the Lorentz force $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$. Quantum mechanically, in the weak-field approximation,

$$\hbar \dot{\vec{k}} = q \left(\vec{E} + \vec{v}_{\text{Bloch}}(\vec{k}) \times \vec{B} \right), \quad \vec{v}_{\text{Bloch}} = \frac{1}{\hbar} \frac{\partial \varepsilon}{\partial \vec{k}},$$

with $\hbar \vec{k}$ the quasi-momentum.

Kinetic equation for f .

$$\underbrace{\frac{\partial f}{\partial t} + \vec{v}_{\text{Bloch}}(\vec{k}) \cdot \nabla f + \frac{q}{\hbar} \left(\vec{E} + \vec{v}_{\text{Bloch}} \times \vec{B} \right) \cdot \frac{\partial f}{\partial \vec{k}}}_{\text{conservative}} = \int \frac{d^3k'}{(2\pi)^3} \left[W_{\vec{k}' \rightarrow \vec{k}} f_{\vec{k}'} (1 - f_{\vec{k}}) - W_{\vec{k} \rightarrow \vec{k}'} f_{\vec{k}} (1 - f_{\vec{k}'}) \right],$$

where $(1 - f)$ is the probability that the final state is empty. By microscopic reversibility, $W_{\vec{k} \rightarrow \vec{k}'} = W_{\vec{k}' \rightarrow \vec{k}}$, so

$$\frac{\partial f}{\partial t} + \vec{v} \cdot \nabla f + \frac{q}{\hbar} \left(\vec{E} + \vec{v} \times \vec{B} \right) \cdot \frac{\partial f}{\partial \vec{k}} = \int \frac{d^3k'}{(2\pi)^3} W_{\vec{k} \rightarrow \vec{k}'} (f_{\vec{k}'} - f_{\vec{k}}) =: -\mathcal{W}[f],$$

with

$$\mathcal{W}[f](\vec{k}) = \frac{2\pi}{\hbar} n_{\text{imp}} \int \frac{d^3k'}{(2\pi)^3} \delta(\varepsilon_{\vec{k}'} - \varepsilon_{\vec{k}}) \left| \hat{\phi}_{\vec{k}\vec{k}'} \right|^2 (f_{\vec{k}'} - f_{\vec{k}}),$$

a linear, positive semi-definite operator: electrons disappear from $\vec{k}$ and appear elsewhere. Approximations:

1. $\mathcal{W}[f] = f/\tau$: the **relaxation-time approximation**;
2. $\mathcal{W}[f] = f/\tau(\vec{k})$: a little better;
3. isotropic models (e.g. Na, Cs): $\varepsilon(\vec{k}) = \varepsilon(|\vec{k}|)$, so $\delta(\varepsilon_{\vec{k}} - \varepsilon_{\vec{k}'}) \rightarrow \delta(|\vec{k}| - |\vec{k}'|)$, and $\hat{\phi}(\vec{k}, \vec{k}') = \hat{\phi}(|\vec{k}|, \hat{k} \cdot \hat{k}')$ depends only on the scattering angle. Using $\delta(g(x)) = \delta(x - x_0)/|g'(x_0)|$ and $v_{\text{Bloch}} = \frac{1}{\hbar} \partial \varepsilon / \partial k$,

$$\int d^3k' \delta(\varepsilon_{\vec{k}} - \varepsilon_{\vec{k}'}) \hat{\phi}(\vec{k}, \vec{k}') = \frac{k^2}{\hbar |v_{\text{Bloch}}|} \int d^2\hat{k}' \hat{\phi}(|\vec{k}|, \hat{k} \cdot \hat{k}'),$$

so

$$\mathcal{W}[f](\vec{k}) = \frac{n_{\text{imp}} k^2}{v(\vec{k}) \hbar^2} \int \frac{d^2\hat{k}'}{(2\pi)^2} \left| \hat{\phi}(k, \hat{k} \cdot \hat{k}') \right|^2 \left[f(k\hat{k}) - f(k\hat{k}') \right].$$

6.3 Electrical conduction

Assume (1) a stationary case, (2) $\vec{B} = 0$ and $\vec{E} = \epsilon \vec{E}_0$ with $\epsilon \ll 1$, (3) a homogeneous response. Then $f(\vec{k}) = f_{\text{FD}}(k) + \epsilon f_1(\vec{k})$, and substituting into the kinetic equation (with $\mathcal{W}[f_{\text{FD}}] = 0$, since f_{FD} is isotropic),

$$\frac{q}{\hbar} \vec{E}_0 \cdot \frac{\partial f_{\text{FD}}}{\partial \vec{k}} = -\mathcal{W}[f_1].$$

In the isotropic model $f_{\text{FD}} = f_{\text{FD}}(\epsilon(k))$ and $\frac{\partial f_{\text{FD}}}{\partial \vec{k}} = \frac{\partial f_{\text{FD}}}{\partial \epsilon} \hbar \vec{v}_{\text{Bloch}}$. Define $h_{\text{FD}}(\epsilon) := -\partial f_{\text{FD}} / \partial \epsilon$, with $\lim_{T \rightarrow 0} h_{\text{FD}} = \delta(\epsilon - \epsilon_F)$. Then

$$\mathcal{W}[f_1] = q h_{\text{FD}}(\epsilon(\vec{k})) \vec{v}_{\text{Bloch}} \cdot \vec{E}_0,$$

which is non-zero only near the Fermi energy. Even without a model, $\mathcal{W}$ contains $\delta(\epsilon - \epsilon')$: it acts “on shell”, and so does $\mathcal{W}^{-1}$ (on the space where it exists; $\mathcal{W}[f_{\text{FD}}] = 0$). Hence $f_1 \propto \mathcal{W}^{-1}[\dots, h_{\text{FD}}(\epsilon(\vec{k})), \dots]$ is non-zero only near μ . For an insulator there is no $\vec{k}$ with $\epsilon(\vec{k}) = \mu$, so $f_1 = 0$ for all $\vec{k}$: there is no response.

Quantitatively, in the approximation $\mathcal{W}[f_1] = f_1 / \tau(\vec{k})$,

$$f_1(\vec{k}) = q h_{\text{FD}} \tau \vec{v} \cdot \vec{E}_0.$$

The current density is $\vec{J} = \frac{1}{V} \sum_{\vec{k}} q \vec{v}(\vec{k}) f(\vec{k})$, with $\sum_{\vec{k}} = V \int \frac{d^3 k}{(2\pi)^3}$ or $V \int d\epsilon g(\epsilon)$:

$$\vec{J} = \frac{q^2}{V} \sum_{\vec{k}} h_{\text{FD}}(\epsilon(\vec{k})) \tau(\vec{k}) \vec{v}(\vec{k}) \vec{v}(\vec{k}) \cdot \vec{E}_0 = \boldsymbol{\sigma} \cdot \vec{E}_0,$$

Ohm’s law with the conductivity tensor $\boldsymbol{\sigma}$. In an isotropic model $\boldsymbol{\sigma} = \sigma \mathbb{K}$, $\sigma_{nn} = \sigma$, so

$$\sigma = \frac{1}{3} \sum_n \sigma_{nn} = \frac{q^2}{3V} \sum_{\vec{k}} h_{\text{FD}} \tau v^2, \quad \sigma = \frac{q^2}{3} \int d\epsilon g(\epsilon) h_{\text{FD}}(\epsilon) \tau(\epsilon) v^2(\epsilon).$$

- *Insulator*: g and h_{FD} do not overlap.
- *Conductor*: g and h_{FD} overlap. A good conductor has large g (many available states), large τ or large v ; a bad conductor fails in one of them.
- *Normal conductor* ($T_F \gg T_{\text{room}}$): $f_{\text{FD}} \simeq \Theta(\epsilon_F - \epsilon)$, $h_{\text{FD}} \simeq \delta(\epsilon - \epsilon_F)$, and

$$\sigma = \frac{q^2}{3} g(\epsilon_F) \tau(\epsilon_F) v^2(\epsilon_F),$$

with $g(\epsilon_F)$ the density of states at the Fermi surface. This is similar to Drude, but Drude says it depends on the carrier density.

- *Insulator at finite T* : h_{FD} has exponential tails reaching the bands across the gap Δ , and $\sigma_{\text{ins}} \sim e^{-\Delta/2k_B T}$. In a semiconductor the tail reaches the band. In industry, more electrons are added without changing the band structure by replacing atoms (e.g. Si by P).

6.4 Coherent transport: the Landauer formalism

We now consider the passage of electrons through a “circuit”. Typically, Ohm’s law reads $V = IR$, or $I = GV$ with G the conductance. Locally $\vec{J} = \sigma \vec{E}$, with $\vec{J} = e \eta \vec{v}$ (η the electron density) and typically $\boldsymbol{\sigma} = \sigma \mathbb{K}$. For a wire of length L and cross section S along $\hat{x}$, multiplying by S :

$$I = JS = \frac{\sigma S}{L} LE = GV, \quad G := \frac{\sigma S}{L}.$$

Drude model. After each collision the electron loses the information about its previous momentum, $\vec{v}(t) = \vec{v}(0) + \frac{e\vec{E}}{m_e}t$ with $\langle \vec{v}(0) \rangle = 0$ after a collision. Then $\langle \vec{v} \rangle = e\vec{E}\tau/m_e$, where τ is not exactly the time between collisions but the time in which momentum correlation is lost, τ_m . In the stationary state, multiplying by $e\eta$,

$$\vec{J} = \left(\frac{e^2\eta\tau_m}{m_e} \right) \vec{E}, \quad \sigma = \frac{e^2\eta\tau_m}{m_e}.$$

Chemical potential gradients. In statistical physics, a gradient of chemical potential generates a current. Two isolated subsystems at the same T with μ_1, μ_2 exchange N particles: $\Delta E_1 = T\Delta S_1 + \mu_1(-N)$, $\Delta E_2 = T\Delta S_2 + \mu_2 N$, $\Delta E_1 + \Delta E_2 = 0$, so $T\Delta S = (\mu_1 - \mu_2)N \geq 0$: if $N > 0$ then $\mu_1 \geq \mu_2$. There is a flow of electrons if the chemical potentials differ. With an electrostatic potential ($\nabla\phi = -\vec{E}$), the energy of a cell with N_l electrons is $\varepsilon_l + e\phi N_l$, and in the grand-canonical distribution

$$p_l = \frac{e^{-\beta(\varepsilon_l - \tilde{\mu}N_l)}}{\Theta}, \quad \tilde{\mu} := \mu - e\phi,$$

the **electrochemical potential**. Locally $\mu(\vec{r}) = \mu(\eta)$, and $\vec{J} \sim \nabla\tilde{\mu}$, with $\nabla\mu = \frac{\partial\mu}{\partial\eta}\nabla\eta$.

A typical experimental situation: 2D electron gas. At the junction between n-doped AlGaAs and p-doped GaAs, when the crystals come into contact the bands bend and a thin layer of “free” electrons forms at the interface, below the chemical potential: a 2D electron gas. There is no scattering at the boundary; the impurities generate “obstacles” for the electrons on the surface. The relevant time scales are the collision time τ_c , the momentum relaxation time τ_m and the energy relaxation time τ_E . For elastic collisions $\tau_c \lesssim \tau_m < \tau_E$; for inelastic ones $\tau_c \sim \tau_m \sim \tau_E$.

Interference. On the 2D surface an electron can take several paths around a defect. The propagator is a sum over paths,

$$K(x, x'; t, t') = \langle x' | U(t', t) | x \rangle = \sum_{\text{paths } x \rightarrow x'} e^{\frac{i}{\hbar} \int_t^{t'} d\tau L(\tau)},$$

and if E is constant, $\int L d\tau = \int p dq - Et$, so $K = e^{-iEt/\hbar} \sum_{\text{paths}} e^{\frac{i}{\hbar} \int p dq}$. Short paths are less likely to suffer an inelastic collision than long ones. With a magnetic flux through a ring, the two arms c_1, c_2 acquire phases $\int_{c_i} \vec{A} \cdot d\vec{l}$:

$$\langle x' | U | x \rangle \simeq e^{-iEt/\hbar} \left[e^{\frac{i}{\hbar} \int_{c_1} (p dq + \vec{A} \cdot d\vec{l})} + e^{\frac{i}{\hbar} \int_{c_2} (p dq + \vec{A} \cdot d\vec{l})} \right],$$

which can be controlled. Why $\vec{A}$ and not $\vec{B}$? The expression can be rearranged to leave $\oint \vec{A} \cdot d\vec{l}$ (the enclosed flux) plus an irrelevant global phase.

Landauer setting. A 2D sample between two electron reservoirs at μ_L and μ_R (same T), with “gates” defining forbidden regions ($\psi = 0$) and defects inside. The current I is driven by $\Delta\mu$, and the transport is fully quantum: only elastic collisions (described by $\hat{H}$) occur in the sample, while inelastic ones (coupling $\hat{V}$ to another system) occur in the reservoirs. Idealize as: reservoir L, a lead (channel) of width W_L , a scattering region, a lead of width W_R , reservoir R.

In a lead, the Schrödinger equation with $\psi(x, y) = \phi(y)e^{ikx}$ gives

$$-\frac{\hbar^2}{2m}[-k^2\phi + \phi''] = E\phi \Rightarrow \left[\frac{2mE}{\hbar^2} - k^2\right]\phi + \phi'' = 0,$$

plus boundary conditions $\phi(0) = \phi(W) = 0$, so $\sqrt{2mE/\hbar^2 - k^2}W = n\pi$ and

$$k_n = \sqrt{\frac{2mE}{\hbar^2} - \left(\frac{n\pi}{W}\right)^2}, \quad \psi_n(x, y) = A \sin\left(\frac{n\pi}{W}y\right)e^{ik_nx},$$

with $W = W_L$ or W_R . A mode n incident from the left, $\phi_n^{(L)}e^{ik_n^{(L)}x}$, is reflected into mode m with amplitude $(r_{LL})_{mn}$ and transmitted into mode m' on the right with amplitude $(t_{RL})_{m'n}$. At fixed energy, k_n is real if $2mE/\hbar^2 > (n\pi/L_y^{(L)})^2$, i.e. for

$$n < \frac{L_y^{(L)}}{\pi} \sqrt{\frac{2mE}{\hbar^2}} =: M_L(E)$$

open channels on the left (analogously on the right). The scattering problem is solved by the scattering matrix

$$S(E) = \begin{pmatrix} r_{LL} & t_{LR} \\ t_{RL} & r_{RR} \end{pmatrix},$$

where $(r_{LL})_{mn}$ ($M_L \times M_L$) is the probability amplitude of being reflected into mode m for an electron incident in mode n . (References: Y. Nazarov and Y. M. Blanter, *Quantum Transport*; S. Datta, *Electronic Transport in Mesoscopic Systems*.)

Example (Adiabatic waveguide). Let the width $L_y = L_y(x)$ vary slowly, $\partial L_y/\partial x \ll 1$ (more precisely, $\frac{\partial L_y}{\partial x} \lambda \ll \lambda$), with $L_y \rightarrow L_y^{(L)}$ as $x \rightarrow -\infty$ and $L_y \rightarrow L_y^{(R)}$ as $x \rightarrow +\infty$. In the 2D Schrödinger equation $-\frac{\hbar^2}{2m}(\partial_x^2 + \partial_y^2)\phi = E\phi$ try $\phi = \varphi(x)\Psi(y, L_y(x))$ with $\Psi = A \sin(\frac{n\pi}{L_y(x)}y)$. The terms involving $\partial L_y/\partial x$ are negligible, and dividing by Ψ ,

$$\left[\frac{2mE}{\hbar^2} - \left(\frac{n\pi}{L_y(x)}\right)^2\right]\varphi + \varphi'' = 0, \quad \text{i.e.} \quad -\frac{\hbar^2}{2m}\varphi'' + \mathcal{V}\varphi = E\varphi, \quad \mathcal{V}(x) = \frac{\hbar^2}{2m}\left(\frac{n\pi}{L_y(x)}\right)^2.$$

Each mode sees an effective potential barrier, highest at the narrowest point $L_{\min}$. A channel is open (propagating) if

$$\frac{2mE}{\hbar^2} > \left(\frac{n\pi}{L_{\min}}\right)^2.$$

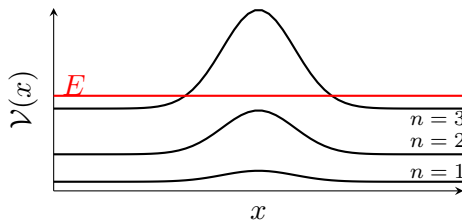


Figure 6.1: Effective potentials of the transverse modes in an adiabatic constriction; here only $n = 1$ propagates (open channel).

Current through an adiabatic guide. The current carried by $\vec{k}$ is $e\vec{v}(\vec{k})$, and each mode is occupied with probability $f(\vec{k})$, so $\vec{J} = \int \frac{d^2k}{(2\pi)^2} e\vec{v}(\vec{k}) f(\vec{k})$, with periodic boundary conditions in the quantization volume. The same current flows through every cross section of the guide. In the guide, with $k_y = n\pi/L_y$ and $dk_y/\pi = 1/L_y$,

$$J_x = \sum_n \int \frac{dk_x}{2\pi} \frac{1}{L_y} e v_x(k_x) f_n(k_x), \quad I = J L_y = 2e \sum_n \int \frac{dk_x}{2\pi} v_x(k_x) f_n(k_x)$$

(the factor 2 from spin). With $v_x = \frac{1}{\hbar} \partial E / \partial k_x$ and $E = \frac{\hbar^2}{2m} (k_x^2 + (n\pi/L_y)^2)$, $\int \frac{dk_x}{2\pi} v_x \rightarrow \int \frac{dE}{2\pi\hbar}$. Right-movers come from L with f_L and left-movers from R with f_R :

$$I = \frac{2e}{2\pi\hbar} \sum_{n \text{ open}} \int dE [f_L(E) - f_R(E)].$$

At the same T with $\mu_L \neq \mu_R$, only the energy window between μ_R and μ_L contributes net current, and

$$I = \frac{2e}{2\pi\hbar} N_{\text{open}} (\mu_L - \mu_R), \quad N_{\text{open}} \simeq \sqrt{\frac{2mE}{\hbar^2}} \frac{L_{\text{min}}}{\pi}.$$

With $I = GV$ and $V = (\mu_L - \mu_R)/e$,

$$G = \frac{2e^2}{h} N_{\text{open}} = G_0 N_{\text{open}}, \quad G_0 := \frac{2e^2}{h},$$

the **conductance quantum**. Experimentally, in a quantum point contact (QPC) controlled by a gate voltage V_g , G/G_0 shows steps.

General case. With the reflection probabilities $R_n^{(L)}(E) = \sum_m |(r_{LL}(E))_{mn}|^2$ and $R_n^{(R)}(E) = \sum_m |(r_{RR}(E))_{mn}|^2$,

$$I = 2e \sum_n \left\{ \int_0^\infty \frac{dk_x}{2\pi} v_x(k_x) f_L + \int_{-\infty}^0 \frac{dk_x}{2\pi} v_x(k_x) [R_n^{(L)}(E) f_L + (1 - R_n^{(R)}(E)) f_R] \right\},$$

and with $k_x \rightarrow -k_x$ in the second integral,

$$I = 2e \sum_n \int_0^\infty \frac{dk_x}{2\pi} v_x(k_x) (1 - R_n(E)) (f_L - f_R).$$

By unitarity of S ($S^\dagger S = S S^\dagger = \mathbb{K}$), and with time-reversal symmetry ($S = S^\top$), $R^{(L)} = R^{(R)}$, and $\sum_n (1 - R_n) = \text{Tr}\{t^\dagger t\}$. This gives the **Landauer formula**

$$I = \frac{2e}{2\pi\hbar} \int_0^\infty dE \text{Tr}\{t^\dagger t\} (f_L(E) - f_R(E)).$$

Heat conduction. Similarly, for a bath, $\Delta S_B = -Q/T_B$, $\Delta E_B = T \Delta S_B + \mu_B \Delta N_B$ and $Q = \Delta E_B - \mu_B \Delta N_B$: in the Landauer formula one replaces the charge e carried by each electron by the energy $E - \mu$ it carries.

Remark. In chaotic scatterers (e.g. a channel with obstacles where trajectories can go back), Landauer's approach makes it possible to measure fractal invariant probability measures.

7 Phase transitions far from equilibrium: instabilities

7.1 From equilibrium to non-equilibrium steady states

In equilibrium we have control parameters X ($P, T, B, \dots$) and order parameters or observables σ ($\rho, M, \dots$). As X varies, σ can develop singularities: these are phase transitions. In equilibrium $\sigma = \sigma[f]$, so the free energy f is singular. For example, $M(T)$ at $B = 0$ vanishes continuously at T_c : in the control-parameter space (T, B) the transition line $B = 0, T < T_c$, ends at the critical point. Near (or at) equilibrium the tool is Landau theory, which replaces a singular function by an analytic one with an additional parameter, $F(T, B) \rightarrow F(T, B, M)$.

What happens far from equilibrium? Example: a fluid between two plates at temperatures T_+ (bottom) $> T_-$ (top), under gravity g ; we measure the heat flux Q between the plates as a function of $\Delta T = T_+ - T_-$. Experimentally, Q grows linearly (conduction) up to ΔT^* , then faster (convection), and changes again when T_+ reaches the boiling point. ΔT^* marks a phase transition between two NESS (non-equilibrium steady states). Another order parameter is the circulation $C := \oint d\vec{l} \cdot \vec{v}$ of the velocity field around a convection roll: $C = 0$ for $\Delta T < \Delta T^*$ and $C = \pm C_0$ beyond, which looks like the equilibrium pitchfork.

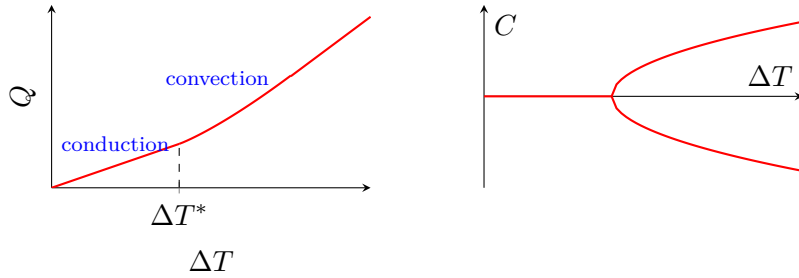


Figure 7.1: Rayleigh–Bénard convection: heat flux and circulation as functions of ΔT .

Goal: find a language and tools. How do we describe convection? There is no free energy F , but we have equations of motion (the Navier–Stokes equations). Starting from $\Delta T < \Delta T^*$, the equations of motion are $\partial_t \psi = \mathcal{N}(\psi)$, with ψ the vector of fields $(\rho, \vec{v}, T)$ and $\mathcal{N}$ a non-linear operator. A NESS ψ_0 satisfies $\partial_t \psi_0 = 0$, i.e. $\mathcal{N}(\psi_0) = 0$. (Reference: the soft-matter book in the bibliography.)

7.2 Linear stability analysis

Let $\psi = \psi_0 + \epsilon \psi_1(t)$ with $\epsilon \ll 1$ (in general $\psi(\vec{r}, t) = \psi_0(\vec{r}) + \psi_1(\vec{r}, t)$). Then

$$\epsilon \frac{\partial \psi_1}{\partial t} = \mathcal{N}(\psi_0 + \epsilon \psi_1) \simeq \mathcal{N}(\psi_0) + \epsilon \left. \frac{\partial \mathcal{N}}{\partial \psi} \right|_{\psi_0} \psi_1 \Rightarrow \frac{\partial \psi_1}{\partial t} = \mathcal{L}_{\psi_0}[\psi_1], \quad \mathcal{L}_{\psi_0} := \left. \frac{\partial \mathcal{N}}{\partial \psi} \right|_{\psi_0},$$

a linear operator. Example: the G–L equation $\partial_t \psi = \psi - \psi^3 + \nabla^2 \psi$, with ψ_0 satisfying $\psi_0 - \psi_0^3 + \nabla^2 \psi_0 = 0$; with $\psi = \psi_0 + \psi_1$,

$$\frac{\partial \psi_1}{\partial t} = \psi_1 - 3\psi_0^2(\vec{r})\psi_1 + \nabla^2 \psi_1 = \mathcal{L}(\vec{r})[\psi_1].$$

$\mathcal{L}_{\psi_0}$ is typically non-Hermitian (its eigenvalues come in complex-conjugate pairs).

Case 1: few degrees of freedom. In a small box, ψ_1 is expanded in Fourier modes or spherical harmonics and truncated. One computes the eigenvalues of $\mathcal{L}_{\psi_0}$, $\mathcal{L}_{\psi_0}\varphi_m = \lambda_m\varphi_m$; with $\psi_1(t) = \sum_n a_n(t)\varphi_n$, $\dot{a}_n = \lambda_n a_n$ and $a_n(t) = a_n(0)e^{\lambda_n t}$. ψ_0 is stable if $\text{Re } \lambda_n < 0$ for all n , and unstable if some $\text{Re } \lambda_n > 0$. The λ_n depend on ψ_0 and therefore on the control parameter X . Varying X , there are two generic scenarios (all others are topological accidents):

1. a single real eigenvalue crosses the imaginary axis, while all others stay at $\text{Re } \lambda < c < 0$ (a gap);
2. a complex-conjugate pair crosses the imaginary axis.

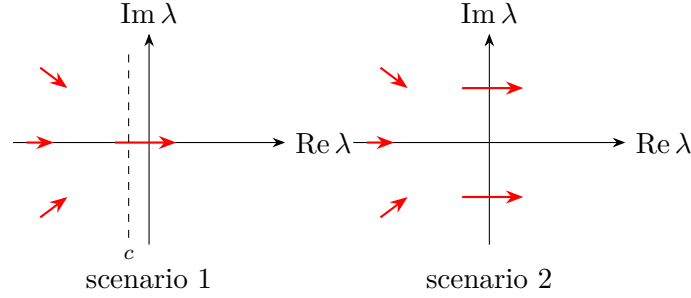


Figure 7.2: Generic ways in which eigenvalues of $\mathcal{L}_{\psi_0}$ cross into the unstable half plane as X is varied.

Case 1, scenario 1: stationary instability ($\lambda^{\text{inst}} \in \mathbb{R}$). Near the instability, $X \simeq X^*$, $\lambda_{n^*} \simeq 0$ while $\text{Re } \lambda_m < c < 0$ for the rest: all other a_m decay on a time $\sim 1/c$, and in comparison a_{n^*} decays or grows slowly. Only one variable matters, $a_{n^*} \rightarrow a$, with $\dot{a} = \lambda a$, λ small and changing sign: $\lambda = \hat{\lambda}(X - X^*)$, so

$$\dot{a} = \hat{\lambda}(X - X^*)a.$$

If $X > X^*$, a grows without bound. To avoid this we expand $\mathcal{N}$ to higher orders in a ; by the symmetry $a \rightarrow -a$,

$$\dot{a} = \hat{\lambda}(X - X^*)a - \mu a^3.$$

The stationary solutions (new NESS) are $a = 0$ and $a = \pm \sqrt{\hat{\lambda}(X - X^*)/\mu}$: a **supercritical bifurcation** (the analogue of a second-order transition). Sometimes $\mu < 0$, and one has to go to the next order,

$$\dot{a} = \hat{\lambda}(X - X^*)a + |\mu|a^3 - \nu a^5,$$

which gives a **subcritical bifurcation** (the analogue of a first-order transition), with unstable branches and hysteresis.

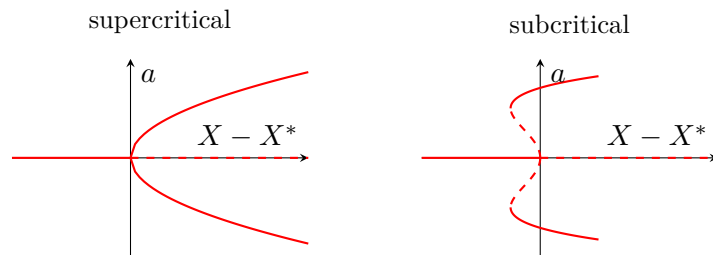


Figure 7.3: Supercritical and subcritical bifurcations (dashed: unstable branches).

Case 1, scenario 2. This has no thermodynamic analogue (examples: a flag fluttering, the wheel of a supermarket trolley, a garden hose). With two complex-conjugate eigenvalues there are two unstable modes, with small $\text{Re } \lambda$ and finite $\text{Im } \lambda$: a **Hopf instability**, in which the system reaches a limit cycle.

Case 2: extended and homogeneous systems. For example, a fluid layer between two plates (at T_+ and T_-), infinite in x and y . Then $\mathcal{L}_{\psi_0} = \mathcal{L}_{\psi_0}(z, \partial_z, \partial_z^2, \dots; \partial_x, \partial_x^2, \dots; \partial_y, \partial_y^2, \dots)$, with an explicit dependence on z and homogeneous in x, y . It commutes with the translation operators, $[\mathcal{L}_{\psi_0}, T_x] = [\mathcal{L}_{\psi_0}, T_y] = 0$ and $[T_x, T_y] = 0$, so (as in quantum mechanics) there is a common basis; the basis of bounded functions for T_x, T_y is the Fourier basis $e^{i\vec{k}_{\parallel} \cdot \vec{r}_{\parallel}}$, $\parallel = x, y$. If the system is isotropic it suffices to take $\vec{k}_{\parallel} = k\hat{x}$, and the basis functions are $\psi_1(\vec{r}, t) = f(z, t)e^{ikx}$. Then

$$\partial_t f(z, t) = \hat{\mathcal{L}}_{\psi_0}(z, \partial_z, \partial_z^2, \dots; k) f(z, t) :$$

each Fourier mode evolves differently, since $\hat{\mathcal{L}}$ depends on k . With eigenfunctions $\hat{\mathcal{L}}_{\psi_0} \varphi_{n,k}(z) = \lambda_{n,k} \varphi_{n,k}(z)$, the solution is $\psi_1(\vec{r}, t) = \sum_n a_n(t) \varphi_{n,k}(z) e^{i\vec{k}_{\parallel} \cdot \vec{r}_{\parallel}}$ with $\dot{a}_{nk} = \lambda_{nk} a_{nk}$: unstable if $\text{Re } \lambda > 0$. The eigenvalues depend on k , on n and on ψ_0 (and hence on X): the spectrum consists of bands $\text{Re } \lambda_n(k)$, and the unstable modes are those with $\text{Re } \lambda > 0$.

Instabilities: three scenarios (as X is varied):

1. The maximum of $\text{Re } \lambda(k)$ is at a finite k^* ; at $X = X^*$ only one wavelength $l = 2\pi/k^*$ becomes unstable: *patterns*.
2. A symmetry imposes $\lambda(k=0) = 0$; the instability happens when the sign of $\partial^2 \lambda / \partial k^2|_{k=0}$ changes (chaotic phenomena).
3. The small- k modes become unstable.

Each comes in two versions depending on whether the imaginary part vanishes or not: 3×2 scenarios.

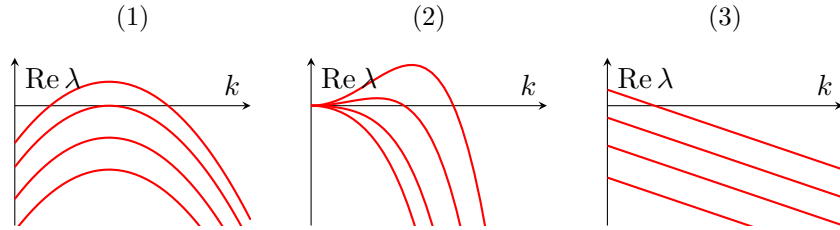


Figure 7.4: Three scenarios for the instability of an extended system, as the control parameter X increases.

8 Critical phenomena and the renormalization group

8.1 Critical exponents and scaling

In magnetism the critical point is at the end of the transition line $B = 0$, $T < T_c$; for the liquid–gas transition, at the end of the coexistence line in the P – T plane. Near it we use the scaled variables t (along the line) and h (perpendicular to it). Experimentally (around 1950–70), near the critical point there are algebraic singularities. In magnetism,

$$c_V \sim |T - T_c|^{-\alpha}, \quad \chi \sim |T - T_c|^{-\gamma}, \quad M \sim \begin{cases} |T - T_c|^\beta, & T < T_c, \\ 0, & T > T_c, \end{cases}$$

with χ the magnetic susceptibility. Surprises:

1. $\alpha, \beta, \gamma, \dots$ are irrational numbers. Landau gives $\beta = \frac{1}{2}$, so Landau theory is wrong.
2. They are universal, independent of the microscopic details (of the ferromagnetic material).
3. They are the same for ferromagnets, the gas–liquid transition and the Ising model.

Scaling hypothesis (near the critical point).

1. Replace T, B by dimensionless variables that vanish at the critical point, $t = (T - T_c)/T_c$ and $h = B/B_{\text{scale}}$.
2. The free energy density is

$$f(t, h) = f_{\text{regular}}(t, h) + \underbrace{|t|^{2-\alpha} g_f\left(\frac{h}{|t|^\Delta}\right)}_{f_{\text{singular}}}.$$

With only two exponents, α and Δ , all the others can be computed.

3. The correlation length ξ , defined by $C(\vec{r}) \sim e^{-r/\xi}/r^{(\dots)}$, diverges near the critical point (experimentally). Hypothesis:

$$\xi(t, h) \sim |t|^{-\nu} g_\xi\left(\frac{h}{|t|^\Delta}\right),$$

with a new exponent ν .

So far three exponents are needed.

Hyperscaling hypothesis. The singular part of f comes from ξ . Divide a system of size L (near, but not at, the critical point) into boxes of size ξ : each box is independent of the rest (cutting a box in two would not be extensive, since there are correlations inside), and f is extensive. Hence

$$\beta F = \left(\frac{L}{\xi}\right)^d g_s + \left(\frac{L}{a}\right)^d g_a,$$

where the first term (dimensionless) contributes to the singular part, and the second, with a a microscopic (analytic) length, to the regular part. With $f = F/L^d$, $f_{\text{reg}} \sim k_B T g_a/a^d$ and $f_{\text{sing}} \sim k_B T g_s/\xi^d$, so

$$|t|^{2-\alpha} g_f\left(\frac{h}{|t|^\Delta}\right) \sim |t|^{d\nu} g_\xi^{-d}\left(\frac{h}{|t|^\Delta}\right) \Rightarrow d\nu = 2 - \alpha,$$

the **hyperscaling relation**. Everything reduces to two exponents, α and Δ . Can the hypothesis be proven? How can α, Δ be computed? Why are there universality classes? The answer is the **renormalization group** (RG) of Kadanoff (in real space) and Wilson (in Fourier space).

8.2 Kadanoff's block-spin transformation

Consider the Ising model $\mathcal{H} = -J \sum_{\text{NN}} S_i S_j - B \sum_i S_i$ (NN: nearest neighbours) on a 2D square lattice. Its critical point (numerical, and exact by Onsager) is at $B_c = 0$, $T_c = 2.269 \dots J/k_B$. At the critical point $\xi \rightarrow \infty$ and

$$C(\vec{r}) \sim \frac{1}{|\vec{r}|^{d-2+\eta}},$$

with η the anomalous dimension exponent. The system looks the same at all scales: domains of $+$ contain domains of $-$, which contain domains of $+$, and so on (a fractal structure).

Block spin transformation (BST). Divide the lattice into boxes of side b (e.g. $b = 3$), each with b^d spins $S_i \in \mathcal{M}$ (the spin space; for Ising $\mathcal{M} = \{-1, +1\}$).

1. Each box takes a single spin $S' \in \mathcal{M}$ (the same space), by some reasonable rule: e.g. the majority, or the central spin (decimation).

2. Look from further away (rescale lengths by b).
3. Go back to 1.

Does it converge to something? Kadanoff: if it converges to the same thing (a fractal structure), we are at a critical point. Formally, with $P[\{S_i\}] = Z^{-1}e^{-\beta\mathcal{H}}$, define

$$H := \beta\mathcal{H} = -\beta J \sum_{\text{NN}} S_i S_j - \beta B \sum_i S_i =: -K_1 \sum_{\text{NN}} S_i S_j - K_2 \sum_i S_i,$$

where K_1, K_2 are functions of T . The BST $\{S_i\} \rightarrow \{S'_i\}$, $S' = S'[S]$ (some rule), reduces the number of degrees of freedom $N \rightarrow N' = N/b^d$. Define $H'(S')$ by

$$e^{-H'(S')} = \sum_S \delta(S' - S'(S)) e^{-H(S)}.$$

Then

$$Z' := \sum_{S'} e^{-H'(S')} = \sum_{S, S'} \delta(S' - S'(S)) e^{-H(S)} = Z :$$

H' generates the same partition function, hence the same thermodynamics. What is H' ? Starting from Ising, in general

$$H' = K'_1 \sum_{\text{NN}} S_i S_j + K'_2 \sum_{\text{2nd neighbours}} S_i S_j + K'_3 \sum_{\square} S_i S_j S_k S_l + \dots,$$

and Wilson showed that the prefactors of the new terms are small.

Formally $H = H(S; \{K_n\})$, with $\{K_n\}$ an (infinite) set of coupling constants. Under the BST,

$$S \rightarrow S'(S), \quad H \rightarrow H'(S', K'), \quad K \rightarrow K' = R(K), \quad \xi \rightarrow \xi' = \xi/b$$

(we look from further away). We look for fixed points, $K^* = R(K^*)$, with the same H . At a fixed point, iterating gives $\xi' = \xi/b$ while the same H requires $\xi' = \xi$, so $\xi^* = 0$ (ideal gas) or $\xi^* = \infty$ (critical point).

8.3 Explicit examples

1. $d = 1$, $B = 0$. $H = -K_1 \sum_i S_i S_{i+1}$, with blocks of $b = 3$ spins and the decimation rule (keep the central spin of each block). Summing over the decimated spins,

$$e^{-H'(S'_1, S'_2, \dots)} = \sum_{\text{decimated}} e^{K_1(S_1 S_2 + S_2 S_3 + S_3 S_4 + \dots)} = e^{\text{const}} e^{K'_1(S'_1 S'_2 + S'_2 S'_3 + \dots)},$$

so $H' = \text{const} - K'_1 \sum_i S'_i S'_{i+1}$, with the mapping $K'_0 = K_0 + \text{const}$ and

$$K'_1 = \text{artanh}(\tanh^3 K_1); \quad x := \tanh K_1 \Rightarrow x' = x^3.$$

The fixed points of $x' = x^3$ are $x = 0$, $x = 1$ and $x = -1$ (the last one requires negative K_1); $x = \pm\infty$ are not admissible since $x = \tanh K_1$. So $K_1 = 0$ and $K_1 = \infty$. Under the BST x flows from 1 towards 0: K_1 decreases and ξ decreases. Any initial K_1^0 flows after infinitely many iterations to $K_1 = 0$ ($T = \infty$ or $J = 0$) with $\xi = 0$. *The 1D Ising model has no critical point.*

2. $d = 2$, $B = 0$: Migdal–Kadanoff approximation. After the BST, H' contains $K'_1, K'_2, K'_3, \dots$; the approximation is $K'_2 = K'_3 = \dots \simeq 0$. Then $K'_1 = R(K_1)$ (and also $K'_0 = R_0(K_0)$), $H' = K_0 - K_1 \sum_i S_i S_j$. With $K := K_1$,

$$e^{2K'} = \frac{e^{4K} + e^{-4K}}{2}, \quad K' = \frac{1}{2} \ln \cosh(4K).$$

There are three fixed points, $K^* \in \{0, 0.305\dots, \infty\}$; the intermediate one is unstable: for $K < K^*$ the flow goes to 0, for $K > K^*$ to ∞ .

Let $K_0 \in (0, K^*)$. Since it is not a fixed point, $\xi(K_0) \neq 0, \infty$ is finite. For $K \in (K_0, K^*)$ the BST takes $K \rightarrow K_0$ in $n > 0$ steps, and in each step $\xi \rightarrow \xi' = \xi/b$, so $\xi(K_0) = \xi(K)/b^n$, i.e. $\xi(K) = \xi(K_0)b^{n(K)}$. Taking $K \rightarrow K^*$, $n(K) \rightarrow \infty$ and

$$\xi(K^*) \rightarrow \infty :$$

$K^* = 0.305$ is a critical point. From K_0 to 0 infinitely many steps are needed, so $\xi(0) = \xi(K)/b^\infty = 0$: a trivial (non-interacting) point; the same for $K = \infty$, $\xi(\infty) = 0$. Recall $K = J/k_B T$: $K = 0$ means $J = 0$ or $T = \infty$, and $K = \infty$ means $J = \infty$ or $T = 0$ (the former is easier to interpret).

Kadanoff: unstable fixed points are critical points; stable fixed points are trivial points (ideal gas). (Reference: J. Cardy, *Scaling and Renormalization in Statistical Physics*.)

How does ξ diverge? Near the unstable fixed point, Taylor expanding the flow $K' = R(K)$,

$$K' \simeq R(K^*) + R'(K^*)(K - K^*) \Rightarrow K' - K^* = R'(K^*)(K - K^*),$$

with $R'(K^*) > 1$ (unstable). Define y by $R'(K^*) = b^y$, so $y > 0$ and $K' - K^* = b^y(K - K^*)$. Since $K = J/k_B T$, $K - K^* = \frac{\partial K}{\partial T}|_{T^*}(T - T^*)$, and

$$T' - T^* = b^y(T - T^*).$$

In addition, the BST gives $\xi' = \xi/b$, while phenomenologically $\xi(T) = \xi_0|T - T^*|^{-\nu}$. Then

$$\xi_0|T' - T^*|^{-\nu} = \frac{\xi_0}{b}|T - T^*|^{-\nu} \Rightarrow \xi_0(b^y|T - T^*|)^{-\nu} = \frac{\xi_0}{b}|T - T^*|^{-\nu} \Rightarrow b^{-\nu y} = b^{-1},$$

$$\nu = \frac{1}{y}.$$

The chain is $R \rightarrow K^* \rightarrow R'(K^*) \rightarrow y \rightarrow \nu$; the weak step in Kadanoff's approach is computing R .

8.4 General theory

Goals: to show universality, to compute the other exponent Δ , and to prove the scaling law. Consider a general Ising model $H = H(S_i, \{K\})$, with the BST $K'_a = R_a(\{K\})$, $a = 1, 2, \dots, \infty$, and fixed points $K^* = R(K^*)$. Near a fixed point,

$$K'_a = R_a(K^*) + \sum_b \underbrace{\frac{\partial R_a}{\partial K_b}}_{T_{ab}} \bigg|_{K^*} (K_b - K_b^*) \Rightarrow K'_a - K_a^* = T_{ab}(K_b - K_b^*).$$

Let ϕ^i be the left eigenvectors of T_{ab} with eigenvalues λ^i : $\phi_a^i T_{ab} = \lambda^i \phi_b^i$. Define the **scaling variables** (linear combinations of the couplings)

$$\mu^i := \sum_a \phi_a^i (K_a - K_a^*) \Rightarrow \mu^{i'} = \sum_a \phi_a^i T_{ab} (K_b - K_b^*) = \lambda^i \mu^i.$$

Define y_i by $\lambda_i = b^{y_i}$. Classification:

- $\lambda_i > 1$, $y_i > 0$: the flow moves away from K^* ; a *relevant* direction (relevant variable or coupling);

- $\lambda_i < 1, y_i < 0$: the flow approaches K^* ; an *irrelevant* direction;
- $\lambda_i = 1, y_i = 0$: *marginal* (not discussed here).

The $\{K_a\}$ live in a space of dimension n' ; let n be the number of relevant directions and $n' - n$ the number of irrelevant ones. The irrelevant directions span the **critical surface** through K^* (e.g. in the plane of $K_1 =$ nearest neighbours and $K_2 =$ second neighbours).

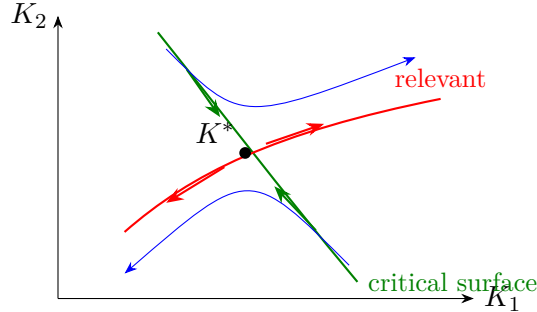


Figure 8.1: RG flow near a fixed point with one relevant and one irrelevant direction.

Properties:

1. Let K_0 lie on the relevant surface: it is not a fixed point, so $\xi(K_0)$ is finite.
2. Infinitely many iterations are needed from K^* to K_0 , so $\xi(K^*) = \infty$.
3. On the critical surface $\xi = \infty$ (infinitely many iterations are needed to reach K^*).
4. How many “knobs” must be tuned to be on the critical surface? One must set the n relevant $\mu^i = 0$.

In magnetism, experimentally $n = 2$ (T and B). For Ising with $B = 0$, experimentally $n = 1$: in the general model $H_{\text{gen}} = -K_1 \sum_{\text{NN}} S_i S_j - K_2 \sum_{\text{NNN}} S_i S_j - K_3 \dots$ the classical Ising model lies on the axis $(K_1, 0, 0, \dots)$. In an experiment starting at $K_1^{\text{ini}} \lesssim K_{1c}$, $K_2^{\text{ini}} = K_3^{\text{ini}} = \dots = 0$, ξ^{ini} is large; after some iterations $K \simeq K^*$ (near the critical surface), so ξ is still large; later the flow moves away and ξ becomes small; finally $\xi = 0$. The correlation function $G(r)$ is algebraic at short distances and exponential at long distances.

Universality. The classical Ising model ($K_1 \neq 0, K_2 = 0$) and the Ising model with second neighbours ($K_1, K_2 \neq 0, K_3 = \dots = 0$), $H = \beta \mathcal{H} = -K_1 \sum_{\text{NN}} S_i S_j - K_2 \sum_{\text{NNN}} S_i S_j$ with $K_1 = J_1/k_B T$, $K_2 = J_2/k_B T$, varying only T , move along lines $K_2/K_1 = J_2/J_1$. These lines intersect the critical surface at different T_c , but seen from far away they look the same: *universality*. The world can be more complicated: there can be two fixed points on the critical surface, and hence *two universality classes*.

Example: gas–liquid. The atomic model $H = \sum_i p_i^2/2m + \sum U(r_i - r_j)$, with excluded volume and an attractive well, has a critical point at the end of the coexistence line. A lattice model: on each site an occupation variable $c_i = 0$ (no atom) or 1 (one atom); $c_i \leq 1$ emulates the excluded volume. With μ the chemical potential and a nearest-neighbour attraction,

$$H = -\mu \sum_i c_i - \varepsilon_0 \sum_{\text{NN}} c_i c_j.$$

With $S_i = 2c_i - 1$ ($c_i = 0 \Rightarrow S_i = -1$, $c_i = 1 \Rightarrow S_i = +1$),

$$H \rightarrow -\mu \sum \left(\frac{S_i + 1}{2} \right) - \varepsilon_0 \sum \left(\frac{S_i + 1}{2} \right) \left(\frac{S_j + 1}{2} \right) = \text{const} + \text{const} \sum_i S_i + \text{const} \sum_{\text{NN}} S_i S_j,$$

equivalent to Ising.

8.5 Scaling from the RG

For Ising with $\vec{B}$, $H = -K_1 \sum S_i S_j - K_h \sum S_i$, there are two relevant dimensions in experiment, μ_t (temperature) and μ_h (magnetic field); the critical surface is μ_{irr} directions, and the critical point is $\mu_t = \mu_h = 0$ at $t = 0$, $h = 0$. This allows a Taylor expansion,

$$\mu_t \simeq at + bh + \dots, \quad \mu_h \simeq ct + dh + \dots,$$

and by symmetry (vector analysis) $\mu_t \sim t/t_0$, $\mu_h \sim h/h_0$. What are μ_t, μ_h ? They are linear combinations of $K_1 - K_1^*$, $K_2 - K_2^*$, $\dots$, so

$$H = K_1 \sum_{\text{NN}} S_i S_j + K_2 \sum_{\text{NNN}} S_i S_j + \dots = K_1 \Theta_1 + K_2 \Theta_2 + \dots$$

$\xrightarrow{\text{BST, approaching } K^*} \mu_t \Theta_t + \mu_h \Theta_h + \text{subdominant},$

with Θ_t a linear combination of $\Theta_1, \Theta_2, \dots$, which there is no general way to compute. (Wilson: the Θ_t are easy if $d = 4 - \epsilon$. Numerical RG is another option.)

Proof of the scaling hypothesis. Under the BST, $S_i \rightarrow S'_i$, $K \rightarrow K'$, but $Z \rightarrow Z' = Z$ with fewer degrees of freedom: $V' = V/b^d$, $F' = F$, so $V'f' = Vf$ with f the free energy density. To leading order $f(\{K_a\}) \rightarrow f(\mu_t, \mu_h)$, and $f = \frac{V'}{V} f'$:

$$f(\mu_t, \mu_h) = b^{-d} f(\mu'_t, \mu'_h) = b^{-d} f(b^{y_t} \mu_t, b^{y_h} \mu_h),$$

using $\mu'_t = \lambda_t \mu_t = b^{y_t} \mu_t$, $\mu'_h = b^{y_h} \mu_h$. Near the critical point, $f(t, h) = b^{-d} f(b^{y_t} t, b^{y_h} h)$. Choosing b such that $b^{y_t} |t| = 1$, $b = |t|^{-1/y_t}$,

$$f(t, h) = |t|^{d/y_t} f\left(1, \frac{h}{|t|^{y_h/y_t}}\right),$$

which proves the scaling hypothesis $f(t, h) = |t|^{2-\alpha} g(h/|t|^\Delta)$, with

$$2 - \alpha = \frac{d}{y_t}, \quad \Delta = \frac{y_h}{y_t}.$$

(Together with $\nu = 1/y_t$ this gives hyperscaling, $d\nu = 2 - \alpha$.) It remains to find $R(K)$: in real space (Kadanoff) or in Fourier space (Wilson).

9 Renormalization group in dynamical systems

We now apply RG ideas to non-equilibrium statistical mechanics.

9.1 Diffusion with a higher-order term

Consider

$$\partial_t \rho = D_1 \nabla^2 \rho + D_2 \nabla^4 \rho.$$

Look from far away: rescale $x = bx'$ with $b > 1$; the equation holds in x , and we ask which equation holds in x' . Time is also rescaled, $t = b^z t'$, so $\partial_x = b^{-1} \partial_{x'}$, $\partial_t = b^{-z} \partial_{t'}$, and

$$b^{-z} \partial_{t'} \rho = b^{-2} D_1 \partial_{x'}^2 \rho + b^{-4} D_2 \partial_{x'}^4 \rho \Rightarrow \partial_{t'} \rho = b^{z-2} D_1 \partial_{x'}^2 \rho + b^{z-4} D_2 \partial_{x'}^4 \rho.$$

Choosing the **dynamical exponent** $z = 2$, $\partial_{t'} \rho = D_1 \partial_{x'}^2 \rho + b^{-2} D_2 \partial_{x'}^4 \rho$. Under the rescaling $D_1 \rightarrow D_1 = b^{y_1} D_1$ and $D_2 \rightarrow b^{-2} D_2 = b^{y_2} D_2$: $y_1 = 0$ (marginal) and $y_2 = -2$ (irrelevant). The flow takes $D_2 \rightarrow 0$ at fixed D_1 .

9.2 KPZ equation

Surface growth (Kardar–Parisi–Zhang). Material is deposited from a gas onto a surface described by its height $h(\vec{x}, t)$. In a time δt the thickness grows by $v \delta t$ (v the deposition speed) perpendicular to the surface. If the surface is tilted by an angle α , the height grows by

$$\delta h = \frac{v \delta t}{\cos \alpha}, \quad \cos \alpha = \frac{\Delta x}{\sqrt{\Delta x^2 + \Delta h^2}} = \frac{1}{\sqrt{1 + (\Delta h / \Delta x)^2}} \simeq \frac{1}{\sqrt{1 + (\nabla h)^2}}$$

(locally), so $\delta h = v \delta t \sqrt{1 + (\nabla h)^2}$. For small gradients,

$$\frac{\partial h}{\partial t} = v \sqrt{1 + (\nabla h)^2} \simeq v \left(1 + \frac{1}{2} (\nabla h)^2 \right).$$

Adding diffusion and noise, $\partial_t h = v + \frac{v}{2} (\nabla h)^2 + \nu \nabla^2 h + \eta(\vec{x}, t)$. With $\tilde{h} = h - vt$ (removing the average growth) and then $\tilde{h} \rightarrow h$, $v \rightarrow \lambda$, we get the **KPZ equation**

$$\partial_t h = \nu \nabla^2 h + \frac{\lambda}{2} (\nabla h)^2 + \eta(\vec{x}, t), \quad \langle \eta(\vec{x}_1, t_1) \eta(\vec{x}_2, t_2) \rangle = 2D \delta(t_1 - t_2) \delta(\vec{x}_1 - \vec{x}_2),$$

with white noise η (it can be mapped onto several other known equations). Simulations show scaling,

$$\langle h(\vec{x}_1, t_1) h(\vec{x}_2, t_2) \rangle \sim |\vec{x}_1 - \vec{x}_2|^{2\chi} g\left(\frac{|t_1 - t_2|}{|\vec{x}_1 - \vec{x}_2|^z}\right),$$

with χ the roughness exponent and z the dynamical exponent, neither integer nor rational.

Power counting. Rescale $x = bx'$, $t = b^z t'$, $h = b^\chi h'$:

$$b^{\chi-z} \partial_{t'} h' = \nu b^{\chi-2} \nabla'^2 h' + \frac{\lambda}{2} b^{2(\chi-1)} (\nabla' h')^2 + \eta,$$

$$\partial_{t'} h' = \nu b^{z-2} \nabla'^2 h' + \frac{\lambda}{2} b^{\chi+z-2} (\nabla' h')^2 + \underbrace{b^{z-\chi} \eta}_{\eta'(\vec{x}', t')}.$$

For the noise, $\delta(\vec{x}_1 - \vec{x}_2) = b^{-d} \delta(\vec{x}'_1 - \vec{x}'_2)$ and $\delta(t_1 - t_2) = b^{-z} \delta(t'_1 - t'_2)$, so

$$\langle \eta'(\vec{x}'_1, t'_1) \eta'(\vec{x}'_2, t'_2) \rangle = b^{2z-2\chi} 2D b^{-d-z} \delta(\vec{x}'_1 - \vec{x}'_2) \delta(t'_1 - t'_2) = b^{z-2\chi-d} 2D \delta \delta.$$

So KPZ maps to KPZ with

$$\nu' = b^{z-2} \nu, \quad \lambda' = b^{\chi+z-2} \lambda, \quad D' = b^{z-2\chi-d} D.$$

Choose z and χ so that the linear dynamics is kept, $\nu' = \nu$ and $D' = D$: $z = 2$, $\chi = 1 - \frac{d}{2}$. Then

$$\lambda' = b^{1-d/2} \lambda :$$

λ is irrelevant for $d > 2$, relevant for $d < 2$, and marginal for $d = 2$. Why does this fail? It would predict rational exponents. There is another relevant scale, the correlation length ξ , which was not considered; one would have to follow the flow to see where it ends (a non-trivial fixed point at finite λ^*).

9.3 Directed percolation

1. Bond percolation. On a square lattice rotated by 45° , each intersection is a node and each line a bond, open (connected) with probability $p \in [0, 1]$. As p increases there is a p_c above which there is a finite probability P_∞ of having an infinite cluster (one that percolates from wall to wall); below it, clusters are absorbed. p_c is a critical point with $\xi \rightarrow \infty$.

2. Directed percolation (DP). Bonds are open with probability p but only point downwards. Reading the vertical direction as time gives a dual interpretation: geometry or dynamics.

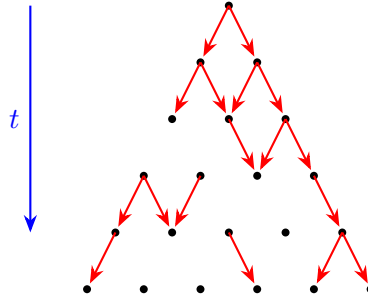


Figure 9.1: Directed bond percolation: open bonds (red) point only forwards in time.

Dynamical interpretation. Sites are 0 (empty/dry) or 1 (wet); bonds are closed (probability $1 - p$) or open (connected, probability p). The rule $S_i(t + 1) = F[S(t)]$ for DP: site i at $t + 1$ (even sites) depends on sites $i - 1$ and $i + 1$ at t (odd sites),

$$S_i(t + 1) = \begin{cases} 1, & \text{if } S_{i-1}(t) = 1 \text{ and (prob. } p) \text{ the bond is open,} \\ 1, & \text{if } S_{i+1}(t) = 1 \text{ and (prob. } p) \text{ the bond is open,} \\ 0, & \text{otherwise.} \end{cases}$$

Interpretation as a chemical reaction: $A \rightarrow \emptyset$ with probability $(1 - p)^2$ (destruction); $\emptyset + A \rightarrow A + \emptyset$ with probability $p(1 - p)$ (diffusion); $A \rightarrow 2A$ (offspring); $2A \rightarrow A$ (coagulation). These are non-conserving chemical reactions, e.g. in catalysis (around 1980: systems with a catalyst reaching an absorbing state).

Domany–Kinzel model (a generalization of DP), with transition probabilities $P[S_i(t + 1) | S_{i-1}(t), S_{i+1}(t)]$:

1. $P(1|00) = 0$;
2. $P(1|10) = P(1|01) = p_1$;
3. $P(1|11) = p_2$;
4. $P(0|\cdot\cdot) = 1 - P(1|\cdot\cdot)$.

Relation with DP: $p_1 = p$, and p_2 has three contributions (both bonds open, p^2 , or only one, $2p(1 - p)$):

$$p_2 = p^2 + 2p(1 - p) = p(2 - p).$$

The phase diagram in the (p_1, p_2) plane has an inactive phase, in which for every initial condition there is a time at which the number of wet sites is zero (an absorbing state), and an active phase, with a finite probability of a cluster that lives forever. They are separated by a critical (universal) transition line, which the DP curve $p_2 = p_1(2 - p_1)$ crosses.

Contact model. A continuous-time model with transition rates

$$w(0 \rightarrow 1 | n) = \frac{\lambda n}{2d}, \quad w(1 \rightarrow 0 | n) = r,$$

with d the dimension of space, n the number of occupied neighbours and $n/2d$ the fraction of occupied neighbours. Interpretation: 0 healthy, 1 infected; $0 \rightarrow 1$ is infection, $1 \rightarrow 0$ recovery. The contact model belongs to the DP universality class.

9.4 Critical properties and scaling of DP

For DP there is $0 < p_c < 1$. Observable: the fraction of occupied sites (averaged over realizations), $\rho(t) := \left\langle \frac{1}{N} \sum_i S_i(t) \right\rangle$, and $\rho_{\text{ness}} = \lim_{t \rightarrow \infty} \rho(t)$:

$$\rho_{\text{ness}} = \begin{cases} 0, & p < p_c, \\ \sim (p - p_c)^\beta, & p > p_c, \end{cases} \quad \beta \simeq 0.277 \text{ (1 + 1 dims)}, \quad \beta \simeq 0.58 \text{ (2 + 1 dims)}.$$

Spreading: a cluster starting from a single seed has longitudinal (time) and transverse (space) sizes

$$\xi_{\parallel} \sim |p - p_c|^{-\nu_{\parallel}}, \quad \xi_{\perp} \sim |p - p_c|^{-\nu_{\perp}} :$$

three exponents, $\beta, \nu_{\perp}, \nu_{\parallel}$. Another observable is P_{∞} , the probability that a site belongs to an infinite cluster, $P_{\infty} \sim (p - p_c)^{\beta'}$ for $p > p_c$ (and 0 below); for DP $\beta = \beta'$ (exercise).

Scaling. With $\Delta := p - p_c$: $x \sim \Delta^{-\nu_{\perp}}$, $t \sim \Delta^{-\nu_{\parallel}}$, $\rho \sim \Delta^{\beta}$, and correlations $C(x, t) \sim t^{\alpha} f(x/t^{1/z})$ with the dynamical exponent $z := \nu_{\parallel}/\nu_{\perp}$.

Continuum field description. From a distance, the discrete $S_i(t)$ becomes a continuous field $\rho(x, t)$, the fraction of occupied sites. For the contact model,

$$\frac{\partial \rho}{\partial t} = \underbrace{-r\rho}_{\text{recovery}} + \underbrace{\lambda\rho(1-\rho)}_{\text{infection (neighbour must be healthy)}} + \underbrace{D\nabla^2 \rho}_{\text{diffusion}} + \underbrace{\xi}_{\text{noise}}$$

(without noise this is Fisher's model), with

$$\langle \xi(\vec{x}, t) \rangle = 0, \quad \langle \xi(\vec{x}_1, t_1) \xi(\vec{x}_2, t_2) \rangle = \Gamma \rho(\vec{x}_1, t_1) \delta(\vec{x}_1 - \vec{x}_2) \delta(t_1 - t_2),$$

by the central limit theorem; the noise proportional to ρ generates the probability of absorbing states. (Exercise: by a change of variables KPZ maps to a similar equation with $\langle \xi \xi \rangle \sim \rho^2$, another equivalence class.)

Power counting. Write $\partial_t \rho = a\rho - \lambda\rho^2 + D\nabla^2 \rho + \xi$ with $a := \lambda - r$; the critical point is at $a = 0$ ($a \simeq \Delta$). Rescale (note: this is the inverse flow, approaching the critical point)

$$a \rightarrow \Lambda a, \quad x \rightarrow \Lambda^{-\nu_{\perp}} x, \quad t \rightarrow \Lambda^{-\nu_{\parallel}} t, \quad \rho \rightarrow \Lambda^{\beta} \rho.$$

Then

$$\begin{aligned} \Lambda^{\beta+\nu_{\parallel}} \partial_t \rho &= \Lambda^{1+\beta} a \rho - \lambda \Lambda^{2\beta} \rho^2 + D \Lambda^{\beta+2\nu_{\perp}} \nabla^2 \rho + \Lambda^{(\beta+d\nu_{\perp}+\nu_{\parallel})/2} \xi, \\ \partial_t \rho &= \underbrace{\Lambda^{1-\nu_{\parallel}} a \rho - \lambda \Lambda^{\beta-\nu_{\parallel}} \rho^2 + D \Lambda^{2\nu_{\perp}-\nu_{\parallel}} \nabla^2 \rho}_{\text{deterministic}} + \Lambda^{-\frac{\beta}{2} + \frac{d\nu_{\perp}}{2} - \frac{\nu_{\parallel}}{2}} \xi. \end{aligned}$$

Requiring the deterministic equation to be invariant gives the mean-field prediction

$$\nu_{\parallel} = 1, \quad \beta = 1, \quad \nu_{\perp} = \frac{1}{2}, \quad z = \nu_{\parallel}/\nu_{\perp} = 2.$$

	mean field	$d = 1$	$d = 2$
$\nu_{\parallel}$	1	1.73	1.29
$\nu_{\perp}$	$\frac{1}{2}$	1.09	0.73
β	1	0.277	0.58

Mean field fails. Why? The exponent of the noise, in mean field, is

$$y_{\xi} = -\frac{\beta}{2} + \frac{d\nu_{\perp}}{2} - \frac{\nu_{\parallel}}{2} = -\frac{1}{2} + \frac{d}{4} - \frac{1}{2} = \frac{d-4}{4}.$$

Since this is the inverse flow, $y_{\xi} > 0$ is irrelevant and $y_{\xi} < 0$ relevant: for $d < 4$ the noise is relevant.

Why does power counting fail? Decimating a fluctuating profile $\rho(x)$ and looking from far away, one gets a smooth profile. But $\langle \rho^2 \rangle \neq \langle \rho \rangle^2$, so the term $\lambda \rho^2$ changes λ : e.g. for three points with $\rho_2 = (\rho_1 + \rho_3)/2$, $\langle \rho_1^2 + \rho_2^2 + \rho_3^2 \rangle > 3\rho_2^2$. Effectively, λ is renormalized by contributions of large k .

9.5 Renormalization of large wavenumbers

Wilson's idea: there is an ultraviolet cutoff Ω ,

$$\rho(\vec{x}, t) = \int_{|\vec{k}| < \Omega} \frac{d^d k}{(2\pi)^d} e^{i\vec{k} \cdot \vec{x}} \tilde{\rho}(\vec{k}, t),$$

with $\Omega = 2\pi/h$ for the starting model, h the lattice spacing. Split

$$\tilde{\rho}(\vec{k}, t) = \begin{cases} \rho_{<}(\vec{k}, t), & k < \Omega/\Lambda, \\ \rho_{>}(\vec{k}, t), & \Omega/\Lambda < k < \Omega, \end{cases}$$

and sum over the shell $\rho_{>}$. Methods: (1) dynamic renormalization (see the reading), (2) field theory.

9.6 Field-theory formulation (Martin–Siggia–Rose)

Stochastic ODE (no space). Consider

$$\dot{\phi} = -\gamma \frac{\partial H}{\partial \phi} + \xi, \quad \langle \xi(t_1) \xi(t_2) \rangle = \Gamma \phi(t_1) \delta(t_1 - t_2),$$

with $H(\phi)$ a free energy. Goal: the propagator $\langle \phi(t_1) \phi(t_2) \rangle_\xi$. Writing it as an integral over all functions, restricted to those that satisfy the equation of motion at every t ,

$$\langle \phi(t_1) \phi(t_2) \rangle_\xi = \left\langle \int \mathcal{D}[\phi] \phi(t_1) \phi(t_2) \delta(\dot{\phi} + \gamma H'(\phi) - \xi) \right\rangle_\xi$$

(the Jacobian is 1 with the Itô prescription). Using $\delta(x) = \frac{1}{2\pi} \int d\omega e^{i\omega x}$, one needs an ω for each t : a function $\omega(t) \rightarrow \tilde{\psi}(t)$. With $\tilde{\phi} = -i\tilde{\psi}$,

$$\langle \phi(t_1) \phi(t_2) \rangle_\xi = \left\langle \int \mathcal{D}[\phi] \mathcal{D}[\tilde{\phi}] \phi(t_1) \phi(t_2) e^{-\int dt \tilde{\phi} [\dot{\phi} + \gamma H' - \xi]} \right\rangle_\xi.$$

The noise is Gaussian with zero mean and functional probability

$$P[\xi] = \prod_i p(\xi(t_i)) \propto f[\phi] e^{-\int dt \frac{\xi^2(t)}{2\Gamma\phi(t)}}, \quad \langle A \rangle_\xi = \int \mathcal{D}[\xi] P[\xi] A.$$

Completing the square in ξ (the normalization $\int e^{-\int \xi^2/2\Gamma\phi} = 1/f[\phi]$ cancels),

$$\begin{aligned} \langle \phi(t_1) \phi(t_2) \rangle_\xi &= \int \mathcal{D}[\phi] \mathcal{D}[\tilde{\phi}] \phi(t_1) \phi(t_2) e^{-\int dt \tilde{\phi} [\dot{\phi} + \gamma H' - \frac{\Gamma}{2} \tilde{\phi} \phi]} \\ &= \frac{\delta}{\delta h(t_1)} \frac{\delta}{\delta h(t_2)} \int \mathcal{D}[\phi] \mathcal{D}[\tilde{\phi}] e^{-S[\phi, \tilde{\phi}]} \Big|_{h=0}, \end{aligned}$$

with the **action** and the **partition function**

$$S[\phi, \tilde{\phi}] := \int dt \left\{ \tilde{\phi} \left[\dot{\phi} + \gamma H'(\phi) - \frac{\Gamma}{2} \tilde{\phi} \phi \right] - h\phi \right\}, \quad Z := \int \mathcal{D}[\phi] \mathcal{D}[\tilde{\phi}] e^{-S[\phi, \tilde{\phi}]}.$$

PDE: the contact process. For

$$\frac{\partial \phi}{\partial t} = r\phi - \lambda\phi^2 + D\nabla^2\phi + \xi, \quad \langle \xi\xi \rangle = \Gamma\phi\delta(t-t')\delta(\vec{x}-\vec{x}'),$$

replace $\phi(t) \rightarrow \phi(\vec{x}, t)$, $\tilde{\phi}(t) \rightarrow \tilde{\phi}(\vec{x}, t)$ and $-\gamma H'(\phi) \rightarrow r\phi - \lambda\phi^2 + D\nabla^2\phi$:

$$S[\phi, \tilde{\phi}] = \int d^d x dt \tilde{\phi} \left[\partial_t \phi - D\nabla^2 \phi - r\phi + \underbrace{\lambda\phi^2 - \frac{\Gamma}{2}\tilde{\phi}\phi}_{\text{cubic}} \right] + \dots$$

To symmetrize, change variables $\phi' = \sqrt{2\lambda/\Gamma}\phi$, $\tilde{\phi}' = \sqrt{\Gamma/2\lambda}\tilde{\phi}$, $\Gamma' = \sqrt{2\Gamma\lambda}$, and allow a coefficient τ in front of ∂_t :

$$S[\phi, \tilde{\phi}] = \int d^d x dt \left\{ \underbrace{\tilde{\phi}[\tau\partial_t - D\nabla^2 - r]\phi}_{S_0} + \underbrace{\frac{\Gamma}{2}\tilde{\phi}(\phi - \tilde{\phi})\phi}_{S_{\text{int}}} \right\}.$$

The operator $\tau\partial_t - D\nabla^2 - r$ is almost self-adjoint, but not quite (irreversible).

Analogy with equilibrium phase transitions. In magnetism, the G-L free energy

$$F[\psi] = \int d^d x \left[\underbrace{\frac{t}{2}\psi^2 + \frac{\kappa}{2}(\nabla\psi)^2}_{F_0} + \underbrace{\frac{u}{4}\psi^4}_{F_{\text{int}}} \right], \quad Z = \int \mathcal{D}[\psi] e^{-F[\psi]},$$

has its critical point at $t = 0$. What is $\int \mathcal{D}[\psi]$? For a real field, $\psi(\vec{x}) = \frac{1}{V} \sum_{\vec{k}} e^{i\vec{k}\cdot\vec{x}} \psi_{\vec{k}} = \int \frac{d^d k}{(2\pi)^d} e^{i\vec{k}\cdot\vec{x}} \psi(\vec{k})$ with $\psi_{\vec{k}}^* = \psi_{-\vec{k}}$: not all modes are independent, and

$$\mathcal{D}[\psi] = \prod_{\vec{k}, k_x > 0} d\text{Re}[\psi_{\vec{k}}] d\text{Im}[\psi_{\vec{k}}]$$

(and in $(\vec{x}, t)$, $\mathcal{D}[\phi] = \prod_{\vec{k}, k_x > 0} \prod_{\omega} d\text{Re}[\phi_{\vec{k}\omega}] d\text{Im}[\phi_{\vec{k}\omega}]$). For the Gaussian part,

$$\int d\vec{x} \psi^2 = \int \frac{d^d k}{(2\pi)^d} |\psi(\vec{k})|^2, \quad \int d\vec{x} (\nabla\psi)^2 = \int \frac{d^d k}{(2\pi)^d} k^2 |\psi(\vec{k})|^2,$$

$$F_0 = \int \frac{d^d k}{(2\pi)^d} \left(\frac{t}{2} + \frac{\kappa}{2} k^2 \right) |\psi(\vec{k})|^2,$$

and the Gaussian integrals factorize (an “ideal gas of Fourier modes”):

$$Z_0 = \int \mathcal{D}[\psi] e^{-F_0} = \prod_{\vec{k}} \frac{2\pi}{t + \kappa k^2}$$

(up to volume factors). The interaction,

$$F_{\text{int}} = \frac{u}{4} \int d\vec{x} \psi^4 = \frac{u}{4} (2\pi)^d \int \frac{d\vec{k}_1 d\vec{k}_2 d\vec{k}_3 d\vec{k}_4}{(2\pi)^{4d}} \delta(\vec{k}_1 + \vec{k}_2 + \vec{k}_3 + \vec{k}_4) \psi(\vec{k}_1) \psi(\vec{k}_2) \psi(\vec{k}_3) \psi(\vec{k}_4),$$

couples different modes (an interacting gas: the energy depends on more than one mode) and is not Gaussian. Wilson’s approach is perturbation theory in $\mathcal{U} := F_{\text{int}}$: $e^{-F_0 - \mathcal{U}} = e^{-F_0} [1 - \mathcal{U} + \frac{\mathcal{U}^2}{2!} - \frac{\mathcal{U}^3}{3!} + \dots]$, each term being a Gaussian moment. *But* power counting of G-L says that $\mathcal{U}$ is relevant for $d < 4$, so perturbation theory does not converge there (for $d > 4$ one could work with F_0 alone).

9.7 Wilson's renormalization

1. *Coarse-graining* (equivalent to Kadanoff's decimation), with the lattice spacing a the smallest length of the system: take $b > 1$ and $\hat{\psi}(\vec{x}) = \frac{1}{(ab)^d} \int d\vec{x}' \psi(\vec{x}')$ over the box of size $(ab)^d$ around $\vec{x}$.
2. Rescale space, $\vec{x}' = \vec{x}/b$.
3. Renormalize the field, $\psi' = \psi/\zeta$.

In Wilson's version, step 1 is done in $\vec{k}$ -space: the ultraviolet cutoff $\Omega = 2\pi/a$ restricts the space of functions, $\mathcal{D}[\psi] = \prod_{|\vec{k}| < \Omega} d\psi_{\vec{k}}$, and we split

$$\psi_{\vec{k}} = \begin{cases} \psi_{\vec{k}}^+, & \Omega/b < |\vec{k}| < \Omega, \\ \psi_{\vec{k}}^-, & |\vec{k}| < \Omega/b. \end{cases}$$

Then $Z = \int \mathcal{D}[\psi^-] \mathcal{D}[\psi^+] e^{-F[\psi^-, \psi^+]}$, and the coarse graining is

$$e^{-F_-[\psi^-]} := \int \mathcal{D}[\psi^+] e^{-F[\psi^-, \psi^+]},$$

very hard to compute in general. With $F = H = F_0 + \mathcal{U}$ and F_0 diagonal in $\vec{k}$ (the $\vec{k}_-$ and $\vec{k}_+$ integrals separate),

$$\int \mathcal{D}[\psi^+] e^{-F_0} = e^{-F_0[\psi^-]} \underbrace{\int \prod_{\vec{k}_+} \frac{d\psi_{\vec{k}_+}}{(2\pi)^d} e^{-\int d\vec{k}_+ [\frac{t}{2} + \frac{\kappa}{2} k^2] |\psi_{\vec{k}}^-|^2}}_{a \in \mathbb{R}, \text{ just a number}},$$

so

$$F_-[\psi^-] = F_0[\psi^-] - \ln a - \ln \left\langle e^{-\mathcal{U}[\psi^+, \psi^-]} \right\rangle_{F_0, \psi^+}.$$

Up to here everything is exact. Wilson's approximation, justified only a posteriori (power counting says u is relevant), is a Taylor expansion in u , i.e. the cumulant expansion

$$\ln \langle e^{-\mathcal{U}} \rangle = -\langle \mathcal{U} \rangle + \frac{1}{2} [\langle \mathcal{U}^2 \rangle - \langle \mathcal{U} \rangle^2] - \frac{1}{6} [\langle \mathcal{U}^3 \rangle - \dots] + \dots,$$

so that $F_-[\psi^-] = F_0[\psi^-] - \ln a + \langle \mathcal{U}[\psi^-, \psi^+] \rangle_{F_0, \psi^+} + \dots$. To evaluate $\langle \mathcal{U} \rangle = \langle \frac{u}{4} \int d\vec{x} \psi^4 \rangle$ write each $\psi_{\vec{k}} = \psi_{\vec{k}_-} + \psi_{\vec{k}_+}$ (each vanishing outside its shell). $\mathcal{U}$ then has 16 terms:

- 1 with only $\vec{k}_+$: a constant;
- 1 with only $\vec{k}_-$: $\mathcal{U}[\psi^-]$;
- 4 with one $\vec{k}_-$ and three $\vec{k}_+$, and 4 with three $\vec{k}_-$ and one $\vec{k}_+$: zero by parity;
- 6 with two $\vec{k}_-$ and two $\vec{k}_+$:

$$\frac{1}{2} \int \frac{d\vec{k}_1 d\vec{k}_2}{(2\pi)^{2d}} \psi_{\vec{k}_1-} \psi_{\vec{k}_2-} \cdot \frac{6u}{2} \left\langle \int d\vec{k}_3 d\vec{k}_4 \psi_{\vec{k}_3+} \psi_{\vec{k}_4+} \delta(\dots) \right\rangle_{F_0} = \frac{t_1}{2} \int d\vec{x} \psi^2,$$

since the average is proportional to $t_1 \delta(\vec{k}_1 + \vec{k}_2)$.

Hence, with $F_0[\psi] = \int d\vec{x} [\frac{t}{2} \psi^2 + \frac{\kappa}{2} (\nabla \psi)^2]$,

$$F_-[\psi^-] = F_0[\psi^-] + \mathcal{U}[\psi^-] + \text{const} + \frac{t_1}{2} \int d\vec{x} \psi^2 :$$

the coarse graining gives $t \rightarrow t + t_1$. This is the **perturbative renormalization group**; one must go to second order so that the flow does not run off to infinity.

9.8 Perturbative renormalization of directed percolation

DP is described by

$$\frac{\partial \phi}{\partial t} = r\phi - \lambda\phi^2 + D\nabla^2\phi + \xi, \quad \langle \xi\xi \rangle = \Gamma\phi\delta(t-t')\delta(\vec{r}-\vec{r}'),$$

or equivalently by the action

$$S[\phi, \tilde{\phi}] = \int dx dt \left\{ \tilde{\phi}(\tau\partial_t - D\nabla^2 - r)\phi + \frac{\Gamma}{2}\tilde{\phi}(\phi - \tilde{\phi})\phi \right\}, \quad Z = \int \mathcal{D}[\phi] \mathcal{D}[\tilde{\phi}] e^{-S[\phi, \tilde{\phi}]}.$$

1. Rescaling (power counting). With $x = bx'$, $t = b^z t'$, $\phi = b^\chi \phi'$, $\tilde{\phi} = b^\chi \tilde{\phi}'$ (omitting the primes),

$$S' = \int b^{d+z} dx dt \left[b^{2\chi} \tilde{\phi}'(b^{-z}\tau\partial_t - b^{-2}D\nabla^2 - r)\phi' + \frac{\Gamma}{2}b^{3\chi}\tilde{\phi}'(\phi' - \tilde{\phi}')\phi' \right],$$

$$\tau' = b^{2\chi+d}\tau, \quad D' = b^{d+z-2+2\chi}D, \quad r' = b^{d+z+2\chi}r, \quad \Gamma' = b^{d+z+3\chi}\Gamma.$$

For an infinitesimal step $b = 1 + l$, $l \ll 1$ (the shell is almost a surface), e.g. $\tau' = (1 + (2\chi + d)l)\tau$.

2. Coarse graining. Split $S = S_0 + S_{\text{int}}$ (quadratic and cubic). Assuming Γ small, $e^{-S} = e^{-S_0}(1 - S_{\text{int}} + \frac{1}{2}S_{\text{int}}^2 - \dots)$, so $Z = Z_0 \left\langle 1 - S_{\text{int}} + \frac{1}{2}S_{\text{int}}^2 + \dots \right\rangle_0$. With $\mathcal{D}[\phi, \tilde{\phi}] = \mathcal{D}[\phi, \tilde{\phi}]_- \mathcal{D}[\phi, \tilde{\phi}]_+$,

$$Z = \int \mathcal{D}[\phi, \tilde{\phi}]_- \underbrace{\int \mathcal{D}[\phi, \tilde{\phi}]_+ e^{-S}}_{=: e^{-S_-[\phi, \tilde{\phi}]}} e^{-S}, \quad S_- = -\ln \left[\left\langle 1 - S_{\text{int}} + \frac{1}{2}S_{\text{int}}^2 - \dots \right\rangle_{0,+} \int \mathcal{D}[\phi, \tilde{\phi}]_+ e^{-S_0} \right],$$

where the last integral is easy. In Fourier space,

$$S_0 = \int \frac{d\vec{k}}{(2\pi)^d} \frac{d\omega}{2\pi} \tilde{\phi}(-\vec{k}, -\omega) (-i\tau\omega + Dk^2 - r)\phi(\vec{k}, \omega) = \frac{1}{2} \int d\vec{k} d\omega \vec{\Phi}(-\vec{k}, -\omega) \mathbb{L}(\vec{k}, \omega) \vec{\Phi}(\vec{k}, \omega),$$

with $\vec{\Phi} := (\phi, \tilde{\phi})$ and the non-Hermitian matrix

$$\mathbb{L}(\vec{k}, \omega) = \begin{pmatrix} 0 & Dk^2 - r + i\tau\omega \\ Dk^2 - r - i\tau\omega & 0 \end{pmatrix} =: \mathbb{G}_0^{-1},$$

$$\mathbb{G}_0 = \begin{pmatrix} 0 & G_0(\vec{k}, -\omega) \\ G_0(\vec{k}, \omega) & 0 \end{pmatrix}, \quad G_0(\vec{k}, \omega) = \frac{1}{Dk^2 - r - i\tau\omega}$$

(there is a basis that diagonalizes it). We need $\langle S_{\text{int}} \rangle_0 = \int \mathcal{D}[\phi] e^{-\int \phi \mathbb{L} \phi} S_{\text{int}} / (\dots)$; since S_{int} is cubic, $\langle S_{\text{int}} \rangle_0 = 0$. At second order,

$$\langle S_{\text{int}}^2 \rangle = \frac{\Gamma^2}{4} \int d1 d2 \phi_1 \tilde{\phi}_1 \phi_2 \tilde{\phi}_2 (\phi_1 - \tilde{\phi}_1)(\phi_2 - \tilde{\phi}_2),$$

which after coarse graining contains $2^6 = 64$ terms; the relevant ones have two $\vec{k}_-$ and four $\vec{k}_+$, and give

$$\langle S_{\text{int}}^2 \rangle \supset \int d\vec{k}_1 d\vec{k}_2 \vec{\Phi}(\vec{k}_1) \begin{pmatrix} 0 & \#\omega + \#k^2 \\ \#\omega + \#k^2 & 0 \end{pmatrix} \vec{\Phi}(\vec{k}_2) \delta(\vec{k}_1 + \vec{k}_2),$$

so $S_- = S_0 + (\text{new terms})$, which renormalize r and τ .